District Level Contingency Plans for Weather Aberrations in India

B. Venkateswarlu, A.K. Singh, Y.G. Prasad, G. Ravindra Chary,
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NATURAL RESOURCE MANAGEMENT DIVISION
INDIAN COUNCIL OF AGRICULTURAL RESEARCH
KRISHI BHAVAN, NEW DELHI - 110 114

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Front Cover
Clockwise : Drought effect, flooded paddy field, denuded grazing lands, cold injury in mango

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कृषि एवं खाद्य प्रसंस्करण उद्योग मंत्री
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13 July, 2011

MESSAGE

Monsoon is the most important factor impacting Indian agriculture, particularly during *kharif* season. Due to aberrations in monsoon behaviour in terms of onset, distribution and withdrawal, farmers are facing hardships in agricultural operations and often experience huge crop losses. Climate change and climate variability are likely to further aggravate this erratic behaviour. Every year, droughts and floods are playing havoc with agriculture sector resulting in loss of production and infrastructure. The Ministry of Agriculture and different State Governments have evolved contingency strategies which are implemented during droughts. However, little information is available on tackling other weather aberrations, particularly in preparedness and advance action. The existing documented contingent plans are of generic nature with broad strategies aimed at state or agro-climatic zone level. For addressing the problems in an effective and targeted manner, it is important that the country should have district level contingency plans covering agriculture and the allied sectors. The Parliamentary Committee on Agriculture has indeed suggested for preparing district level contingency plans for the entire country and put them into use during droughts, floods, cyclones, heat and cold waves, etc., to minimize losses to the farmers.

I am happy that the Indian Council of Agricultural Research at the behest of the Department of Agriculture and Cooperation has prepared district level contingency plans covering all aspects of weather aberrations like droughts, floods, un-seasonal rains, cyclones, heat wave and cold wave. It is the first such attempt to prepare plans upto the district level. The document captures the major farming systems in different districts, the type of natural hazard the district is prone to, and detailed strategies for each of the above contingencies for major food and horticulture crops in the district. For the first time, livestock, poultry and fishery sectors are also covered. Management practices to cope with extreme weather events are described for key commodities like wheat, rice, onion, mango and apple. Lastly, operational guidelines for implementing the plans are also included.

I hope the document will be highly useful to all the stakeholders both at central and state level in planning and implementation of contingency plans during weather aberrations. I complement the Department of Agriculture and Cooperation and Indian Council of Agricultural Research, particularly the Central Research Institute for Dryland Agriculture (CRIDA), Hyderabad for coming out with such an useful and timely publication.

(SHARAD PAWAR)

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Preface

Weather plays an immense role in agriculture. Timely onset and good distribution of rainfall are critical for achieving optimum crop yields by farmers, particularly during *kharif* season along with other factors like inputs, labour and technology. Temperature plays a key role during *rabi* season. Rainfall during the monsoon season not only determines the success of rainfed crops but also influences water availability to irrigated agriculture as most rivers in India are fed by rainfall unlike in many temperate countries where substantial waterflows are contributed by snow melt. Any deviation from normal monsoon pattern affects crop production, fodder availability to livestock and causes huge losses to farmers. Over the years, Indian agriculture has become somewhat resilient to climate variability mainly due to the contribution of nearly 50% of food production in the *rabi* season which is less vulnerable to rainfall variation. Even *kharif* production is achieving some degree of stability due to adoption of improved technologies by farmers, better preparedness by the government and consistent investments on land development and water conservation by the Government. However, monsoon rains wield a large influence on Indian agriculture production, domestic and international trade in food grains and consumer prices.

Whenever there is delay in monsoon or mid-season breaks, central and state governments are implementing contingency plans, particularly for droughts. The current plans are of generic nature implemented at state or a particular agro-climatic zone within the state. These strategies are not specific to the distinct farming situations or current weather conditions in a particular district to enable the district administration to take up focused and timely action. Since the administrative machinery to implement the contingency plans is organized at the district level, there has been a demand for preparing such plans for each district, separately.

In pursuance of the recommendations of the Parliamentary Committee on agriculture, the Department of Agriculture and Cooperation (DAC)entrusted the responsibility to ICAR. The authors have worked on this project for more than one year involving the cooperation of a large number of scientists from ICAR institutes and state agricultural universities. The authors tried to compile all relevant information on managing different weather related contingencies in a given district, including extreme events. For the first time, information on livestock, poultry and fisheries has been included.

We acknowledge the technical guidance and encouragement received from Dr S. Ayyappan, Secretary, DARE and Director General, ICAR particularly in facilitating the participation of all the agricultural universities and Shri P.K. Basu, Secretary, Department of Agriculture and Cooperation for conceptualizing the project, monitoring and writing to all the state governments for feedback on the template. We also thank all the Vice Chancellors and Directors of Research for guiding and permitting the nodal officers to have wide consultations within the SAU for preparing the plans. We earnestly hope that this document and the completed district plans provided as a soft copy will be useful to all the stakeholders at the central, state and district levels. The plans need to be revised continuously with updated information incorporating new farming practices coming out of research.

Authors

Hyderabad

14 July 2011

Chapter I

Background

Agriculture is the source of livelihood for nearly two-thirds of the population in India. The sector currently accounts for 14.2% of the GDP and employs 55% of country's total workforce. Agriculture growth rate in GDP was low in 2009-10 due to the unprecedented and widespread drought while it picked up again in 2010-11 due to a good monsoon. The sharp fluctuations in agricultural growth are mainly attributed to the vagaries of monsoon. Some part of the country or the other experiences monsoon failure almost every year but most states encounter drought once in 2-4 years (Table 1).

Table 1: Periodicity of deficit rainfall in various parts of the country

Frequency of deficit rainfall	Meteorological sub-division
Once in 2.5 years	West Rajasthan, Rayalaseema, Telangana, Haryana, Chandigarh and Delhi
Once in 3 years	East Rajasthan, Gujarat region, Jammu and Kashmir, Tamil Nadu and Pondicherry, West Uttar Pradesh
Once in 4 years	North Interior Karnataka, Uttarakhand, Vidarbha
Once in 5 years	Bihar, Coastal Andhra Pradesh, East Uttar Pradesh, Gangetic West Bengal, Jarkhand, Kerala, Orissa, South Interior Karnataka, Madhya Maharashtra, West Madhya Pradesh
Once in 15 years	Arunachal Pradesh, Assam, Meghalaya, Nagaland, Manipur, Mizoram and Tripura

*Source: Ministry of Agriculture, GOI,

The south west monsoon accounts for nearly 75% of the precipitation received in the country and therefore exerts a strong influence on kharif food grains production (Fig.1) and the economy in terms of agricultural output, farmers' income and price stability. Both the amount of rainfall and its distribution are crucial factors influencing performance of agriculture. The probability of monsoon rains being erratic is 40% of the time which implies that in 4 out of 10 years

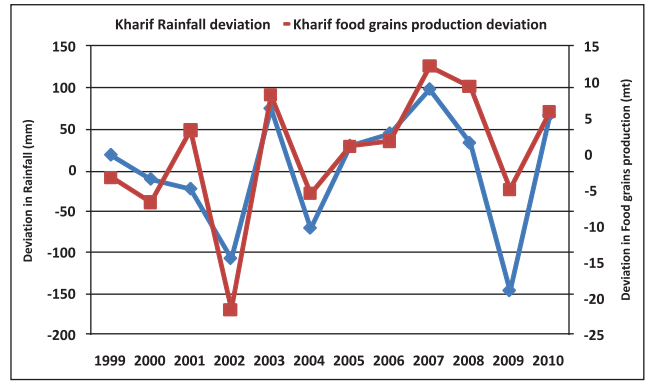


Fig.1 : Rainfall vs food grains production in Kharif season

there would be an adverse impact on crop production in the absence of appropriate strategy to deal with such eventualities.

Monsoon failures result in drought which has serious implications for small and marginal farmers and livelihoods of the rural poor. Rainfed areas constitute nearly 58% of the net cultivated area and account for 40% of the country's food production and support 40% of human and 60% of the livestock population and are the most vulnerable to monsoon failures. It has been estimated that even after full irrigation potential is realized, nearly 70 million ha of cultivated area will continue to be under rainfed farming. For this reason, performance of rainfed agriculture is key to achieve growth, equity and sustainability of agricultural production in the country. The demand for water from agriculture and allied sectors is ever increasing. Insufficient rainfall and the growing disregard for prudent use of water resources within the renewable limits has compounded the problem of water shortages which is being felt even in the irrigated regions. Increase of variability in precipitation being witnessed at the regional levels is forcing us to enhance our preparedness to face monsoon aberrations even in the irrigated regions.

Extreme weather events like heat wave, cold wave, untimely and high intensity rainfall, hailstorm and frost are increasingly being experienced in different parts of the country. For example heat wave during February-March in north India caused an estimated loss of 6 million tonnes of wheat in 2002 - 03. A decline in production of 39% in cocoa, 60% in rapeseed and 50% in linseed was observed in Himachal Pradesh due to heat wave in March 2004. Cold wave conditions in 2002-03 caused an estimated loss of 10-100% in different crops. In Punjab, 40-100% damage was recorded in mango and litchi in the Hoshiarpur district. In Rajasthan, about 20-30% damage in tomato and in Haryana 15-25% damage in mustard were recorded. Other extreme events like heavy rains accompanied by hailstorm during March 2007 damaged wheat, sugarcane and oilseed crops in thousands of hectares in Punjab and Haryana. In Madhya Pradesh, entire pigeonpea crop in an area of 7000 ha was damaged due to frost and extreme cold conditions. In 2009, heavy rainfall in Raichur district of Karnataka, Kurnool and Mahabubnagar districts of A.P damaged standing crops in lakhs of hectares due to floods and sand casting on river banks of Krishna and Tungabhadra rivers. During the same year, the areas affected by drought initially were also affected by flood later in the season resulting in contingency measures being taken up by respective district authorities. It is to be noted that the maximum expected flood limit in 100 years for Krishna river exceeded during the same year due to very intense rainfall events of more than 250 mm per day in the catchment.

Drought during 1987 in Barmer district of Rajasthan (Livestock Census, 1989) resulted in reduction in population of cattle by 52%, buffaloes by 15%, sheep by 58% and goats by 44%. The super cyclone in Orissa (ODMM, 1989) during October 1989 killed more than 3 lakhs each of large and small ruminants and more than 18 lakhs of poultry birds in 12 affected districts. The per day milk collection under the impact of the cyclone saw a 25% fall. The Flood of 1996 affected all the coastal districts of AP resulting in a loss of more than 66,000 cattle. Summer floods in 2004 caused flare up of diseases such as foot and mouth and bovine diarrhea in north eastern states (Sanjoy, 2006). Tsunami in 2004 affected shrimp hatcheries, loss of equipment and infrastructure in the east coast. Alappuzha and Kollam districts were severely hit by the tsunami and the estimated loss to the marine fisheries sector was assessed at Rs.1000 million with the mechanized sector accounting for 64% of the total loss. Nearly 10,880 fishing craft (18% of the craft operating in the state) were destroyed or damaged. Incessant rains during 2010 monsoon resulted in severe blue tongue disease outbreak in coastal districts of Tamil Nadu, Karnataka and Andhra Pradesh due to heavy breeding of the vector, *Culicoides* sp.

The widespread failure of the monsoon in 2009 and increasing frequency of occurrence of extreme events in recent years have reinforced the need for better preparedness, planning and response to mitigate the adverse impacts of such events. Four coping strategies for preparedness for climatic risks are disaster preparedness; mitigation practices; contingency planning and responses, and disaster risk mainstreaming (Rathore and Stigter, 2007). Contingency planning is one of the major strategies of preparedness for what should be done after the aberrant weather events that could not be prevented.

Need for district level contingency plans

- District is the administrative unit for operationalisation of any action plan or developmental plan by the State or Central Government under various sponsored schemes / programmes
- In many States, the contingency plans are currently available either at the agro-climatic zone of ICAR or at the State level prepared by respective SAUs or by other agencies in case of natural calamities. These plans are not directly amenable for implementation at the district level. Further, the existing plans do not contain information on allied sectors of agriculture such as horticulture, livestock, poultry and fisheries
- The existing 126 agro-climatic zones in the country have been delineated by ICAR under the National Agricultural Research Project (NARP) on the basis of topography, rainfall, soils, cropping pattern and irrigation availability. The zones comprise of relatively homogeneous districts or part of districts and are useful for regional level planning. However, despite similarities in dominant cropping pattern, agricultural practices adopted by the local farming community tend to vary according to the location within these zones. These variations are best captured and optimized to resource endowments while planning at the district level.

Many agro-climatic zones include whole or parts of at least two to three districts. Hence, the mismatch and pitfalls between the plan recommendations, their implementation and impact can be overcome by preparing district plans which cater to a contiguous unit.

- Many times the major weather aberrations happen at the micro level for example a district may experience drought and flood simultaneously which may not be the case at the agro-climatic zone level. Planning at the district level will address these climatic anomalies for quicker response mechanisms and help the administration to channelize the resources appropriately for effective mitigation of the adverse impacts of such eventualities
- Currently there are several developmental initiatives / schemes / programs like MGNREGS, RKVY, NFSM, NHM, ISOPOM, NLM, IWMP, NAIS, and WBCIS etc. being implemented in a district to provide both short term and long term support during natural calamities which can be leveraged to implement the district plans for better outcome
- Agriculture and allied sectors are basically input driven. Timely availability of inputs in right quantities is crucial in a normal season and more so in an abnormal season. Planning and projections for input requirements are done at the district level and hence contingency plans at the district level will provide the requisite technical information for projecting the demand like seeds, fertilizers, pesticides, feed, fodder and vaccines.
- Besides the above, awareness, sensitivity and capacity building of all stakeholders are integral to implementation of the contingency plans and this is best done when the plans are prepared through a participatory bottom-up process. Over a time these processes will lead to a self reliant planning and implementation mechanism at the local level
- Eventually aggregation of these district level plans will help plan contingency strategies at the state level and finally at the national level. Based on two to three years of operational experience, we can evolve computer based decision support systems (DSS) though which all districts in the country can be linked through the states and such implementation can be monitored at the national level on a real time basis.

Chapter II

Process of Preparation

- The Parliamentary Consultative Committee on Agriculture, Food, Civil supplies and Consumer Affairs, Government of India (GOI) has suggested the preparation of district level contingency plans for the entire country for droughts, floods, heat wave, cold wave and pest outbreaks etc., covering crops, livestock and fishery sectors based on the available research information
- Department of Agriculture and Cooperation (DAC), Ministry of Agriculture requested the Indian Council of Agricultural Research (ICAR) in the ICAR-DAC interface meeting to take up the responsibility of preparing contingency plans at district level for all the 126 Agro climatic zones of the country to deal with weather related aberrations
- The Secretary, DARE and Director General, ICAR entrusted the task to the NRM division which in turn identified CRIDA, Hyderabad to act as the nodal institute at the national level with the overall responsibility of planning, coordination and submission of the district level plans in a time frame. Regional level coordination was entrusted to seven more institutes of NRM division viz., NBSS & LUP, Nagpur for the western region; DWM, Bhubaneswar, for the eastern region, PDFSR, Modipuram for the northern region and ICAR Research Complex for NEH, Barapani for the north-eastern region while CRIDA, Hyderabad has the additional responsibility for the southern region. Further, scrutiny of draft plans was entrusted to three more NRM institutes viz., CAZRI, Jodhpur; CSWCR & TI, Dehradun and ICAR RCER, Patna.
- CRIDA, Hyderabad developed a draft template along with guidelines for preparing the district plans in a uniform manner across the country which was discussed first in the Directors meeting of NRM division on 15th January 2010 at New Delhi followed by consultations with the DAC
- The template was presented by the NRM division during the Vice Chancellor's Conference in February, 2010 and suggestions were invited for its improvement
- Vice Chancellors of all SAUs on the request of DG, ICAR identified Nodal Officers in each SAU at Associate Director level for coordinating the work involving scientists/officials from various districts under the SAU. In case of more than one SAU in a State, each SAU was required to prepare plans for the districts under its jurisdiction. In case of separate universities for agriculture, horticulture, veterinary and fisheries, the responsibility of the university was to provide respective information for all the districts in the State in their subject. However, the final integration of information on all sectors was done by CRIDA, Hyderabad.
- The final template comprised of two parts (see page 10). The first part provides information on the resources in the district at a glance while the second part deals with the strategies for weather related contingencies in respect of agriculture, livestock, poultry and fisheries sectors

- CRIDA, Hyderabad has constituted a dedicated multi-disciplinary team of scientists to exclusively work on this task
- The first step was to provide orientation to nodal officers about the template. In this regard five regional orientation workshops were organized at CRIDA, NBSS & LUP, DWM, PDFSR and ICAR Research Complex for NEH Region as given in Table 2

Table 2 : Details of Orientation Workshops conducted for preparation of district level contingency plans

Sl. No.	Region	States	Location of the meeting	Date(s)	Nodal Institute for organizing the workshop and vetting of plans
1	South	AP Tamil Nadu Karnataka Kerala	CRIDA Hyderabad	17 April,2010	CRIDA for all the 4 States
2	West	MP Gujarat Maharashtra Rajasthan	NBSS & LUP Nagpur	19 May, 2010	NBSS & LUP for Maharashtra and MP CAZRI, Jodhpur for Gujarat and Rajasthan
3	East	West Bengal Bihar Orissa Chhatisgarh Jharkhand	DWM, Bhubaneswar	24 May, 2010	DWM, Bhubaneswar for Orissa, West Bengal, Chattisgarh ICAR RCER for Bihar and Jharkhand
4	North	UP Punjab Haryana HP J & K Uttarakhand	PDFSR, Modipuram	3 June, 2010	PDFSR for UP, Punjab and Haryana CSWCR & TI, Dehradun for HP, J&K, Uttarakhand
5	North -East	Assam Meghalaya Manipura Mozoram Nagaland Sikkim Arunachal Pradesh Tripura	ICAR Research Complex for NEH Region, Umiam, Meghalaya	8 June, 2010	ICAR Res. Complex for NEH for all North Eastern States

- The template was further refined taking into account the regional requirements on the basis of inputs provided by the nodal officers at the workshops. This was followed-up with smaller orientation meetings at the SAU level where the template was explained to district teams
- The template was filled at the district level by multi-disciplinary teams drawing information from all available sources. The first draft of district plan was sent to the nodal officer of SAUs for preliminary scrutiny and compilation. The drafts were then sent to the nodal officer of the concerned ICAR institute and CRIDA for scrutiny and vetting
- CRIDA developed a check list for vetting of draft plans submitted by SAUs. The check list was explained and further improved in a vetting workshop conducted for nodal officers of ICAR institutes at Hyderabad on 28th August 2010.
- The vetted district plans were sent to back to the respective SAUs for revision of shortcomings and incorporation of suggestions
- The plans were again revised by the district teams and submitted to the SAU nodal officer. Vetting workshops were facilitated by CRIDA team and were organized at the SAU headquarter for finalization of the plans involving all the district teams, nodal officer of SAU and his team, nodal officer of ICAR and his team, invited experts from SAU and AICRPs. The plans were presented by the district team leaders, revised and finalized during the workshop (Table 3).
- The finalized plans were submitted to DAC. The finalized plans were also shared with the respective SAU and State department of agriculture for operational use, testing and further refinement. User feed back over time will help in revising and further improving these plans on a continuous basis

Table 3: Details of Vetting Workshops for preparation of district level contingency plans

S.No.	States	Location of the workshop	Date(s)
1	Maharashtra & Madhya Pradesh	NBSS & LUP, Nagpur	4 October, 2010
2	Gujarat & Rajasthan	CAZRI, Jodhpur	8 October, 2010
3	Karnataka	UAS, Bengaluru	20-21 October, 2010
4	AP	ANGRAU, Hyderabad	5-6 January, 2011
5	Orissa	OUAT, Bhubaneswar	19 February, 2011
6	Punjab	PAU, Ludhiana (KV Rao)	14-15 March, 2011
7	Maharashtra	MPKV, Solapur	15-15, March, 2011
8	West Bengal	BCKV, Kalyani	12-3 April, 2011
9	Maharashtra	PDKV, Akola	24-25, May, 2011

Dr. A. K. Singh, DDG (NRM), ICAR reviewing the progress at CRIDA

Orientation Workshop at DWM, Bhubaneshwar

Orientation Workshop at NBSS & LUP, Nagpur

Vetting Workshop at CAZRI, Jodhpur

Vetting Workshop at ANGRAU, Hyderabad

Chapter III

Standard Template

As desired by the Parliamentary Consultative Committee, the district-level contingency plans are to be prepared for all the major weather related aberrations including extreme events viz., droughts, floods, heat wave, cold wave, untimely and high intensity rainfall, frost, hailstorm, pest and disease outbreaks. These plans need to integrate information for agriculture and allied sectors like horticulture, livestock, poultry and fishery. In this regard, there was a need to develop a standard template to ensure a uniform format for providing contingency strategies across the country. The challenge is to bring uniformity in format so that it can be computerized in future with options to retrieve information for any district and at the same time accommodate the regional variations in climate and cropping patterns. The template consists of two parts viz., 1) District agricultural profile with information on resource endowments such as rainfall, land use, soil types, irrigation sources, five most dominant crops and cropping systems and their sowing windows; livestock, poultry and fisheries resources; production and productivity statistics; major contingencies faced by the district and digital soil and rainfall maps 2) Detailed strategies for weather related contingencies in the case of crops/cropping systems starting with delay in onset of monsoon and mid season breaks resulting in drought both in rainfed and irrigated situations, and strategies for untimely rains, floods, extreme events. This is followed by strategies for contingency situations, in the case of livestock, poultry and fisheries.

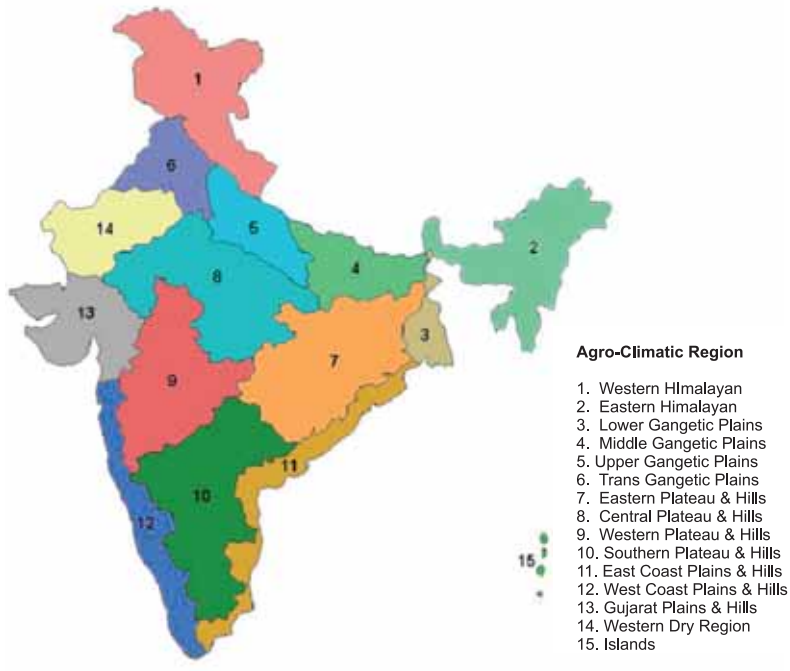
1.0 District Agricultural Profile

1.1 Agro Climatic / Ecological Zone for the district (nomenclature)

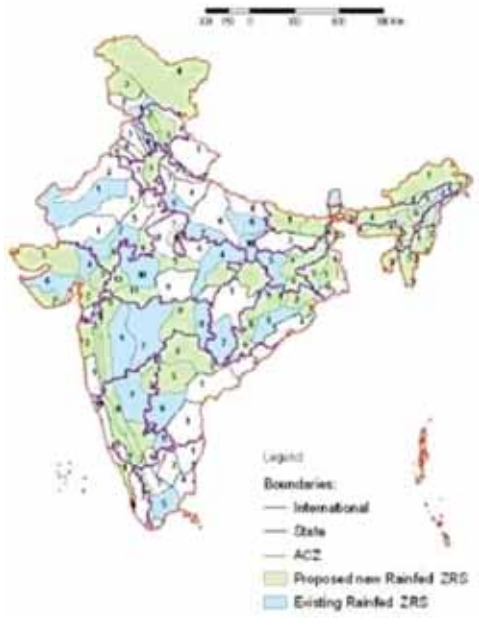
Agro Ecological Sub Region (AESR): The National Bureau of Soil Survey & Land Use Planning (ICAR) has delineated the country into 20 agro-ecological regions and further delineated into 60 AESRs based on bio-climate, soils, length of growing period and rainfall. An AESR gives details of physiography, climate, soils, State and districts etc.

Agro Climatic Region: The Planning Commission has demarcated the geographical area of India into 15 agro-climatic regions

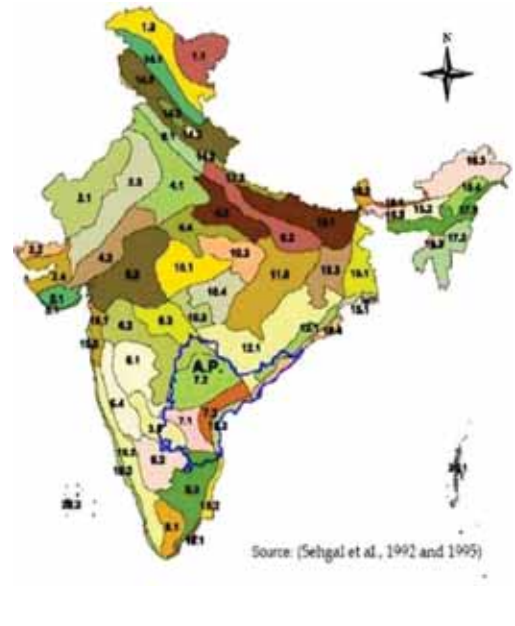
Agro Climatic Zone: The ICAR under the National Agricultural Research Project (NARP) identified 126 agro-climatic zones (ACZs) based on topography, rainfall, soils, cropping patterns and irrigation availability in 17 major states (now 20) and in 6 smaller states of North East Hill region. Few agro-climatic zones comprise one agro-climatic region of the Planning commission.



(Source : Guha and Basu, 1996)



(Source : Ghosh, 1991)



(Source : Sehgal *et al.,* 1999)

Listing of all the districts falling under the NARP Zone: Generally more than one district falls under a NARP zone and hence other districts which partly fall under the zone are listed here

Geographical co-ordinates of district headquarter: Latitude, longitude and the altitude of the district headquarter is given for geographical identity

Name and address of the concerned ZRS / ZARS / RARS / RRS / RRTTS: For every NARP zone of ICAR, a Zonal Research Station was started with a view to develop location specific management practices for existing production systems and districts within that zone. In order to collect relevant research and technology information pertaining to a district, the contact details of ZRS/ZARS/RARS/RRS/RRTTS are given. This also helps in sourcing technical back stopping by the district administration for implementation of the plan on a continuous basis.

Krishi Vigyan Kendra: Krishi Vigyan Kendra is a nodal extension institution in a district for technology demonstration, capacity building and allied activities. KVKs can serve as information source and further for dissemination and validation of contingency plans. The contact details of the KVK(s) within the district are given

Name and Address of Agromet Field Unit: Agromet field (AMFU) unit of SAUs are a repository of the local weather information and render agro advisory services, hence are useful to the local /district administration for any need based information, particularly during aberrant weather conditions.

1.2 Rainfall: Agriculture in most of the country is mainly dependent on the south west monsoon. In some parts of the country, the agricultural activities are also based on north east monsoon. Hence, information on the amount of rainfall and its distribution, number of rainy days during monsoon and post-monsoon periods is of much value in assessing the scenario and developing appropriate contingency plan for the district

1.3 Land Use Classification: A nine fold land use classification viz., geographical area, cultivable area, forest area, land under non agricultural use, permanent pastures, cultivable wasteland, land under miscellaneous tree crops and groves, barren and uncultivable land, current fallows and other fallows for the latest available data is given to understand the arable and non-arable land use scenario in the district.

1.4 Soils: Soil is the natural resource base that influences cropping pattern. Information on dominant soil types along with depth and their extent is essential for advocating contingency interventions. Drought in light textured soils will have quick negative effect and hence such reasons with in a district should receive higher priority in taking up contingency measures.

1.5 Gross and net sown area: The details on net sown area, area sown more than once and gross cropped area are useful to estimate the cropping intensity

1.6 Irrigation: The details on net irrigated area and gross irrigated area provide information on the extent of rainfed and irrigated area in the district. Sources of irrigation vary from district to district. Hence this information on extent of irrigation through canals, tanks, open wells, bore wells, lift irrigation schemes, micro irrigation, medium irrigation projects and other sources is important for a assessing the reliability of irrigation and developing contingency strategies in irrigated tracts. There is also need to identify the groundwater irrigated area within canal command areas as the recharge in these areas is dependent on canal irrigation systems rather than incident rainfall. In case of rainfed areas, groundwater irrigation systems are dependent on rainfall for recharge purpose

Groundwater availability and use: The information on groundwater availability and use including the quality of water (critical/safe) is important to understand not only the groundwater potential but also its efficient and sustainable use during scarcity periods.

1.7 Major field and horticultural crops in the district: The information on season-wise irrigated and rainfed area under five major field crops in the district are given. Similarly information is given for the dominant horticultural crops (fruits and vegetables), medicinal and aromatic crops, plantation crops and fodder crops along with the area under grazing land and other enterprises such as sericulture

1.8 Livestock: The statistics on livestock resources of the district viz., cattle, buffaloes, goats, sheep and others such as pig, camel and yak etc. indicates the importance of the livestock in the district

1.9 Poultry: Both commercial and backyard poultry statistics are given for developing contingency strategies related to extreme events and this sector is most vulnerable

1.10 Fisheries: Information on capture (marine and inland) and culture (brackish and fresh water) is given to indicate the dominant system in a district for developing suitable contingency strategies related to extreme events

1.11 Production and productivity of major crops: The production and productivity of five major crops in the district indicates average levels and will serve as a benchmark for assessing the losses during abnormal/extreme events scenario and further develop suitable strategies for bringing stability. This helps in focusing on the major corps which are key to the district economy

1.12 Sowing window: The normal start and end of sowing period for various crops in the district is generally evolved over a period of time based on monsoon rainfall, irrigation, soil type, susceptibility to pests and diseases research experimentation and farmers own experiences. Any delay beyond this window warrants contingency crop planning. This information is provided for 5 major crops for both rainfed and irrigated situations for different seasons i.e., *kharif*, *rabi* and summer.

1.13 Major contingency experienced in the district: In rainfed districts, drought is frequent. Floods are common in river basin/catchment/canal command areas when heavy rains occur. Cyclones are devastating in nature in coastal areas. Hailstorms are basically physiographic, weather and season related. Heat wave, cold wave, frost are localized temperature extreme events for short periods causing extensive crop damage and casualities in livestock, poultry and fisheries. Sea water inundation into cultivated area and main lands in coastal regions not only damages the crops but also deteriorates and degrades the soil making it saline. This can occur due to over draft of ground water in low lying coastal districts or in any coastal areas during tsunamis. Certain pests and diseases are triggered by weather events and result in outbreaks causing severe losses to both crops and animals. It is pertinent to identify the proneness of the district to one or more major weather related aberrations (regular i.e. > 6 out of 10 years; or occasional) so as to focus in developing appropriate contingency strategies.

1.14 Maps: Location map of the district within the state and soil map indicating spatial distribution of soil types are given. Mean annual rainfall is given month-wise in the form of a histogram. This helps in district level implementation team where to focus more attention in case of deficit rainfall.

2.0 Strategies for weather related contingencies

2.1 Drought: Drought is a recurrent phenomenon (Fig. 3) resulting from deficit in soil moisture and or water both in rainfed and irrigated areas. The probability (percent) of occurrence of severe drought is depicted in Fig. 4.

Fig. 3 : Drought is the most common contingency in rainfed areas

2.1 The effect is pronounced due to delays in onset of monsoon, prolonged breaks in monsoon and deficit rainfall, poor inflows into water bodies or poor recharge leading to reduction in cropped area, cropping intensity, dip in crop and livestock productivity and sometimes total crop failures adversely impacting rural livelihoods and economy.

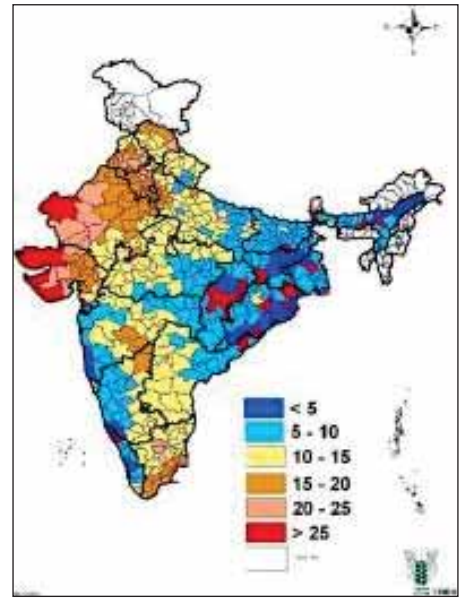


Fig.4: Percent probability of occurrence of severe drought

2.1.1 Rainfed situation: Contingent planning for rainfed crops is important when the onset of monsoon is delayed. Generally the south west monsoon covers the whole of the country in 8 weeks time starting first of June. Historical monsoon behaviour indicates the delays by 2 to 6 weeks across the country and in exceptional cases by 8 weeks. Hence the template gives scope for developing contingency strategies for 2, 4, 6 and 8 weeks delay in onset of monsoon for major farming situations prevalent in the district. The contingency measures suggested against the normal crop / cropping systems are in the form of alternate choice of crop / cropping systems, appropriate cultivars, and changes in agronomic practices along with suggested linkages with ongoing governmental schemes / programmes in the district

A. Early season drought (delay in onset of monsoon by 2, 4, 6 and 8 weeks)

a. Major farming situation: Provides information on growing environment (rainfall and soil information - colour, depth & texture) such as low rainfall shallow red sandy loam soils, high rainfall deep black soils, uplands, medium lands, eroded hill slopes etc.

b. Normal crops or cropping systems: The normal crops / cropping systems adopted in a given farming situation

c. Change in crop / cropping system including variety: Based on the extent of delay in monsoon and the sowing window for the major crops for a given farming situation, changes in crop / cropping systems are suggested. These suggestions are based on research conducted by the local agricultural research stations.

d. Agronomic measures: Again based on the extent of delay in monsoon and major farming situation, appropriate agronomic measures are suggested for alternate crops / cropping systems. These measures are mostly related to sowing operations such as increase in seed rate, changes in spacing, intercropping in case of sole crops and inter terrace land

management practices like ridges and furrows, dust mulching, broad bed furrow method of planting etc.

e. Remarks on implementation: Linkages with seed agencies are indicated for source of the seed for normal crop (short duration varieties) or suggested alternate crop. Further linkages are indicated for convergence with ongoing central or state government schemes like MGNREGS, IWMP, RKVY, NFSM, ISOPOM, NHM, NFSM, CLDP to meet the cost of materials, labour or implements etc.

B. Normal onset on monsoon followed by early, mid-season and terminal drought

The crop is sown under normal onset of monsoon. However, dry spells occur at various stages during the cropping season resulting in early, mid-season and terminal droughts adversely impacting performance of standing crops.

i) Early season drought due to 15-20 days dry spell after sowing

In case of normal onset followed by early season dry spell immediately after sowing leading to poor germination and poor crop establishment, re-sowing, gap filling, thinning may be necessary depending upon the crop and soil type. In addition, moisture conservation measures, nutrient management and life saving irrigation wherever possible are suggested.

ii) Mid-season drought at vegetative and reproductive stages of crop

In case of mid season drought, crop, soil nutrient management and moisture conservation measures are suggested. These include intercultivation for weed management, mulching and other *in situ* moisture conservation measures, protective irrigation, split or postponement of fertilizer application, foliar sprays of nutrients or anti-transpirants etc.

iii) Terminal drought

In case of terminal drought due to dry spells or early withdrawal of the south west monsoon, crop management measures such as life saving irrigation are suggested as in the case of mid season drought wherever feasible. In case of impending crop failure, harvesting at physiological maturity or early harvest of grain crops and use as fodder for livestock are suggested. In case of total crop failure, early rabi crop planning with suitable crops/varieties is the only option.

Adopt recommended location specific soil and water conservation measures based on soil type and rainfall even during a normal season and more particularly in an abnormal year as mandatory.

2.1.2 Drought in irrigated situation

Contingent planning for irrigated crops is also important under conditions such as : i) delayed release of water due to low rainfall in catchment areas

ii) limited release of water in canals due to low rainfall

iii) non-release of water in canals under delayed onset of monsoon in catchment area

iv) lack of inflows into tanks due to insufficient / delayed onset of monsoon and

v) insufficient groundwater recharge due to low rainfall

Hence, the template gives scope for developing contingency strategies for all these conditions. The contingency measures suggested against the normal crop / cropping systems are in the form of alternate choice of crop / cropping systems, appropriate cultivars, and changes in agronomic measures along with suggested linkages with ongoing governmental schemes / programmes in the district.

a. Major Farming Situation: Provides information on farming situations describing physiography, soils and source of irrigation such as tank fed medium or deep black/loamy/red soils, tube well irrigated red soils, canal irrigated red soils, well irrigated black soils etc. This is important as there are distinct soil-climatic regions which require different strategies within a district.

b. Normal crops or cropping systems grown in a given irrigated situation

c. Change in the crop, variety or cropping system is indicated in view of delay in release of irrigation water, less water availability etc.

d. Agronomic measures like improved methods of irrigation (skip row etc.), micro irrigation (drip/sprinkler/sub-surface), deficit irrigation, limited area irrigation and mulching that improve water use efficiency and make best use of limited water including methods of ground water recharge and sharing are indicated.

e. Linkages with seed agencies are indicated for source of the seed for normal crop (short duration varieties) or suggested alternate crop. Further linkages are indicated for convergence with ongoing central or state government schemes like MGNREGS, IWMP, RKVY, NFSM, ISOPOM, NHM, NFSM, CLDP to meet the cost of materials, labour or implements etc.

2.2 Unusual rains (untimely, unseasonal) both in rainfed and irrigated situations

In the recent past, continuous high rainfall in a short span leading to water logging and heavy rainfall coupled with high speed winds in a short span are being experienced at various growth stages of annual and perennial crops leading to serious crop losses (Fig. 5 & 6), outbreak of pests and diseases and sometimes total crop failure. These events at post-harvest stages lead to huge economic losses due to low prices and marketing failures of poor quality or damaged produce. The livestock and poultry sector suffer due to short supply of quality feed and fodder.

Fig.5: Damage to standing crop due to untimely rains

Fig.6: Submergence due to heavy rains

Suggested contingency measures include re-sowing, providing surface drainage, application of hormones/nutrient sprays to prevent flower drop or promote quick flowering/fruiting and plant protection measures against pest/disease outbreaks with need based prophylactic / curative interventions. At crop maturity stage suggested measures include prevention of seed germination and harvesting of produce. Post harvest measures include shifting of produce to safer place for drying and maintaining the quality of grain/fodder and protection against pest/disease damage in storage etc

2.3 Floods

Floods are common in river basins and coastal areas of the country (Table 4) leading to physical loss of crop (Fig. 7) human and livestock (Fig. 8) population. Also serious land degradation is an after effect which requires considerable effort to reclaim the land for cultivation. Heavy rainfall results in flash floods in streams and rivers, breach of embankments leading to transient water logging and continuous submergence of crop lands and entry of sea water into cultivated fields and intrusion of seawater into groundwater in coastal districts. Crop/field management depends on nature of material (sand or silt) deposited during floods.

Fig. 7: Compltely devastated banana crop in Thippalakatta village in Guntur district, A.P.

Fig.8: Heavy casuality of livestock due to flood in Kurnool district, A.P.

Table 4 : Flood years and their category

Year	Area affected ($1x10^6$ sq.km)	% of the area affected	MFI Value	Category
1961	1.795	57.166	3.614	Exceptional
1971	1.427	45.446	2.668	Exceptional
1878	1.513	48.185	2.889	Exceptional
1975	1.268	40,382	2.260	Exceptional
1884	1.175	37.420	2.021	Exceptional
1892	1.162	37.006	1.987	Exceptional
1933	1.145	36.465	1.943	Exceptional
1959	1.135	36.146	1.918	Exceptional
1983	1.030	32.803	1.648	Exceptional
1916	1.025	32.604	1.635	Exceptional

(Source : De *et al.,* 2005)

In sand deposited crop fields/ fallows, ameliorative measures suggested include early removal or ploughing in of sand (depending on the extent of deposit) for facilitating *rabi* crop or next *kharif*. In silt deposited indo-gangetic plains, early *rabi* crop plan is suggested in current cropped areas and current fallow lands. Other measures include drainage of stagnant water and strengthening of field bunds etc. In diara (flood prone) land areas, measures include alternate crop plans for receding situations. Usually rice cropped areas are flood prone causing loss of nurseries, delayed transplanting or damage to the already transplanted fields etc. Suggested measures include community nursery raising, scheduling bushenings, re-transplanting in damaged fields and transplanting new areas or direct seeding depending on seed availability so that the season is not lost. Other steps include prevention of pre-mature germination of submerged crop at maturity or of harvested produce by spray of salt solution.

2.4 Extreme events

Extreme weather events like heat wave, cold wave, frost, hailstorm and cyclone are climatic anomalies which have major impact on food, commercial and horticultural crops. In recent times the frequency of these events is increasing causing enormous damage not only to agriculture but also to other sectors like livestock, poultry and fisheries. Proper documentation of occurrence, impacts and traditional coping mechanisms is crucial for developing contingency strategies for such events at different crop stages viz., seedling / nursery stage, vegetative and reproductive stages and at harvest.

2.4.1 Heat wave:

In regions where the normal maximum temperature is more than 40^{0}C, if the day temperature exceeds 3^{0}C above normal for 5 days it is defined as heat wave. Similarly, in regions where the normal temperature is less than 40^{0}C, if the day temperature remains 5^{0}C above normal for 5 days, it is considered to be experiencing heat wave. The historical heat wave episodes in India are given in Table 5.

Table 5 : Heat wave epochs in India

State	Epochs				
	1911-67	1968-77	1978-99	2000-06	1901-2006
West Bengal	76	9	28	2	115
Bihar	105	6	23	12	146
U.P	27	3	42	32	104
Rajasthan	43	1	7	39	90
Gujarat (Saurashtra & Kutch)	-	2	-	9	11
Punjab	-	1	-	20	20
H.P	-	-	-	3	3
J&K	26	5	35	9	75
Maharashtra	82	4	13	14	113
M.P	32	4	15	38	89
Orissa	25	8	18	13	64
A.P	21	-	3	14	38
Assam	-	4	19	1	24
Haryana	-	1	2	28	31
Tamilnadu	5	-	2	6	13
Karnataka	-	-	-	6	6
A.P. (Telangana)	-	-	-	9	9
A.P. (Rayalaseema)	31	2	28	5	66

Source: 1911-67 : Bedekar FMU
1968-77: Disastrous weather events reports-Annual
1978-99: WMO Bulletin October 2000, 49, 4, pp.340-348
Source: Chattopadhyay and Lal, 2007

Eastern U.P, Punjab, east Madhya Pradesh, Saurashtra & Kutch in Gujarat are highly heat prone areas and heat waves were experienced in recent years during 1998, 2002, 2003, 2004 and 2007. The impact of heat wave on wheat and poultry are shown in Fig. 9 and 10, respectively.

Fig. 9: Heat wave impact on wheat in Punjab

Fig. 10: Huge mortality of poultry birds due to heat wave in A.P.

Generally affected crops due to heat wave are wheat, mustard, rapeseed, linseed and vegetables. Wheat production showed a loss of 4-6 million tonnes in recent years due to increased temperature in February-March. A decline of 39% in annual cocoa, 60% in rapeseed and 50% in linseed yield was noticed in Himachal Pradesh due to heat wave in March 2004. Ameliorative measures include conserving soil moisture through appropriate tillage practices, frequent and light irrigations, use of efficient water harvesting techniques, choosing heat tolerant varieties provide shelter to young fruit plants and Shelter management for livestock and poultry to alleviate heat stress.

2.4.2 Cold wave:

In regions where normal minimum temperature remains 10^0C or above, if the minimum temperature remains 5^0C lower than normal continuously for 3 days or more it is considered as cold wave. Similarly in regions where normal minimum temperature is less than 10^0C, if the minimum temperature remains 3^0C lower than normal it is considered as cold wave. The adverse impacts observed are on growth, flowering, fruiting, delay in ripening and mortality of young and aged orchard plants. Poor growth rate is observed and disease outbreaks are experienced in case of fisheries. Jammu & Kashmir, Rajasthan, U.P, Haryana, Punjab are identified as frequent cold wave prone areas. Recently cold wave was experienced during 2000, 2001 and 2003. The numbers of cold wave epochs experienced by India are as shown in the Table 6. The severity of cold wave in mango orchard is seen in Fig. 11 (Samra *et al.,* 2003).

Fig. 11: Mango orchard destroyed by severe cold wave in 2002-03

Crop damage estimates due to cold wave during 2002-03 is 10-100% depending upon crop and variety within the crop. In Hoshiarpur, Punjab about 40-100% damage was recorded in mango and litchi. In Jodhpur, Rajasthan about 20-30% damage in tomato and 5-10% damage in chilli was recorded. In Hisar 15-25% damage was recorded in mustard. Mostly horticultural crops (eg. mango, papaya, banana, litchi, pomegranate etc.) are affected by cold wave. Suggested measures include proper selection of fruit species /varieties which are cold tolerant, use of wind breaks or shelter belts, frequent smoking in the orchards and covering young fruit plants with thatches or plastic shelter.

Table 6: Cold wave epochs experienced in India during 1901-2006

State	Epochs					
	1901-10	1911-67	1968-77	1978-99	2000-06	1901-2006
West Bengal	2	14	3	28	3	50
Bihar	7	27	8	67	14	123
U.P	21	51	8	47	28	155
Rajasthan	11	124	7	53	28	223
Gujarat (Saurashtra & Kutch)	2	85	6	6	7	106
Punjab	3	34	4	19	48	108
H.P	-	-	4	18	31	53
J&K	1	189	6	15	43	254
Maharashtra	-	60	4	18	-	82
M.P	9	88	7	12	24	140
Orissa	4	5	-	-	12	21
A.P	2	-	-	-	-	2
Assam	1	1	-	-	-	2
Haryana	-	-	4	15	22	41
Tamilnadu	-	-	-	-	2	2
Karnataka	-	10	-	-	17	27
A.P. (Telangana)	-	5	1	-	8	14
A.P. (Rayalaseema)	-	3	-	3		6

Source: Chattopadhyay and Lal, 2007
1968-77: Disastrous weather events reports- Annual
1978-99: WMO Bulletin October 2000, 49, 4, pp.340-348

2.4.3 Frost:

It is a condition that exists when the temperature of the earth's surface and earthbound objects falls below zero degree (freezing). Frost is mostly experienced in the month of January. Himachal Pradesh, Punjab, Haryana, Madhya Pradesh are frost prone areas (De *et al.,* 2005). Crop damage due to frost in Madhya Pradesh was 100% in pigeonpea, sown in about 6990 hectare areas in 2011. Suggested measures include preference of frost tolerant varieties, change in planting time to avoid sensitive stages coinciding with frost periods, adopting shelter belts, shade trees, and use of mulches as ground cover to prevent loss of heat etc.

Fig. 12: Apple and Pear affected due to frost

Fig.13 : Maize crop damaged by frost

2.5.3 Hailstorm:

Hailstorm frequencies are highest in Assam valley, Uttaranchal, Jharkhand and Vidarba and its occurrence noticed during 1997-98, 2005-06, 2007 and 2011. About of 0.46 million ha cropped area in the states of Haryana, Punjab, Himachal Pradesh, Rajasthan, Uttar Pradesh, Maharashtra and Andhra Pradesh was badly hit during the year 1994-95 by hail storms (Fig. 14, 15 and 16). In Andhra Pradesh, hailstorm caused a huge loss in 77,000 ha area in 2005-06. In March, 2007 heavy rains accompanied by hailstorm damaged wheat, sugarcane and oilseed crops in thousands of hectares in Punjab and Haryana. Generally affected crops are apple, litchi, wheat and other fruit crops. Suggested measures include use of anti-hail guns and anti-hail nets, damaged fruits can be used in preparation of processed foods, crop insurance etc.

Fig. 14: Wheat crop damaged by hail

Fig. 15: Mustard crop damaged by hail in Gondkhairi watershed, Nagpur district, M.S.

Fig. 16: Papaya crop damaged by hail

2.4.4 Cyclone

Cyclone starts as a tropical low pressure depression, created by oceanic temperature rising above 26^{0}C and usually occurs between April-May and also between Oct-Dec in the country. The entire coast is affected by cyclones with varying frequency and intensity. Thirteen coastal states and Union Territories (UTs) in the country are affected by tropical cyclones (Table 7). Four states viz., Tamil Nadu, Andhra Pradesh, Orissa and West Bengal and one UT (Pondicherry) on the east coast and Gujarat on the west coast are more vulnerable to cyclone hazards. Plantation and horticultural crops are most vulnerable as the cyclones cause extensive uprooting leading to total crop loss and irreparable physical damage. Loss of livestock is common while in case of the fisheries sector, loss of lives, damage to infrastructure and equipment loss of stock and harvestable produce is observed. Inundation of cropped areas and sea water intrusion are associated with cyclonic events. Proper documentation of occurrence, impacts and traditional coping mechanisms is crucial for developing contingency strategies for such events at different crop stages viz., seedling / nursery stage, vegetative and reproductive stages and at harvest. Suggested measures include cultivation of storm resistant crops wherever feasible (e.g. ginger and pineapple) in highly prone districts, and planting of forestry windbreaks or shelter belts. Other measures include providing field drainage; staking and propping of plantation crops; cleaning and drying of harvested field crops; immediate harvest of fruits from damaged trees, grading and marketing.

Table 7. Most vulnerable districts for cyclone in the country

State	Degree of vulnerability	
	High	Very high
Tamilnadu	Kanyakumari, Sivaganga, Pudukottai, Puduchery and Karaikal	Thanjavur, Cuddalore, Kanchipuram, Thiruvallur, Tiruvanamalai, Viluppuram, Ramanathapuram, Nagapattinam, Thoothukudi, Tirunelveli,
Andhra Pradesh	West Godavari	East Godavari, Krishna, Guntur, Prakasam, Vizianagaram, Nellore, Visakhapatnam, Srikakulam
Orissa	Nayagarh, Dhenkanal	Cuttack, Ganjam, Jagatsinghpur, Kendrapara, Khordha, Puri, Baleswar, Bhadrak, Jajpur
West Bengal	Baradaman,Kolkata	Hugli, North 24 Parganas, South 24 Parganas, Midnapur
Gujarat	Narmada, Porbandar	Kutch, Junagadh
Maharashtra	Mumbai, Thane	-

Source: IMD report on cyclone hazard prone districts of India

2.5 Contingent strategies for livestock, poultry and fisheries

As desired by the Parliamentary Consultative Committee, the district plans integrate information on contingency measures for livestock, poultry and fisheries as these sectors are also prone to weather related anomalies and extreme events. The template gives scope for providing information for each sub-sector as detailed below:

2.5.1 Livestock

Rainfall deficit or drought is likely to affect livestock and crop sectors differently. In areas where crops and livestock occupy the same ecological niche, cropping is likely to be a more risky business. Livestock are able to compensate for localized rainfall shortages by movement to better favored areas. Drought impact on livestock sector manifests after a series of years with below average rainfall, causing a mounting imbalance in the availability of grazing resources mainly dry fodder. Floods and cyclones result in loss of livestock and stored fodder reserves and also cause serious injury to the animals and destroy animal shelters. Heat wave and cold wave also impact livestock. Higher ambient temperature substantially reduces feed intake, which results in loss of weight and production. Exposure to extremes and particularly unexpected cold can cause pneumonia or death of livestock. Extreme heat or cold waves are a threat to surviving animals after natural disasters due to non-availability of feed and shelter and outbreak of disease epidemics.

Possible contingency measures should address the primary effects of natural disasters on livestock such as :

- Loss of livestock and animal sheds
- Less or no fodder for the livestock as the crops and stored fodder are adversely affected
- High cost of fodder with reduced availability of fodder within the farm vicinity
- Increased expenses for transporting the fodder from neighburing areas
- Unavailability or high cost of drinking water for livestock
- Reduced milk and meat production
- Progressive decline in health status of livestock
- High incidence of contagious diseases
- Higher mortality rates
- Distress sale of livestock

A) Effect of drought on livestock

Drought affects the animal and animal production systems in different ways and leads to loss of productive animals, draft animals and livelihoods of the rural poor.

i) ***Feed and fodder availability*****:** A plunge in green fodder availability due to low rainfall is the first major effect of drought on livestock production systems. Low rainfall causes poor pasture growth and may also lead to a decline in fodder supplies from crop residues. During drought, plant growth and metabolism slows down as the available soil moisture declines progressively. Even under mild water deficit conditions, there is a reduction in cell division and protein synthesis and more lignification. This results in reduction in the total amount of plant biomass and its quality as animal fodder in a given growing season. Water stress in addition to over grazing substantially affects the pasture growth in grazing lands and in severe cases, grazing lands completely dry up without a single blade of grass (Fig. 17).

Fig. 17: Grazing Lands without a blade of grass during drought

ii) ***Drinking water:*** In majority of the cases, low rainfall also causes a drop in drinking water availability for livestock, precluding the effective grazing of certain pastures especially dry fodders. Further, frequent drinking of stagnant water, contaminated with snails which act as a reservoir and release endoparasites into water leads to severe worm burden in livestock.

iii) **Animal productivity**: As the access to fodder is limited during drought, malnutrition prevails leading to lower fertility and conception rates. Thus, drought in a year leads to lower calving rates in the subsequent year and finally leads to fall in production and also number of new animals added to the herd. This fall in production affects not only the offspring nutritional status but also the consumption level of the farmer's family. Animal's live weight falls as grazing becomes scarce, reducing the value of stock as meat animals. This loss of condition also makes them less valuable as draught animals and become more susceptible to contagious diseases like FMD, PPR, HS, BQ etc., leading to heavy morbidity and mortality. Mortality is comparatively high in small ruminants (sheep, goats) than large ruminants (cattle and buffaloes) as the former category is more vulnerable to drought. Young and elderly stock and pregnant animals are the most vulnerable within any particular species. As the period of drought continues, nutritional stress intensifies leading to high morality in livestock. In order to cope with drought situation, livestock farmers often migrate (Fig. 18) along with the herd to reduce number of family members dependent on the herd and also in search of other grazing sources. Long distance migration to unfamiliar areas also poses risks to the animals from exhaustion and from the change in pasture species and diet for incoming stock. In addition, movement into higher rainfall or high humidity zones often brings herds into contact with new range of diseases and parasites.

Fig. 18: Migration in search of grazing resources

iv) **Livelihoods**: Distress sale increases sharply as the farmers seek to salvage some value from their animals before they die due to lack of fodder. Initially, less important animals like males and aged females with poor calving and production history will be disposed off by the farmer. As the drought period lengthens and further intensifies, the farmer then is forced to sell breeding animals too. This results in rapid fall in alternative sources of income and purchasing power of the stakeholder. Drought also results in reduction in draft capacity of the animals, leading to lower crop output and income in the subsequent season. Loss of livestock also reduces the household's supply of dung, a product of considerable economic importance both as a fuel and as a means to retain soil fertility.

Strategies for livestock management during drought

Highest livestock density is seen in the most drought prone districts (Fig. 19). The impact of drought can be minimized and livestock productivity can be maintained at optimum level in the district after making a careful assessment of the situation for the prevailing livestock resources.

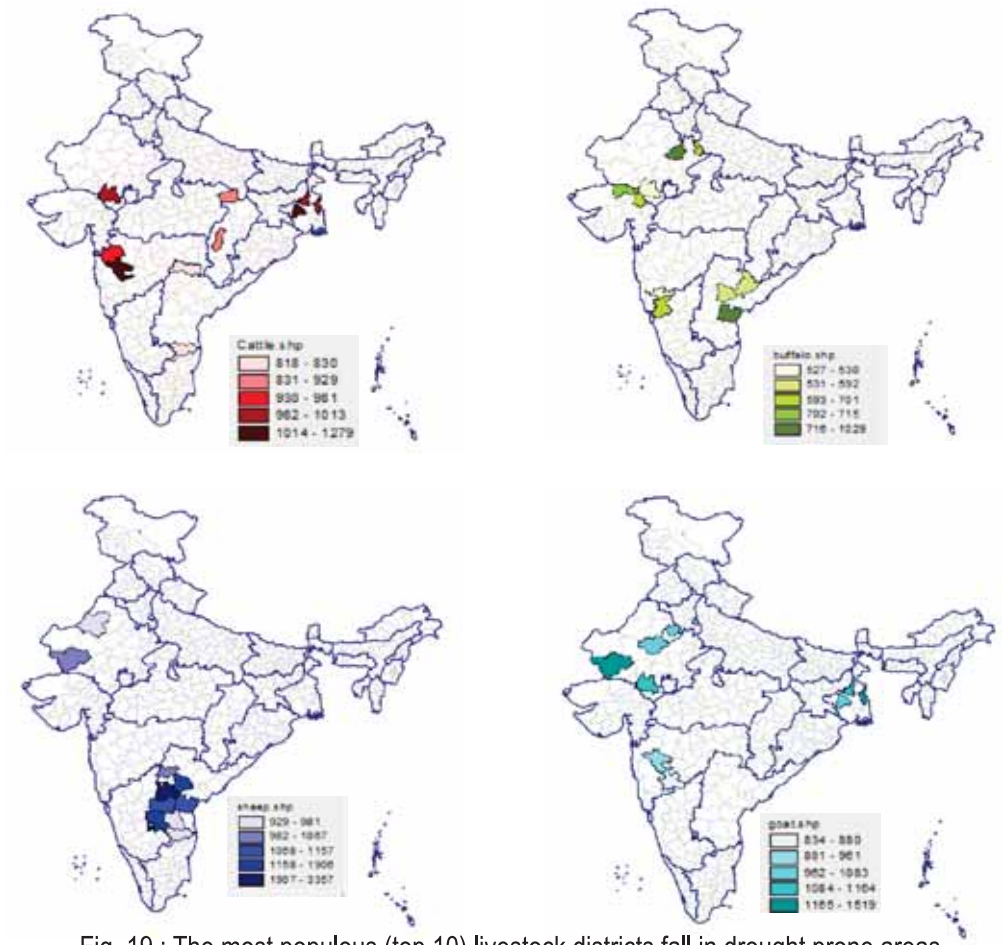


Fig. 19 : The most populous (top 10) livestock districts fall in drought prone areas

Measures for strengthening fodder resource base

- Proper preservation of available crop residues (Fig. 20) at individual farmer level and establishment of community fodder banks at village level with available dry fodder and crop residues
- Harvesting and use of dried up biomass of crops as fodder
- Early *rabi* crop planning with fodder yielding cereals like sorghum/ maize/bajra and leguminous crops like lucerne, horsegram and cowpea taking advantage of rains in North-East monsoon prevalent areas
- Available green maize fodder and sugarcane tops can be preserved as silage by the end of February for use during summer and early drought period
- Silvopastoral models with suitable tree and understory grass component can be developed in common property resources (CPRs)
- Fodder production with short-term fodder crops like sunhemp, pilli pesara (*Phaseolus trilobus)*, horsegram etc. in private lands

Fig. 20 : Stocking of fodder in the village

Measures for feeding management

- Emergency grazing/feeding facilities (cow-calf camps or other special arrangements to protect high productive & breeding stock) may be arranged during severe drought. Also concentrate ingredients such as grains, bran, chunnies and oilseed cakes, low grade grains etc. unfit for human consumption can be procured from government godowns and supplied to the livestock owners for feeding as a supplement
- Optimum ration to be provided for high producing, pregnant and breeding animals while un-productive, sick and aged animals can be culled

- Harvesting the entire top fodder available in and around the village for feeding the livestock as supplement during drought
- Enrichment of crop residues with 2% urea-molasses solution or 1% common salt solution and supplementation with mineral and vitamins as green fodder is scarce
- During prolonged drought, arrangements may be made for mobilization of small ruminants across the districts which are free from drought

Measures for grazing land management

- During drought, strict adherence to the practice of cut and carry system ensures protection of the growing grass from trampling. It also reduces the nutrient requirement of the animals as the animal does not spend any energy in search of grazing resources
- In drought prone areas, before event measures like reseeding or planting of high yielding perennial grass species and bush clearing can be followed to enhance forage availability from grazing lands. Seeding of legumes like *Stylo* with onset of monsoon and application of P as basal dose should be practiced for faster growth of pastures. Silvopastoral models with suitable tree and understory grass component should be developed in range lands with initial protection from grazing

B) Effect of floods and cyclones on livestock

Floods result in loss of livestock, stored fodder reserves and also cause serious injury to the animals. In case of severe floods cattle get stranded and suffer (Fig. 21). Due to the prolonged inundation during floods, all the crops including fodder is destroyed or seriously damaged. Floods cause destruction of animal shelters. Massive loss of livestock after floods or cyclones severely impacts the livelihoods of farmers.

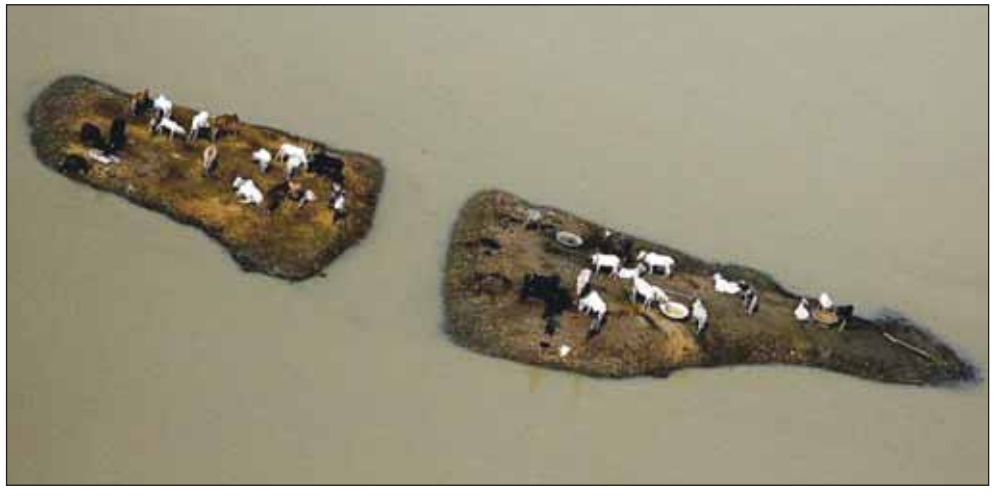


Fig. 21: Cattle stranded on small islands surrounded by flood waters in Madhepura district, Bihar

High intensity rainfall often results in flash floods. This results in loss of livestock and stored fodder reserves. Drowning and washing away of animals in floods is the most common impact. Wet conditions after floods or cyclones enhance chances of infection by internal parasites like round worms, tape worms, liver flukes as well as many epidemic diseases like FMD, HS, BQ etc. There may be non-specific water borne infections causing diarrhea and other enteric diseases in livestock. Animals that have been standing in water for prolonged periods of time may have pneumonia or foot rot.

Measures for livestock management in case of floods and cyclones

- In case of early forewarning (EFW), harvest all the crops that can be useful as feed or fodder in future and properly stored from inundation by flood waters
- Drowned crop material is to be collected, dried and stored properly for future use
- Sufficient dry fodder is to be transported to the flood affected villages to meet the fodder requirements. Apply urea (20-25 kg/ha) in the inundated areas and CPRs to enhance the biomass production
- In case of EFW of severe floods, animals are not sent for grazing. However, animals are un-tethered or let loose in shelters to escape drowning in rising flood waters
- Sick, injured and affected animals are to be treated through arrangement of mobile emergency veterinary hospitals or transporting rescue animal health workers
- Sufficient stock of bleaching powder and lime supply is to be maintained for application in sheds and relief camps inundated by flood waters. Similarly emergency medicines and vaccines are to be procured and stocked for important endemic diseases of the area
- Arrangements may be made for transportation of animals to elevated areas

C) Effect of heat or cold waves on livestock

Higher ambient temperature accompanied by feeding dry fodder reduces feed intake, resulting in loss of weight, production, anoestrus, premature calving etc. High yielding and crossbred animals are more affected with heat wave. Increase in temperature by 5°C results in decrease in livestock production by 10 percent. During hot summer, water consumption increases drastically often causing shortage of drinking water. Lack of water in most of the wells, ponds, tubewells and other water bodies during hot summer season has an adverse effect on animal production. Exposure to extreme and particularly unexpected cold can cause pneumonia or death of livestock. If cold wave is accompanied by heavy and persistent snow, animals unable reach grazing areas and die of hypothermia and starvation. Extreme heat or cold waves kill the plants and lead to famines. This situation often necessitates the farmers to purchase feed at considerable cost to feed their animals. High yielding animals are most severely affected. Cold stress reduces milk production and increases intake requirements in milch animals.

Measures for livestock management during heat / cold waves

- Protect the animals from heat wave by providing well ventilated shed with plantation around
- Allow the animals for grazing depending upon the season and avoid exposing to extreme cold and hot conditions
- Vaccinate as per the schedule

Measures for health and disease management in livestock for weather anomalies

- Procure and stock emergency medicines and vaccines for important endemic diseases of the area
- Surroundings of the animal shed must be kept hygienic
- Rapid Action Veterinary Force may be constituted for immediate action and control of diseases in livestock
- Restrict the movement of livestock in case of any epidemics and perform ring vaccination (8 km radius)
- Fly repellants should be sprayed in animal sheds and relief camps to control the vectors which spread diseases
- Provide wholesome clean drinking water thoughout the year
- Observe for anoestrus conditions in livestock and provide mineral and vitamin supplements. Sprinkle water during hot periods of the day, particularly in case of buffaloes to bring the animals to heat
- Dead animals or carcasses are properly disposed off by burning / burying (4-8 feet deep) with lime powder (1 kg for small ruminants and 5 kg for large ruminants) in pits

2.5.2 Poultry

A) Effect of drought on poultry

Drought results in malnutrition in backyard poultry as availability of insects in scavenging areas and grain at household level gradually declines. This creates stress and impairs heath of the birds and makes them more vulnerable to diseases. The diseases like Ranikhet (RD) will affect the backyard poultry and cause heavy mortality. Further, as the availability of drinking water becomes scarce, birds die due to heat stroke. Drought results in low weight gain, lesser egg production and low quality eggs not suitable for hatching in backyard poultry. Drought also affects the commercial poultry industry as the availability of grains becomes problem and prices go up. High feed costs result in loss of small holders' livelihoods. It results in low chicken and table egg production and increases the cost of chicken and eggs.

B) Effect of cyclones / floods on poultry

Cyclones and floods severely impair poultry production and cause destruction of sheds and flare up disease epidemics. This results in loss of poultry and stored feed reserves. Drowning and washing away of birds is the most common problem in floods. Birds are blown away during

cyclone and high winds. Inundation of water in feed godowns leads to wetting of feed and stored grain with subsequent mold growth (eg. aflatoxin producing) making the material unfit for use. Ammonia accumulates in the poultry sheds due to dampness, which causes irritation and results in low production. Higher humidity during cyclones/floods increases the incidence of respiratory infections in poultry. Wet conditions after floods or cyclones enhance chances of infections like coccidiosis and non-specific water borne infections causing diarrhoea in poultry. Cyclones/floods also interrupt the daily farm operations like egg collection, storage and incubation. Power failure results in mortality of chicks in brooders due to low ambient temperature.

C) Effect of heat/cold wave on poultry

Sudden and severe heat wave results in heavy mortality in broilers and low egg production in layers. The feed intake and weight gain is drastically reduced. Water requirement increases and litter becomes more moist leading to accumulation of ammonia in the sheds.This causes irritation and results in low production. Hatching percentage declines and increases mortality. Heat stress makes the birds more vulnerable to disease epidemics. Prolonged and severe cold wave impairs poultry production. Respiratory infections prevail and growth is hampered. Under cold and foggy weather, the chances of chickens falling sick and dying are high. The litter becomes wet and leads to accumulation of ammonia in the sheds. Infections like coccidiosis may severely affect health and production of poultry.

Measures to mitigate effect of extreme events on poultry

- Provide proper shelter and optimum nutrition to the birds
- Follow regular vaccination schedule
- Dispose poultry manure to prevent accumulation of ammonia in the sheds
- Sprinkle lime powder in sheds to prevent ammonia accumulation
- Provide cool and clean drinking water with electrolytes and Vit. C in hot summer
- In severe heat waves, arrange coolers/foggers and in severe cold waves, arrange heaters in the shed
- Dispose dead birds by burning / burying with lime powder in pit

2.5.3 Fisheries

A) Effect of drought on fisheries

Fish or prawn in ponds become more crowded and confined in a smaller volume of water and are stressed as the water from ponds get evaporated during drought. Adding to this stress is the deterioration of water quality in the pond. Evaporation concentrates waste metabolites like ammonia, carbon dioxide, and nitrites in the water. Increased concentrations of these metabolites can kill entire stock. Further, there will be less total amount of oxygen available for

the fish to breath as warmer water holds less dissolved oxygen. The sea surface temperature (SST) of the Indian coast increased by about 0.2-0.3°C during the period 1960 to 2005. Warming of water may impact fish diversity, distribution, abundance and phenology. Changes in species composition of phytoplankton may occur at higher temperatures. Shifts in the spawning season of fish are now evident in the Indian seas due to climate change. Some species may be found in deeper waters as well and phenological changes may occur. Distributional changes in fish species might lead to novel mixes of organisms in a region, leaving species to adjust to new prey, predators, parasites, diseases and competitors and result in considerable changes in ecosystem structure and function.

B) Effect of cyclones / floods on fisheries

Cyclones and floods severely impact fisheries sector. They cause destruction of ponds and heavy loss of nets, pumps, boats etc. Decline in the marine fish landings and subsequent catches due to low production was observed after cyclones in A.P. The immediate effect of floods on individual fish species appears to be largely dependent on fish size, life stage and habitat complexity. Floods can washout larval and juvenile stages while having little impact on adults particularly in complex habitats. Flash floods, alter stream channel morphology, kill or displace biota downstream or remove potential resources.

C) Effect of heat/cold wave on fisheries

Heat waves severely reduce dissolved oxygen level in water and affect growth and spawning behavior of fish and prawns. Heat or cold waves adversely affect yield of Indian major carps and increase their vulnerability to diseases. The effect on overall health of stocked fishes may be more when such conditions prevail for a longer duration. Significant decline in feed intake takes place below 14-18°C among Indian major carps resulting in malnutrition. In acute cold conditions, bacterial septicemia and fungal infections can occur resulting in heavy mortality in fish. In case of prawns, the mortality could be due to low oxygen and high nutrient load in the pond in prolonged hot or cold conditions.

Measures to mitigate the effects of extreme events

- Shifting the farming time because of shifting of monsoon season
- Promotion of diversified psciculture resistant to weather anomalies and extreme events
- Constructing and renovating ponds and canals for rain water harvesting and use in winter cultivation
- Constructing mud-barrages around the islands to protect them from saline water intrusion
- Reforesting of mangroves on the mud barrage
- Set up early warning systems and capacity building (Prakash Rao and Vivekanandan, 2008):

Chapter IV

Model District Plans

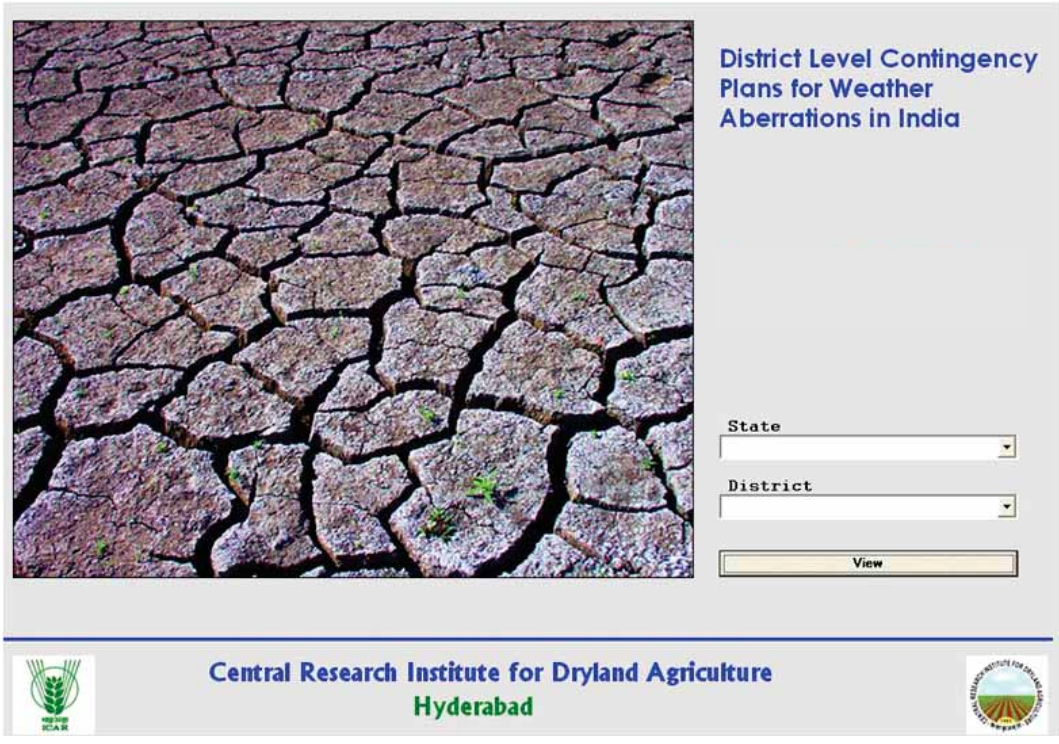
The district level contingency plans are prepared as a bottomup process as detailed in Chapter II. The plans are prepared as per the standard format explained in Chapter III. The current status on the preparation of the plans is given in Table 8. Till June 2011, 209 out of 576 districts have been completed. Highly urbanized districts are omitted where agriculture is not a significant activity.

Table 8. Progress in Preparation of District level contingency plans*

State	Plans envisaged	Plans completed	In process	Yet to be prepared
Andhra Pradesh	22	22	0	0
Karnataka	28	24	4	0
Kerala	14	14	0	0
Tamilnadu	31	31	0	0
Gujarat	25	23	1	1
Maharashtra	33	21	10	2
Madhya Pradesh	50	0	20	30
Rajasthan	32	9	21	2
Haryana	21	19	0	2
Himachal Pradesh	12	0	4	8
Jammu & Kashmir	22	0	2	20
Punjab	20	19	1	0
Uttar Pradesh	69	0	50	19
Uttarakhand	13	0	0	13
Bihar	38	0	26	12
Chhattisgarh	18	0	0	18
Jharkhand	24	0	7	17
Orissa	30	27	3	0
West Bengal	18	0	18	0
Assam	21	0	12	9
Manipur	6	0	5	1
Tripura	2	0	2	0
Sikkim	4	0	4	0
Nagaland	11	0	10	1
Arunachal Pradesh	5	0	4	1
Mizoram	4	0	3	1
Meghalaya	3	0	3	0
TOTAL	**576**	**209**	**210**	**157**

*As on 30th June, 2011

Each district plan runs into 40 to 50 pages. Hence, printing all district plans (576) is beyond the scope of any publication. Two model plans one for rainfed district (Amravati in Maharashtra) and another for irrigated district (Krishna in Andhra Pradesh) are included *in toto* in the following pages. However, all the finalized district plans (209) are provided in the enclosed compact disc. Users can access and retrieve the district plans as illustrated below.



Model Plan for Rainfed situation

District : Amravathi
State : Maharashtra

1.0 District Agriculture profile

1.1	Agro-Climatic/Ecological Zone	
	Agro Ecological Sub Region (ICAR)	Western Maharashtra Plateau, hot moist semi-arid eco- sub region (6.3)
	Agro-Climatic Region (Planning Commission)	Western Plateau and Hills Region (IX)
	Agro Climatic Zone (NARP)	Central Maharashtra Plateau Zone (MH-7)
	List all the districts or part thereof falling under the NARP Zone	Amravati, Akola, Buldhana, Washim
	Geographic coordinates of district headquarter	Latitude: 20° 55' 53.82" N; Longitude: 77° 45' 32.57" E; Altitude: 374 m above MSL
	Name and address of the concerned ZRS/ ZARS/ RARS/ RRS/ RRTTS	Regional Research Center (Dr. PDKV), Morshi Road, Amravati-444603
	Mention the KVK located in the district with full address	KVK, Durgapur, Tq. Badnera Distt. Amravati- 444701 KVK, Ghatkhed, Tq. Chandur Rly. Distt. Amravati "Chirantan" Madhuban Colony,Camp,Amravati-444602
	Name and address of the nearest Agromet Field Unit for agro-advisories in the Zone	AMFU Station, Akola- 444 104

1.2	Rainfall	Normal RF(mm)	Normal Rainy days (number)	Normal Onset	Normal Cessation
	SW monsoon (June-September	775.2	40	2nd week of June	1st week of October
	NE Monsoon(October-December)	69.6	4	-	-
	Winter (January- February)	21.7	3	-	-
	Summer (March-May)	19.9	1	-	-
	Annual	886.4	48	-	-

Source: IMD

1.3	Land use pattern of the district	Geographical Area	Cultivable area	Forest area	Land under non agricultural use	Permanent pastures	Cultivable waste land	Land under miscellaneous tree crops & groves	Barren & uncultivable land	Current fallows	Other fallows
	Area ('000 ha)	1304	766	321	16	29	20	8	28	19	97

Source: DACNET 2005-06

1. 4	Major soil types	Area ('000 ha)	Percent (%) of total geographical area
	Deep black soils	653.7	55.9
	Medium deep black soils	13.1	1.1
	Shallow black soils	501.2	42.9
	Others:	Saline soils in Purna river valley area	

Source: NBSS & LUP, Nagpur

1.5	Agricultural land use	Area ('000 ha)	Cropping intensity %**
	Net sown area	602	118.3
	Area sown more than once	110	
	Gross cropped area	712	

Source: DACNET 2005-06

1.6	Irrigation	Area ('000 ha)		
	Net irrigated area	51.3		
	Gross irrigated area	63.8		
	Rainfed area	540.7		
	Sources of Irrigation	Number	Area ('000 ha)	Percentage of total irrigated area
	Canals		4.5	8.8
	Tanks	-	-	-
	Open wells	55382	46.7	91.2
	Bore wells	38	-	-
	Lift irrigation schemes	3	-	-
	Micro-irrigation		-	-
	Other sources (please specify)	-	-	-
	Total Irrigated Area		51.2	
	Pump sets	40835		
	No. of Tractors	7077		

Source: DACNET 2005-06

	Groundwater availability and use (Data source: State/Central Ground water Department /Board)	No. of blocks / Tehsils	Quality of water (specify the problem such as high levels of arsenic, fluoride, saline etc)
	Over exploited	3	Morshi, Warud and Daryapur tehsils
	Critical	1	RSC > 2.5, unsuitable for irrigation purpose in well water of Chandur Bazaar tehsil
	Semi- critical	-	-
	Safe	10	-
	Wastewater availability and use	-	-
	Ground water quality	-	Good and suitable for drinking and irrigation purpose except the saline areas of Purna Alluvium. The areas of Purna River Alluvium covering southern parts of Anjangaon and Achalpur talukas and entire Daryapur taluka are affected by inland salinity problem.
*over-exploited: groundwater utilization > 100%; critical: 90-100%; semi-critical: 70-90%; safe: <70%			

Source: CGWB, 2004

1.7 Area under major field crops & horticulture etc. (average of last 3 years)

1.7	Major Field Crops cultivated	Area ('000 ha)						Summer	Total
			Kharif			*Rabi*			
		Irrigated	*Rainfed*	Total	*Irrigated*	Rainfed	Total		
	Soybean	-	317.6	317.6	-	-	-	-	317.6
	Cotton	-	179.4	179.4	-	-	-	-	179.4
	Pigeonpea	-	102.1	102.1	-	-	-	-	102.1
	Sorghum	-	49.1	49.1	-	-	-	-	49.1
	Greengram	-	37.1	37.1	-	-	-	-	37.1
	Chickpea	-	-	-	-	79.0	79.0	-	79.0
	Wheat	-	-	-	45.4	-	45.4	-	45.4

Source: District Social and Economic Abstract 2010, Directorate of Finance and Statistics, Govt. of Maharashtra

	Horticulture crops – Fruits	Total area ('000 ha)
	Mandarin (Santra)	80.8
	Mosambi	4.2
	Mango	5.1
	Sapota	0.1
	Guava	0.3
	Aonla	0.3

Horticulture crops – Fruits	Total area ('000 ha)
Kagzi lime	1.1
Custard apple	0.3
Ber	0.7
Tamarind	0.01
Banana	0.4
Pomegranate	0.2
Papaya	0.1
Total	**93.6**

Horticulture crops – Vegetables	Total area (000'ha)
Onion	1.7
Tomato	0.4
Brinjal	0.5
Cabbage	0.2
Cauliflower	0.2
Cluster bean	0.1
Dolichos bean	0.1
Lady's finger	0.2
Spinach	0.2
Fenugreek	0.05

Horticulture crops – Vegetables	Total area (000'ha)
Ridge gourd	0.03
Cucumber	0.05
Bitter gourd	0.05
Other	0.1
Total	**3.8**

Medicinal and Aromatic crops	Total area (000'ha)
Piper longum	0.8
Safed musli	0.01
Palmarosa, Aromatic grasses	0.04
Plantation crops	**-**
Total fodder crop area	**-**
Grazing land	29
Sericulture etc (Mulbery)	0.06

Source: Project Manager (Special Project), Maharashtra State Horticulture, Medicinal Plants Board, Pune and State Department of Agriculture

1.8	Livestock	Male ('000)	Female ('000)	Total ('000)
	Non descriptive Cattle (local low yielding)	257.8	262.5	520.4
	Crossbred cattle	8.0	20.7	28.7
	Non descriptive Buffaloes (local low yielding)	12.8	102.0	114.0
	Graded Buffaloes	0.6	3.6	4.2
	Goat	64.9	222.6	291.5
	Sheep	4.5	17.7	22.2
	Others (Camel, Pig, Yak etc.)			
	Commercial dairy farms (Number)			
1.9	**Poultry**	**Total No. of birds ('000)**		
	Commercial	22.2		
	Backyard	409.4		

Source : *Livestock Census 2007, GOI

1.10	Fisheries						
	A. Capture						
	i) Marine	**No. of fishermen**	**Boats**		**Nets**		Storage facilities (Ice plants etc.)
			Mechanized	Non-mechanized	Mechanized (Trawl nets, Gill nets)	Non-mechanized (Shore Seines, Stake & trap nets)	
		Not applicable					
	ii) Inland	**No. Farmer owned ponds**		**No. of Reservoirs**		**No. of village tanks**	
	B. Culture	5		41		279	
		Water Spread Area ('000 ha)		**Yield (t/ha)**		**Production ('000 tons)**	
	i) **Brackish water**	Not applicalbe					
	ii) **Fresh water** (Data Source: Fisheries Department)	94.36		0.4		3.79	
	Others						

Source: State Fisheries Department

1.11 Production and Productivity of major crops (Average of 5 years: 2004, 2005, 2006, 2007, 2008)

1.11	Name of crop	*Kharif*		*Rabi*		Summer		Total	
		Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)
	Major field crops								
	Soybean	242.8	764	-	-	-	-	242.8	764
	Cotton Lint	238.3	1328	-	-	-	-	238.3	1328

	Name of crop	*Kharif*		*Rabi*		Summer		Total	
		Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)
	Greengram	14.8	399	-	-	-	-	14.8	399
	Pigeonpea	76.9	753	-	-	-	-	76.9	753
	kharif Sorghum	92.8	1890	-	-	-	-	92.8	1890
	Wheat	-	-	47.5	1046	-	-	47.5	1046
	Chickpea	-	-	57.6	729	-	-	57.6	729
	Major Horticultural crops								
	Banana	-	-	-	-	-	-	23.6	6000
	Orange	-	-	-	-	-	-	501.1	6197
	Onion	-	-	-	-	-	-	33.4	2000
	Others	-	-	-	-	-	-	0.2	200

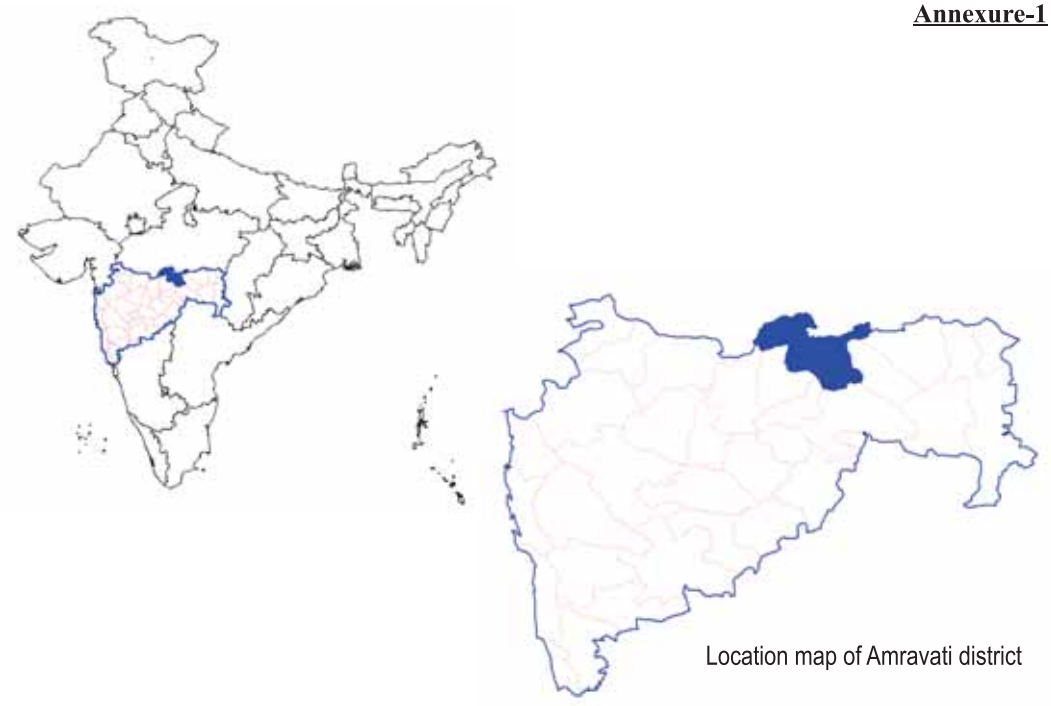
Source: District Social and Economic Abstract 2010, Directorate of Finance and Statistics, Govt. of Maharashtra

1.12	Sowing window for 5 major field crops	Cotton	Soybean	Greengram & Blackgram	Pigeonpea	*Kharif* Sorghum
	Kharif - Rainfed	15th – 30th June	15th June – 15th July	15th – 30th June	15th June – 10th July	15th June – 10th July
	Kharif - Irrigated	20th - 30th May (Pre-monsoon) 1st – 30th June	--	--	--	--
	Major *rabi* crops	**Wheat**	**Chickpea**	**Safflower**	--	--
	Rabi - Rainfed	--	1st -15th October	25th September - 7th October	--	--
	Rabi - Irrigated	1st – 15th November	15th October - 15th November	1st - 30th October	--	--

1.13	**What is the major contingency the district is prone to** (Tick mark)	Regular	Occasional	None
	Drought	--	✓	--
	Flood	--	--	✓
	Cyclone	--	--	✓
	Hail storm	--	✓	--
	Heat wave	✓	--	--
	Cold wave	--	✓	--
	Frost	--	--	✓
	Sea water intrusion	--	--	✓
	Pests and disease outbreak 1) Soybean - *Spodoptera litura*, Semilooper, *Helicoverpa*, girdle beetle, Soybean Mosaic Virus 2) Cotton - Jassids, Mealybug, BLB, red leaf disorder 3) Pigeonpea - Pod borer complex, Sterility mosaic disease, wilt 4) Citrus - *Phytopthora* (gummosis)/CTV/FSM 5) Chickpea - wilt, gram caterpillar 6) Safflower - aphid	✓	--	--

1.14	**Include Digital maps of the district for**	Location map of district within State as Annexure 1	Enclosed: Yes
		Mean annual rainfall as Annexure 2	Enclosed: Yes
		Soil map as Annexure 3	Enclosed: Yes

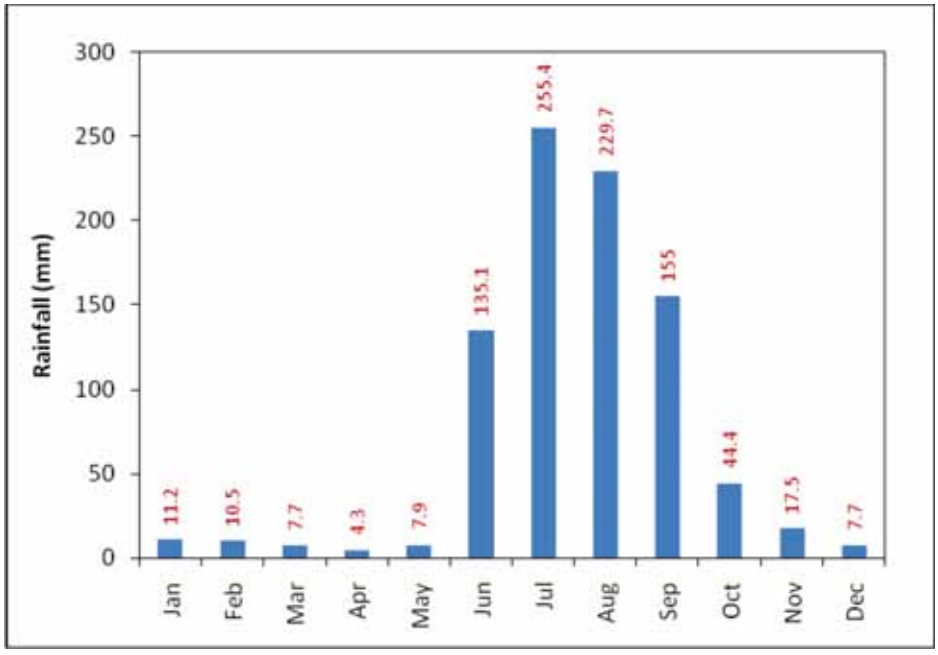
Annexure-1



Location map of Amravati district

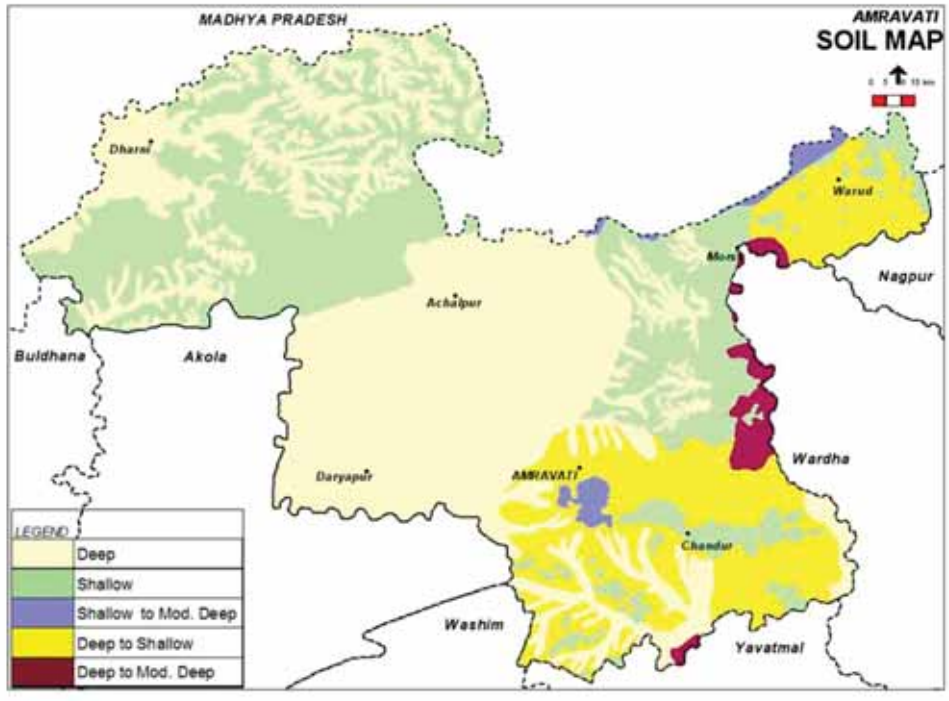
Tehsils and road network of Amravati district

Annexure 2



Mean monthly rainfall (Source: IMD) (1941-1990)

Annexure 3



Soil Map of Amravati District (Source: NBSS & LUP, Nagpur)

2.0 Strategies for weather related contingencies

2.1 Drought

2.1.1 Rainfed situation

Condition				Suggested Contingency measures	
Early season drought (delayed onset)	**Major Farming situation**	**Normal Crop / Cropping system**	**Change in crop / cropping system including variety**	**Agronomic measures**	**Remarks on Implementation**
Delay by 2 weeks **4th week of June**	Deep and medium deep black soils	Cotton	No change	Normal recommended package of practices of Dr.PDKV, Akola	Linkage with Dr.PDKV, MSSC, NSC
		Cotton + Pigeonpea	No change	• Normal recommended package of practices of Dr. PDKV, Akola • Cotton + Pigeonpea (6:2) • Cotton + Greengram/ Blackgram (1:1)	
		Soybean	No change	• Normal recommended package of practices of Dr. PDKV, Akola • Test seed for higher germination percentage • Adopt seed rate of 75-80 kg/ha • Seed treatment with Thiram 3 g Carbendazim 1g / kg seed, dried and then treated with bio -inoculants such as *Rhizobium* 200g + PSB 200g and *Trichoderma* 40 g for every 10 kg seed • Prefer intercropping with one row of pigeonpea after every 4 or 6 rows of soybean • Open the furrow after every 3 or 6 rows of soybean	

Condition				Suggested Contingency measures	
Early season drought (delayed onset)	Major Farming situation	Normal Crop / Cropping system	Change in crop / cropping system including variety	Agronomic measures	Remarks on Implementation
	Shallow black soils*	Pigeonpea	No change	• Normal recommended package of practices of Dr. PDKV, Akola • Prefer intercropping : pigeonpea + soybean (4:2 or 6:2 ratio) or pigeonpea + cotton (8:1 or 6:2 ratio)	
		Sorghum (*kharif*)	No Change	• Normal recommended package of practices of Dr. PDKV, Akola • Seed treatment with Imidacloprid 70 WS @ 7g/ kg seed and Thiram @ 3g / kg seed	
		Soybean	No change	• Normal recommended package of practices of Dr. PDKV, Akola • Test seed for higher germination • Adopt a seed rate of 75-80 kg/ha. • Seed treatment as above	
		Greengram	No Change	Normal recommended package of practices of Dr. PDKV, Akola	
		Blackgram	No change	Normal recommended package of practices of r.PDKV, Akol a	

*Farmers do cultivate cotton in shallow black soils also; however, the productivity is low

Condition Early season drought (delayed onset)	Major Farming situation	Normal Crop / Cropping system	Change in crop / cropping system including variety	Suggested Contingency measures Agronomic measures	Remarks on Implementation
Delay by 4 weeks **(2nd week of July)**	Deep and medium deep black soils	Cotton	Soybean (varieties JS-335, JS-93 -05) Pigeonpea (varieties AKT- 8811, Vipula , PKV- Tara, BSMR-736)	• Normal recommended package of practices of Dr. PDKV, Akola • Test seed for higher germination percentage • Adopt seed rate of 75-80 kg/ha • Seed treatment with Thiram 3 g + Carbendazim 1g /kg seed, dried and then treated with bio -inoculants such as *Rhizobium* 200g + PSB 200g and *Trichoderma* 40 g for every 10 kg seed • Prefer intercropping with one row of pigeonpea after every 4 or 6 rows of soybean • Open the furrow after every 3 or 6 rows of soybean • Weed free condition at critical stages of crop growth • Sowing at wider spacing (90x90 cm)	Linkage with Dr.PDKV for breeder seed / MSSC & NSC for foundation & certified seed
		Cotton + Pigeonpea	Prefer early varieties/hybrids of American cotton / desi cotton No change in varieties for Pigeonpea	• In cotton adopt 20-25% more seed rate than recommended and reduce fertilizer dose by 25% • Replace hybrids with improved varieties in cotton (American Cotton: AKH-8828, PKV Rajat, AKH-081, Desi Cotton: - AKA-5, AKA-7, AKA-8) • Avoid intercropping of Greengram and Blackgram in cotton • Adopt intercropping system of cotton + sorghum + pigeonpea + sorghum (6:1:2:1) to reduce the risk due to delayed sowing • Maintain weed free condition at critical stages of crop growth	Linkage with PDKV / MSSC, NSC for seed
		Soybean	No Change	• Maintain weed free condition at critical stages of crop growth • Normal recommended package of practices	
		Pigeonpea	Prefer varieties: AKT 8811,Vipula, PKV-Tara, BSMR-736	• Adopt weed free condition at critical stages of crop growth • Adopt a spacing of 90 x 20 cm instead of 90 x 30 cm	

Condition Early season drought (delayed onset)	Major Farming situation	Normal Crop / Cropping system	Change in crop / cropping system including variety	Suggested Contingency measures Agronomic measures	Remarks on Implementation
		Sorghum (*kharif* Jowar)	Replace sorghum with soybean (varieties JS-335, JS-93 -05) or with pigeonpea (varieties AKT 8811, Vipula, PKV- Tara, BSMR-736)	• Follow normal recommended package of practices • Maintain seed free condition at critical stages of crop growth	Source of breeder seed : Dr.PDKV / MAU/ MPKV Source of foundation & certified seed : MSSC & NSC
	Shallow black soils	Soybean	No change in varieties	• Normal recommended package of practices of Dr. PDKV, Akola • Test seed for higher germination percentage • Adopt seed rate of 75-80 kg/ha • Seed treatment with Thiram 3 g + Carbendazim 1g /kg seed, dried and then treated with bio - inoculants such as *Rhizobium* 200g + PSB 200g and *Trichoderma* 40 g for every 10 kg seed • Soil test based application of fertilizers is recommended	
		Greengram	Replace Greengram with Blackgram	Seed treatment as above	
		Blackgram	No change	As above	

Condition			Suggested Contingency measures		
Early season drought (delayed onset)	**Major Farming situation**	**Normal Crop / Cropping system**	**Change in crop / cropping system including variety**	**Agronomic measures**	**Remarks on Implementation**
Delay by 6 weeks **(4th week of July)**	Deep and medium deep black soils	Cotton	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tara, BSMR-736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi) • Pearlmillet + pigeonpea inter-cropping(2:1 or 4:2 row ratio)	• Adopt closer spacing (60 x 30 cm) in pigeonpea • Frequent interculture for *in situ* moisture conservation and for weed free conditions	Source of breeder seed : Dr.PDKV / MAU/ MPKV Source of foundation & certified seed : MSSC & NSC
		Cotton + Pigeonpea	Sole pigeonpea / sunflower / sesame / castor / pearlmillet, pearlmillet + pigeonpea as above	As above	
		Soybean	Sole pigeonpea / sunflower / sesame / castor / pearlmillet, pearl millet + pigeonpea as above	As above	
		Pigeonpea	No change. Prefer varieties AKT-8811, Vipula, PKV Tara, BSMR-736.	• Seed hardening • Soil test based fertilizer application is recommended • Frequent interculture for *in situ* moisture conservation and for weed free conditions	

Condition			Suggested Contingency measures		
Early season drought (delayed onset)	Major Farming situation	Normal Crop / Cropping system	Change in crop / cropping system including variety	Agronomic measures	Remarks on Implementation
		Sorghum	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tar a, BSMR-736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi) • Pearlmillet + pigeonpea inter-cropping(2:1 or 4:2 row ratio)	• Adopt closer spacing(60x30 cm)for pigeonpea • Follow *insitu* moisture conservation measures. • Apply 2% urea as foliar spray for millets	Linkage with MSSC and NSC for seed and Dr.PDKV , KVK for agro-techniques
	Shallow black soils	Soybean	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tara, BSMR -736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi) Pearlmillet + pigeonpea inter-cropping (2:1 or 4:2 row ratio)	• Seed hardening • Follow *in situ* moisture conservation measures • Soil test based fertilizer application • Weed free condition at critical stages of crop growth	
		Greengram	Same as above	Same as above	
		Blackgram	Same as above	Same as above	

Condition			Suggested Contingency measures		
Early season drought (delayed onset)	Major Farming situation	Normal Crop / Cropping system	Change in crop / cropping system including variety	Agronomic measures	Remarks on Implementation
Delay by 8 weeks **(2nd week of August)**	Deep and medium deep black soils	Cotton	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tara, BSMR-736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi) • Pearlmillet + pigeonpea inter-cropping(2:1 or 4:2 row ratio)	• Adopt closer spacing (60x30 cm)for pigeonpea • Follow *in situ* moisture conservation measures	Linkage with MSSC and NSC for seed and Dr.PDKV , KVK for agro-techniques
		Cotton + Pigeonpea	Same as above	-do-	
		Soybean	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tara, BSMR-736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi)	-do-	
		Pigeonpea	Prefer varieties: PKV Tara and BSMR-736	-do-	
		Sorghum (*kharif*)	• Sole pigeonpea (varieties AKT-8811, Vipula, PKV Tara, BSMR-736) • Sunflower (hybrids) / sesame (variety AKT64) • Castor (varieties / hybrids: AKC-1, GCH-4, 5, 6 and DCH-117, DCH 32) • Pearlmillet (varieties PKV Raj, Shradha, Saburi)	-do-	
	Shallow black soils	Soybean	• Sunflower (hybrids) • Sesame (AKT64) • Pearlmillet (PKV Raj Shradha, Saburi)	-do-	

Condition	Suggested Contingency measures				
Early season drought (Normal onset)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moisture conservation measures**	**Remarks on Implementation**
Normal onset followed by 15-20 days dry spell after sowing leading to poor germination/crop stand etc.	Deep and medium deep black soils	Cotton / Cotton + Pigeonpea	• Give protective irrigation wherever possible • Raise cotton seedlings in nursery & transplant when sufficient soil moisture is available • Gap filling aided with pot watering 7-10 days after sowing when crop stand is less than 80% • In case of less than 30% germination, take up re-sowing with a wider spacing of 45 cm when sufficient soil moisture is available	• Avoid applying fertilizer till sufficient soil moisture is available • Mulching with crop residue • Apply organic matter / FYM / Compost for better moisture retention • Making of conservation furrows for moisture conservation. • Sowing on broad bed furrow (BBF)	-
		Soybean	• Give protective irrigation wherever possible • Gap filling with Maize/Sesame • If germination is less than 50% re-sowing immediately after receipt of rains • Take up one hoeing for weed management	-do-	
		Pigeonpea	• Gap filling either with Sesame or Maize • Provide protective irrigation, wherever possible. • Take up one hoeing	-do-	
		Sorghum (*kharif*)	Take up thinning to maintain optimum plant population	-do-	
	Shallow black soils	Soybean	• Give protective irrigation wherever possible • Gap filling with Maize/Sesame • If germination is less than 50% re-sowing immediately after receipt of rains Take up one hoeing for weed management	-do-	
		Greengram	• Protective irrigation wherever possible Take up thinning to maintain optimum plant population and one hoeing	-do-	
		Blackgram -	do-	-do-	

Condition	Suggested Contingency measures				
Mid season drought (long dry spell, consecutive 2 weeks rainless (>2.5 mm) period)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moisture conservation measures**	**Remarks on Implementation**
At vegetative stage	Deep and medium deep black soils	Cotton / cotton + pigeonpea	Avoid applying fertilizer till sufficient soil moisture is available Interculture for weeding	• Opening of alternate furrows • Mulching with crop residue • Take up intercultivation to create soil mulch to conserve moisture • Give protective irrigation, if possible	Linkage with on-going government schemes to encourage adoption of micro-irrigation for better use efficiency of scarce water Linkage with on-going farm ponds programme and IWMP for rainwater harvesting and efficient use of water with micro-irrigation techniques like sprinklers
		Soybean	-do-	-do-	
		Pigeonpea	-do-	-do-	
		Sorghum (*kharif*)	-do-	-do-	
	Shallow black soils	Soybean	-do-	• Foliar spray of 2 % urea or DAP • Other measures as above	
		Greengram	Intercultivation	• Spraying of 2 % urea or DAP. • Protective irrigation if possible.	
		Blackgram	-do-	-do-	

Condition	Suggested Contingency measures				
Mid season drought (long dry spell)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moisture conservation measures**	**Remarks on Implementation**
At flowering/ fruiting stage	Deep and medium deep black soils	Cotton / cotton + pigeonpea intercropping	Protective irrigation, if possible	• Apply foliar spray of 2 % urea or DAP • Adopt soil moisture conservation measures like ridges and furrows • Supplemental irrigation (10 mm depth) with harvested rain water in ponds by adopting micro-irrigation (sprinklers)	Linkage with on-going government schemes to encourage adoption of micro-irrigation for better use efficiency of scare water
		Soybean	-do-	-do-	
		Pigeonpea	-do-	-do-	
		Sorghum (*kharif*)	-do-	• Protective irrigation, if possible • In case of poor grain filling harvest for fodder. • Adopt soil moisture conservation measures like ridges and furrows	
	Shallow black soils	Soybean	Protective irrigation	-do-	
		Greengram	-do-	-do-	
		Blackgram	-do-	-do-	

Condition	Suggested Contingency measures				
Terminal drought (Early withdrawal of monsoon)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Rabi Crop planning**	**Remarks on Implementation**
	Deep and medium deep black soils	Cotton / Cotton + Pigeonpea Intercropping	• Give life saving supplemental irrigation, if available • Picking / harvesting at physiological maturity	-	Linkage with NFSM or ISOPOM for seed supply and other inputs
		Soybean	-do-	Plan for *rabi* crops Chickpea/Safflower	
		Pigeonpea	-do-	-do-	
		Sorghum (*kharif*)	-do-	-do-	
	Shallow black soils	Soybean	-do-	-do-	
		Greengram	-do-	Prepare for *rabi* sowing provided irrigation is available	
		Blackgram	-do-	-do-	

2.1.2 Irrigated situation:

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
Delayed release of water in canals due to low rainfall	Deep and medium deep black soils	Wheat and Chickpea	Wheat to be replaced by Chickpea/Safflower/Mustard	Follow alternate row / micro-irrigation Irrigate at critical crop growth stages	Linkage with on-going government schemes to encourage adoption of micro-irrigation for better use efficiency of scare water
	Shallow black soils	Chickpea	Safflower / Mustard	-do-	-do-
Limited release of water in canals due to low rainfall	Deep and medium deep black soils	Wheat and Chickpea	Wheat to be replaced by Chickpea/Safflower/Mustard/ Linseed/Sesamum	Follow alternate row or micro-irrigation Irrigate at critical crop growth stages	As in previous condition
	Shallow black soils	Chickpea	Safflower / Mustard	-do-	
Non release of water in canals under delayed onset of monsoon in catchment	Not applicable				
Lack of inflows into tanks due to insufficient /delayed onset of monsoon	Not applicable				
Insufficient groundwater recharge due to low rainfall	Open well irrigated *rabi* cropping situation	Wheat, Chickpea, Safflower	Chickpea / Safflower	Adopt micro-irrigation (sprinkler)	Linkage with on-going IWMP Encourage percolation tanks for groundwater recharge

2.2 Unusual rains (untimely, unseasonal etc) (for both rainfed and irrigated situations)

Condition	Suggested contingency measure			
Continuous high rainfall in a short span leading to water logging	**Vegetative stage**	**Flowering stage**	**Crop maturity stage**	**Post harvest**
Cotton	• Open field channels to drain excess water and avoiding surface ponding • Apply 2% urea foliar spray after cessation of rains • Interculture at optimum soil moisture to improve soil aeration	• Open field channels to drain excess water and avoid surface ponding • Apply multi-nutrient or hormonal spray to promote flowering	• Open field channels to drain excess water and avoid surface ponding • Timely picking of cotton	• Protect picked cotton in storage from drenching and soiling • Drying of wet cotton and marketing
Soybean	Provide drainage	Provide drainage	Timely harvesting of produce	Shifting to safer place and drying of produce
Greengram	As above	-do-	-do-	-do-
Blackgram	As above	-do-	-do-	-do-
Pigeonpea	• Open field channels to drain excess water and avoid surface ponding • Interculture at optimum soil moisture to improve soil aeration	Open field channels to drain excess water and avoid surface ponding	-do-	Stacking and drying of produce followed by threshing
Horticulture				
Acid Lime and orange	Opening of field channels to drain out excess water and avoid surface ponding in the orchard Interculture at optimum soil moisture to improve soil aeration	• Mrig bahar crop is unaffected • For Ambi bahar crop, open field channels to drain out excess water and avoid surface ponding, • Nutrient spray of NAA 10 ppm + 1% urea to prevent flower drop	Timely harvest to avoid losses	Grading of fruits, cleaning of mold affected ones followed by washing and waxing

Condition	Suggested contingency measure			
Heavy rainfall with high speed winds in a short span	**Vegetative stage**	**Flowering stage**	**Crop maturity stage**	**Post harvest**
Cotton	Open field channels to drain excess water and avoiding surface ponding	Opening of field channels to remove surface ponding,	Timely picking in case of early forewarning of rains	Shifting to safer place for drying
Soybean Greengram Blackgram Pigeon pea	Opening of field channels to remove surface ponding	Opening of field channels to remove surface ponding	Timely harvesting in case of early forewarning of rains	Shifting to safer place for drying
Horticulture				
Nagpur Mandarin Acid lime and sweet orange	Provide bamboo staking to less than 3 year aged plants to avoid lodging	• Provide bamboo staking to less than 3 year aged plants to avoid lodging • Opening of field channels to drain out excess water and avoid remove surface ponding	Opening of field channels to remove surface ponding,	Collection and grading of fallen fruits followed by washing, waxing and marketing

Outbreak of pests and diseases due to unseasonal rains	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Cotton	Provide drainage for removing stagnant water and drench plant base with copper oxy chloride 0.3% or carbendazim 0.1% particularly in low lying patches in the fields to prevent *Fusarium* wilt incidence	• Incessant rains trigger grey mildew incidence-apply foliar spray of sulphur @ 25 g/10 litres water • Wet spells aggravate bacterial leaf blight incidence, protect with streptocycline sulphate 6 g / 10 litres + copper oxychloride 30 g/ l0 litres • Drench plant base with carbendazim 0.1% or COC 0.3% after rains to prevent wilt incidence in low lying patches in field • Remove water logging as excess soil moisture leads to leaf reddening. Go for soil application of $MgSO_4$ @ 20-25 kg/ha or foliar spray of 0.5-1.0% $MgSO_4$ and 1% urea as soon as the reddening symptoms appear • Timely correction of N status either by optimum supply in the soil or through foliar application of 2% urea or DAP at boll development stage reduces leaf reddening	Incessant rains trigger grey mildew incidence – prevent with foliar spray of sulphur @ 25 g/10 litres water Protect boll rot with carbendazim 0.1% spray immediately after cessation of rains	Drying of wet cotton to prevent molds

Outbreak of pests and diseases due to unseasonal rains	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Soybean	• Early planted soybean is likely to be attacked by gridle beetle and green semilooper due to copious rains. Watch for drooping and drying of leaves. • Manually remove the infested plants or plant parts from below the girdles Protect against semilooper when density reaches 2-4 larvae per m row with foliar spray of NSKE 5% or dimethoate 30 EC 1 ml/l	Monitor adult moth activity of *Spodoptera* through pheromone traps (10 traps /ha) and observe egg masses and gregarious larvae. Wet spell followed by a dry spell of 7-10 days during flowering or up to two weeks after flowering is likely to trigger severe pest incidence in endemic areas. When density crosses ETL of 1-2 larvae m row length, apply quinalphos 25 EC 20 ml/10 litres or Emamectin benzoate 5 SG @ 4 g/10 litres or Profenofos 50 EC @ 25 ml/10 liters or Lambda cyhalothrin 5 EC @ 6 ml/10 litres	-	-
Green gram	Protect against powdery mildew with foliar application of penconozol 5 ml or dinocap 10 ml or tridemorph 5 ml or sulphur spray @ 30 g/10 litre of water.	To control powdery mildew penconozol 5 ml or dinocap 10 ml or triadomorph 5 ml or sulphur spray @ 30 g/10 litre of water.	-	-
Black gram	-do-	-do-	-	-
Pigeon pea	Improved field drainage of excess water and drenching with copper oxy chloride @ 25g/10 liters of water to avoid incidence of wilt and root rot	Improved drainage and drenching at plant base with copper oxy chloride @25g/10 liters of water to avoid incidence of wilt and root rot	Protect against pod borer complex with dichlorvos at 1 ml / litre or quinalphos 2 ml / litre	-

Outbreak of pests and diseases due to unseasonal rains	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Horticulture				
Mandarin Orange	Protect against Citrus *psylla* with foliare spray of malathion 50 EC 10 ml or quinalphos 25EC 10 ml or cypermethrin 25 EC 4 ml/10 litres	Protect against Citrus *psylla* with foliar spray of malathion 50 EC 10 ml or quinalphos 25 EC 10 ml or cypermethrin 25 EC 4 ml/10 litres	-	-
Sweet Orange	-do-	-do-	-	-

2.3 Floods: Not Applicable

Condition	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Transient water logging/ partial inundation; Continuous submergence for more than 2 days; Sea water intrusion	Not Applicable			

2.4 Extreme events: Heat wave / Cold wave/ Frost/ Hailstorm / Cyclone:

Extreme event type	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Heat wave				
Oranges (Mandarin and Sweet orange)	• Increase the frequency of irrigation, • Use of temporary shade net • Mulching	• Increase the frequency of irrigation • Pruning of damaged branches / twigs	• Increase the frequency of irrigation • Mulching to reduce soil temperature • Pruning of damaged parts and apply Bordeaux paste 1% to cut ends	Immediate harvesting of fruits, grading and marketing
Cold wave				
Oranges (Mandarin and Sweet orange)	Protect with polythene sheet	Smoking, frequent and light irrigation during evening hours, basin mulching, apply supplementary dose of fertilizer	Smoking, frequent and light irrigation during evening hours, basin mulching, apply supplementary dose of fertilizer	Not applicable
Frost	Not applicable			
Hailstorm				
Wheat, chickpea, safflower	Re-sowing in case of severe damage	Light and frequent irrigation.	• Apply 10% additional nitrogen • Light and frequent irrigation	Timely harvesting and shifting of produce to safer place in case of early forewarning
Oranges (Mandarin and Sweet Orange)	Not applicable	Prune damaged branches and twigs and apply Bordeaux paste 1% to avoid fungal infections	• Prune damaged branches and twigs and apply Bordeaux paste 1% to avoid fungal infections • Apply hormonal spray NAA 20 ppm + 1 % urea to prevent flower drop	Immediate harvesting, grading and marketing of produce
Cyclone	Not Applicable			

2.5 Contingency strategies for Livestock, Poultry & Fisheries

2.5.1 Livestock

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Drought			
Feed and fodder availability	As the district is occasionally prone to drought the following measures are to be taken to mitigate the fodder deficiency problem: • Sowing of cereals (sorghum/bajra) and leguminous crops (lucerne, berseem, horsegram, cowpea) during North-East monsoon under dry land system • Collection of soya meal waste for use as feed supplement during drought • Preserving the green maize fodder as silage • Establishment of fodder bank at village level with available dry fodder (wheat straw, sorghum/ bajra stover etc.) • Development of silvopastoral models with *Leucaena*, *Glyricidia*, *Prosopis* as fodder trees and Marvel, Madras Anjan, Stylo, Desmanthus, etc., as under storey grass • Encourage fodder production with Sorghum – *Stylo* - sorghum on rotation basis and also to cultivate short-term fodder crops like sunhemp • Promote *Azola* cultivation in backyards • Formation of Village Disaster Management Committee • Capacity building and preparedness of the stakeholders and official staff for the drought/floods	• Harvest and use biomass of dried up crops (wheat/ sorghum/ bajra,/ maize / horse gram/ greengram /soyabean) and wetted grain as feed for livestock • Use of unconventional and locally available cheap feed ingredients especially soya meal waste for feeding of livestock during drought • Harvest all the top fodder available (*Subabul*, *Glyricidia*, *Prosopis* etc) and feed the livestock during drought • Concentrate ingredients such as grains, brans, chunnies and oilseed cakes, low grade grains etc. unfit for human consumption may be procured from Government godowns for feeding as supplement for high productive animals during drought • Promotion of horsegram as contingent crop and harvesting at vegetative stage as fodder • Hay should be enriched with 2% Urea molasses solution or 1% common salt solution and fed to livestock • Continuous supplementation of minerals to prevent infertility. • Encourage mixing available kitchen vegetable waste with dry fodder while feeding to the milch animals	• Encourage progressive farmers to grow multi cut-fodder crops of sorghum/ bajra/ maize (UP chari, MP chari, HC-136, HD-2, giant bajra, L-74, K-677, Ananad /African Tall, Kisan composite, Moti, Manjari, B1-7 on their own lands with input subsidy • Supply of quality seeds of COFS 29, *Stylo* and fodder slips of Marvel, Yaswant, Jaywant, Napier, Guinea grass well before monsoon • Flushing the stock to recoup • Replenish feed and fodder banks

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Drinking water	• Adopt water conservation methods at village level to improve the ground water level for adequate water supply • Identification of water resources • Desilting of ponds • Rainwater harvesting and create water bodies/ watering points • Construction of drinking water tanks in herding places/village junctions/relief camp locations • Community drinking water trough can be arranged in shandies /community grazing areas	• Adequate supply of drinking water. • Restrict wallowing of animals in water bodies/resources • Add alum in stagnant water bodies	• Watershed management practices should be promoted to conserve rainwater. Bleach (0.1%) drinking water / water sources • Provide clean drinking water
Health and disease management	• Procure and stock emergency medicines for important endemic diseases • All the stock must be immunized for endemic diseases like Hemorrhagic septicemia (HS), Black quarter (BQ) (large ruminants) and Enterotoxaemia (ET) (small ruminants) etc. before the onset of monsoon • Surveillance and disease monitoring network to be established at Joint Director (Animal Husbandry) office in the district • Procure and stock multivitamins & area specific mineral mixture	• Carryout de-worming to all animals entering into relief camps • Identify and quarantine sick animals • Perform ring vaccination (8 km radius) in case of any disease outbreak • Restricting movement of livestock in case of any epidemic • Tick control measures be undertaken to prevent tick borne diseases in animals • Organize with community, daily lifting of dung from relief camps	• Keep close surveillance on disease outbreak • Undertake vaccination depending on need • Keep the animal houses clean and spray disinfectants • Farmers should be advised to breed their milch animals during July-September so that peak milk production does not coincide with mid summer

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Floods	Not applicable in the district		
Cyclone			
Heat wave	• Plantation with MPTs around the shed • Water sprinklers / foggers in the shed • Application of white paint on the roof to reflect light • Thatched sheds should be provided as a shelter to animal to minimize heat stress	• Allow the animals early in the morning or late in the evening for grazing • Allow for grazing between 10 AM to 3 PM during cold waves • Feed green fodder/ maize silage / concentrates during day time and paddy straw during night time in case of heat waves • Put on the foggers / sprinklers during severe heat weaves wherever possible • In severe cases, vitamin 'C' and electrolytes should be added in water during heat waves	• Feed the animals as per routine schedule • Allow the animals for grazing (normal timings)
Cold wave	• Covering all the wire meshed walls / open area with gunny bags/ polyethylene sheets (with a mechanism for lifting during the day time and pulling down during night time)	• Add 25-50 ml of edible oil in concentrates and feed to the animal during cold waves • Put on heaters during prolonged cold waves, wherever possible • Apply / sprinkle lime powder in the animal shed during cold waves to neutralize ammonia accumulation	• Feed the animals as per routine schedule • Allow the animals for grazing (normal timings)
Insurance	Encouraging insurance of livestock	Listing out the details of the dead animals	Submission for insurance claim and availing insurance benefit

2.5.2 Poultry

	Suggested contingency measures		
	Before the event[a]	**During the event**	**After the event**
Drought			
Shortage of feed ingredients	Store broken rice and other grains such as maize, sorghum, bajra unfit for human consumption for use as feed later	• Supplementation only for productive birds with house hold grain • Supplementation of shell grit (calcium) for laying birds • Culling of weak birds	Supplementation to all surviving birds
Drinking water		Use water sanitizers or offer cool hygienic drinking water	
Health and disease management	• Culling of sick birds. • De-worming and vaccination against Ranikhet Disease (RD) and Infectious Bursal Disease (IBD)	• Mixing of Vitamins A, D, E, K, B-complex and Vitamin C in drinking water (5ml in one litre water)	• Hygiene and sanitation of poultry house • Disposal of dead birds by burning / burying with lime powder in pits
Floods	Not applicable in the district		
Cyclone	Not applicable in the district		
Heat wave			
Shelter/environment management	Provision of shelter with proper ventilation	• In severe cases, foggers/ water sprinklers/ should be arranged or wet gunny bags should be hung to reduce heat stress • Don't allow for scavenging during mid day	Routine practices are followed

	Suggested contingency measures		
	Before the event[a]	**During the event**	**After the event**
Health and disease management	De-worming and vaccination against RD and fowl pox	• Supplementation of household grain • Provide cool and clean drinking water with electrolytes and Vitamin C • In hot summer, add anti-stress probiotics in drinking water or feed	Routine practices are followed
Cold wave			
Shelter/ environment management	• Provision of proper shelter • Arrangement for brooding • Ensure supply of continuous electricity	• Close all openings with polythene sheets • In severe cases, arrange heaters • Don't allow for scavenging during early morning and late evening	Routine practices are followed
Health and disease management	• Hygiene and sanitation of poultry house • Apply lime in poultry houses	• Supplementation of household grain • Mix antibiotic supplements in feed to prevent non-specific enteric and respiratory infections	Routine practices are followed

[a]based on forewarning wherever available

2.5.3. Fisheries

1. Drought	Suggested Contingency Measures		
	Before the event	During the event	After the event
A. Capture			
Marine	Not applicable	Not applicable	Not applicable
Inland			
Shallow water depth due to insufficient rains / inflow	Stocking of advanced fingerlings in half or even less than the normal stocking density or stocking of common carp seed	Immediate harvesting or decreasing the density commensurate with the water quantity.	Weeding and deepening of tank to ensure retention of water for a longer period and provision of employment under MGNREGP
Changes in water quality	Regular monitoring of water quality parameters and application of geolites, soil probiotics, etc to maintain water qaulity	Immediate harvesting or changing the water quality by application of sanitizers.	Removal of top layer, deep ploughing of tank and application of lime
B. Aquaculture			
(i) Shallow water in ponds due to insufficient rains / inflow	Crop holiday or going for stocking of yearlings by reducing the density according to availability of water	Harvesting of fish and leaving the pond fallow till next season	Removal of top layer, deep ploughing of tank and application of lime
(ii) Impact of salt load build up in ponds / changes water quality	Stocking of salinity tolerant fish / shrimp, application of geolites and other buffers	Frequent change of water with fresh water	Frequent draining of the pond with fresh water, removal of top layers
2. Floods	Not applicable in the district		
3. Cyclone / Tsunami	Not applicable in the district		
4. **Heat and Cold wave conditions**			

	Suggested Contingency Measures		
	Before the event	**During the event**	**After the event**
A. Capture			
Marine	Not applicable		
Inland	No intervention		
B. Aquaculture			
Changes in water quality (fresh water / brackish water ratio)	Reduction of biomass by partial harvest in the event of heat as the dissolved oxygen (DO) levels will be very low.	Avoid fishing	Compensatory stocking of seed and restoration of all physical and chemical parameters
Health and disease management	Removal of stress causing factors to maintain the health of the animal	Removal of stress causing factors to maintain the health of the animal	Compensatory stocking of seed and restoration of all physical and chemical parameters

Model Plan for Irrigated situation

District : Krishna

State : Andhra Pradesh

1.0 District Agriculture profile

1.1	Agro-Climatic/Ecological Zone			
	Agro Ecological Sub Region (ICAR)	Eastern Coastal plain, hot, sub-humid to semi arid eco sub-region (7.3, 18.3)		
	Agro-Climatic Region (Planning Commission)	East Coast Plains and Hills region (XI)		
	Agro Climatic Zone (NARP)	Krishna – Godavari Zone (AP-1)		
	List all the districts or part thereof falling under the NARP Zone	Guntur, Krishna, East Godavari (excluding upland areas), West Godavari, parts of Nalgonda, Khammam and Prakasam		
	Geographic coordinates of district headquarters	Latitude 16^0 10'N	Longitude 81^0 08'E	Altitude 14 m above MSL
	Name and address of the concerned ZRS/ ZARS/ RARS/ RRS/ RRTTS	Regional Agricultural Research Station, ANGRAU, Lam, Guntur - 522 034		
	Mention the KVK located in the district	Krishi Vigyan Kendra, Garikapadu, Krishna District – 521 175		
	Mention the AMFU located in the district for agro-advisory in the Zone	Integrated Agromet Advisory Services, Regional Agricultural Research Station, Lam, Guntur 522 034		

1.2	Rainfall	Normal RF (mm)	Normal Rainy days (no.)	Normal Onset (specify week and month)	Normal Cessation (specify week and month)
	SW monsoon (June-September)	686	35	1st week of June	1st week of October
	NE Monsoon(October-December)	250	10	2nd week of October	4th week of December
	Winter (January- February)	15	1	-	-
	Summer (March-May)	83	3	-	-
	Annual	1034	50	-	-

1.3	Land use pattern of the district	Geographical Area	Cultivable area	Forest area	Land under non-agricultural use	Permanent pastures	Cultivable wasteland	Land under Misc. tree crops and groves	Barren and uncultivable land	Current fallows	Other fallows
	Area ('000 ha)	872.7	476.8	76.18	178.3	10.7	27.5	9.6	38.0	28.5	27.1

Source: DACNET. 2007-08

1. 4	Major Soils (common names like shallow red soils etc.,)	Area ('000 ha)	Percent (%) of total geographical area
	Red soils	401.36	46.2
	Black soils	313.22	36.0
	Saline soils	65.3	7.5
	Alluvial soils	42.13	4.9
	Sodic soils	16.94	1.9

Source: NBSS & LUP Regional Centre, Bengaluru

1.5	Agricultural land use	Area ('000 ha)	Cropping intensity %
	Net sown area	463.0	154.7
	Area sown more than once	253.3	
	Gross cropped area	716.3	

1.6	**Irrigation**	Area ('000 ha)		
	Net irrigated area	318.8		
	Gross irrigated area	427.9		
	Rainfed area	144.2		
	Sources of Irrigation	Number	Area ('000 ha)	Percentage of total irrigated area
	Canals	KE canal System	228.6	71.7
	Tanks	913	22.3	7.0
	Open wells	15552		
	Bore wells	13549	54.1	17.0
	Lift irrigation			
	Micro-irrigation			
	Other sources	64	13.7	4.3
	Total Irrigated Area		318.8	100.0
	Pump sets	30000		
	No. of Tractors	8469		
	Groundwater availability and use*	No. of blocks/ Tehsils	Quality of water (specify the problem such as high levels of arsenic, fluoride etc.)	
	Over exploited	3	Fluoride level >1.5 mg/l; iron >1 mg/l, nitrate >45 mg/l	
	Critical	1		
	Semi- critical	1		
	Safe	52		
	Wastewater availability and use	-		
	Ground water quality	In parts of district, groundwater quality is saline		

*over-exploited: ground water utilization > 100%; critical: 90-100%; semi-critical: 70-90%; safe: <70%

Source: SGWB, CGWB

1.7. Area under major field crops & horticulture crops

1.7	Major Field Crops cultivated	Area ('000 ha)			
		Kharif	*Rabi*	Summer	Grand Total
		Total	Total		
	Paddy	228	108	-	336
	Blackgram	3	9	0.5	12.5
	Maize	3	5	-	8
	Cotton	47	-	-	47
	Greengram	9	2	-	11
	Sugarcane		9	-	9
	Groundnut	3	3	-	6
	Tobacco	-	7.1	-	7.1
	Redgram	3	-	-	3
	Horticulture crops - Fruits	**Total area ('000 ha)**			
	Mango	376.5			
	Banana	2.7			
	Guava	1.2			
	Acid lime	0.7			
	Sapota	0.3			
	Horticultural crops - Vegetables	**Total area ('000 ha)**			
	Chillies	7.9			
	Bhendi	2.0			
	Tomato	1.6			
	Gourds	1.1			
	Cucumber	1.1			
	Leafy Vegetables	0.3			

	Medicinal, Aromatic crops, Spices and Condiments	Total area ('000 ha)
	Turmeric	2.0
	Plantation crops	**Total area ('000 ha)**
	Coconut	1.7
	Oil palm	5.4
	Betel vine	0.2
	Fodder crops	**Total area ('000 ha)**
	Pillipesara (*Phaseolus trilobus*)	6.2
	Jowar	1.7
	Maize	1.1
	Para napier	1.3
	Total fodder crop ar ea	10.2
	Grazing land	-
	Sericulture etc	-
	Others (Specify)	-

1.8	Livestock	Male (number)	Female (number)	Total (number)
	Non descriptive Cattle (local low yielding)	67996	31027	104459
	Crossbred cattle		5436	
	Non descriptive Buffaloes (local low yielding)	423925	133501	919509
	Graded Buffaloes		362083	
	Goat			156043
	Sheep			488657
	Others (Camel, Pig, Yak etc.)			19318
	Commercial Dairy farms (Number)			50

Source : 18th Livestock, Census 2007, GoI

1.9	Poultry	No. of farms	Total No. of birds
	Commercial	960	4532139
	Backyard	-	1971630

Source: 18th Livestock Census 2007, GOI

1.10	Fisheries									
	A. Capture									
	i) Marine	No. of fishermen	Boats			Nets				Storage facilities (Ice plants etc.)
			Mechanized	Non-mechanized		Mechanized		Non-mechanized		
				Motorised	Traditional	Trawl nets	Gill nets	Shore Seines stake	Trap nets	
		5774	96	781	137	214	72375	366	10	32
	ii) Inland	No. Farmer owned ponds		No. of Reservoirs				No. of village tanks		
		9131		1				219		
	B. Culture									
		Water Spread Area (ha)		**Yield (t/ha)**		**Production ('000 tons)**				
	i) Brackish water	4750		-		6.4				
	ii) Fresh water	23958		-		10.2				
	Others			-		394.3				

Source: State Fisheries Department

1.11	Production and Productivity of major crops (average of 5 years)	*Kharif*		*Rabi*		Summer		Total	
		Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)	Production ('000 t)	Productivity (kg/ha)
	Major Field crops								
	Paddy	784	3437	425	3919	-	-	1209	3678
	Maize	16	4600	140	8283	-	-	156	6441
	Blackgram	1	213	145	1313	-	-	146	763
	Groundnut	6	2326	5	1655	-	-	11	1990
	Sugarcane	829 (cane yield)	87360	-	-	-	-	829 (cane yield)	87360
	Cotton	141 ('000 bales)	510	-	-	-	-	141 ('000 bales)	510
	Tobacco	-	-	19	2771	-	-	19	2771
	Redgram	2	723	-	-			2	723
	Greengram	1	130	1	807			2	468
	Others	-	-	-	-	-	-	-	-
	Major Horticultural crops								
	Mango			-	-	-	-	2544.5	7088
	Oil palm			-	-	-	-	25.6	4667
	Banana			-	-	-	-	78.2	30000
	Guava			-	-	-	-	17.4	14333
	Lemon			-	-	-	-	11.3	14667

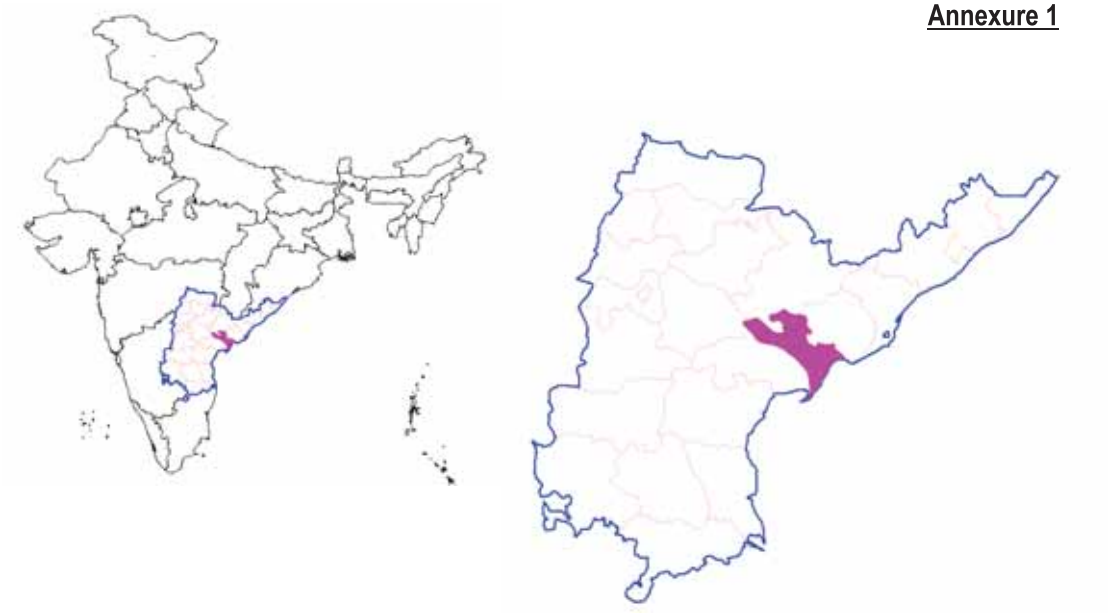
Source: Agriculture Situation in India 2009-10

1.12	Sowing window for 5 major field crops (start and end of normal sowing period)	Paddy	Maize	Blackgram & Greengram	Groundnut	Cotton & Red gram
	Kharif- Rainfed	-	1st week to end of July	1st fortnight of June to end of July	-	1st fortnight of June to end of July
	Kharif-Irrigated	1st fortnight of June to end of July	1st week to end of July	-	-	-
	Rabi- Rainfed	-	-	September 1st fortnight – 1st fortnight of October	2nd fortnight of October– 1st fortnight of November	-
	Rabi-Irrigated	2nd fortnight of December – 1st fortnight of January	2nd fortnight of November–1st fortnight of December	2nd fortnight of October	2nd fortnight of November–1st fortnight of December	-

1.13	What is the major contingency the district is prone to? (Tick mark and mention years if known during the last 10 year period)	Regular	Occasional	None
	Drought	-	✓	-
	Flood	-	✓	-
	Cyclone	✓	-	-
	Hail storm	-	-	-
	Heat wave	-	-	-
	Cold wave	-	-	-
	Frost	-	-	-
	Sea water intrusion	-	✓	-
	Pests and diseases (specify)	Rice: Blast, Redgram: Pod borers Cotton: Sucking pest complex Blackgram : YMV	-	-

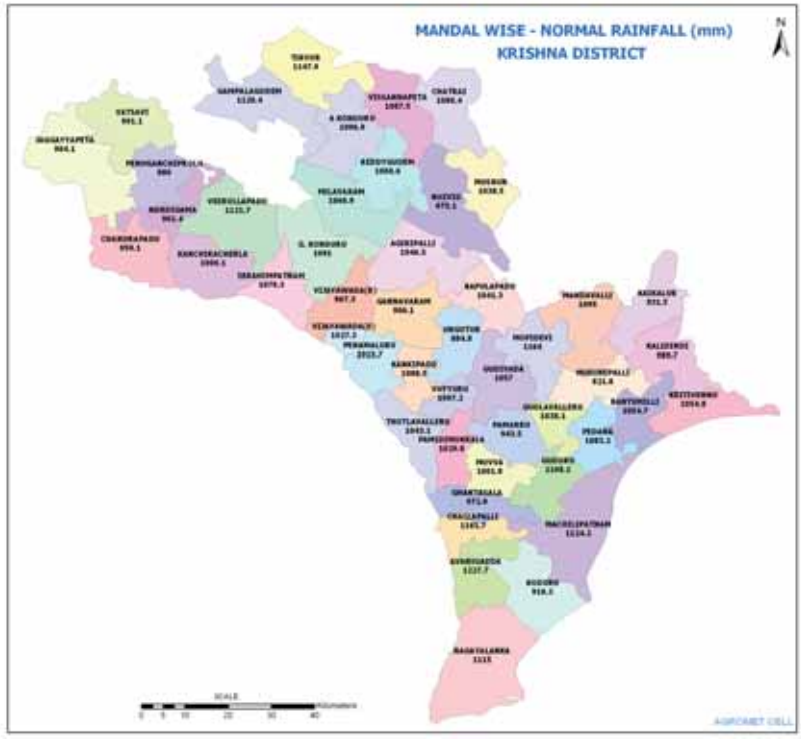
1.14	Include Digital maps of the district for		
		Location map of district within State as Annexure 1	Enclosed: Yes
		Mean annual rainfall as Annexure 2	Enclosed: Yes
		Fertility Status as Annexure 3	Enclosed: Yes

Annexure 1



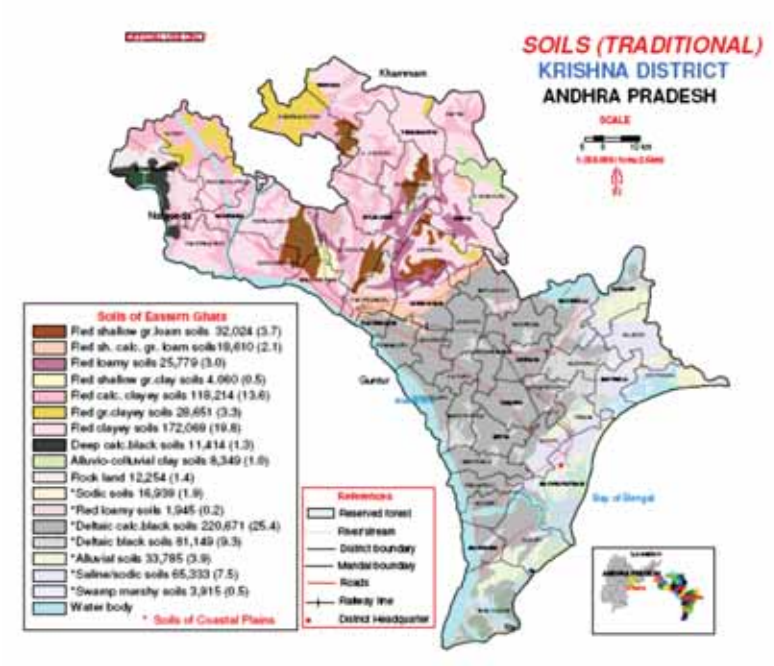
Location map of Krishna district

Annexure 2



Rainfall distribution at Tehsil level in Krishna district (Source: Agromet Cell, ANGRAU)

Annexure 3



Soil map of Krishna district (Source: NBSS & LUP, Regional Centre, Bengaluru)

2.0. Strategies for weather related contingencies

2.1 Drought

2.1.1 Rainfed situation

Early season drought (delayed onset)	Major Farming situation	Normal crop/cropping system	Suggested Contingency measures		
			Change in crop/cropping system	Agronomic measures	Remarks on implementation
Delay by 2 weeks (3rd week of June)	Black soils	Cotton	No change	Normal recommended practices of ANGRAU	Source of foundation and certified seed: APSSDC and NSC
		Greengram			
		Redgram			
	Red soils	Cotton			
		Redgram (Sole crop)			
		Greengram/ Groundnut + Redgram (7:1)			
Delay by 4 weeks (1st week of July)	Black soils	Cotton	No change	Normal recommended practices of ANGRAU	
		Greengram			
		Redgram			
	Red soils	Cotton			
		Redgram (Sole crop)		Reduce redgram inter row spacing from 180 to 150 cm	
		Greengram/ Groundnut + Redgram (7:1)		Normal practices	

Early season drought (delayed onset)	Major Farming situation	Normal Crop/cropping system	Suggested Contingency measures		
			Change in crop/cropping system	Agronomic measures	Remarks on implementation
Delay by 6 weeks (3rd week of July)	Black soils	Cotton	No change	• Adopt closer spacing of 90 x 45 cm • Basal application of 1/3rd N and 1/3rd K_2O along with P • Frequent intercultivation	Source of Breeder seed: Short duration pulses: RARS, Lam farm, Guntur Redgram: RARS, Palem
		Greengram	No change	Prefer varieties like LGG 460, PDM 54 LGG 450	
		Redgram	Prefer short duration varieties like PRG-100, ICPL 84031	Normal practices	
	Red soils	Cotton	No change	Adopt closer spacing of 90 x 60 cm	
		Redgram (Sole crop)		Reduce redgram inter row spacing from 180 to 150 cm	
		Greengram/ Groundnut + Redgram (7:1)		Normal practices	
Delay by 8 weeks (1st week of August)	Black soils	Cotton	No change	Adopt closer spacing of 90 x 45 cm spacing	
		Greengram			
		Redgram			
	Red soils	Cotton	No change	• Adopt closer spacing of 75 x 30 cm spacing • Top dress at 20 days interval at sufficient soil moisture	
		Redgram (Sole crop)	No change	Reduce redgram inter row spacing from 180 to 150 cm	
		Greengram/ Groundnut + Redgram (7:1)	Redgram	As above	

Condition	Suggested Contingency measures			
Early season drought (Normal onset)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moi sture conservation measures**
Normal onset followed by 15 -20 days dry spell after sowing leading to poor germination/crop stand etc.	Black soils	Cotton	Gap filling to be done by pot watering 7 - 10 days after sowing if the crop stand is poor	• When the crop is 2 weeks old, take up intercultivation to conserve moisture • Spray 2 % urea solution or 1 % water soluble fertilizers like 19 -19-19, 20-20-20, 21_21_21 to supplement nutrition • Two sprays of 2% KNO3 at weekly interval when the crop is encountered reddening symptoms
		Greengram	-	Spray 2 % urea solution
		Redgram	Thinning	Frequent intercultivation to conserve moisture
	Red soils	Cotton	Gap filling to be done by pot watering 7 - 10 days after sowing if the crop stand is poor	• When the crop is 2 weeks old, take up intercultivation to conserve moisture • Spray 2 % urea solution or 1 % water soluble fertilizers like 19 -19-19, 20-20-20, 21-21-21 to supplement n utrition
		Redgram (sole crop)	Thinning	• Inter cultivation to be done after 2 weeks of sowing to conserve soil moisture
		Greengram/ Groundnut + redgram (7:1)	-	• Foliar spray of 2% urea to supplement nutrition

Condition	Suggested Contingency measures			
Mid season drought (long dry spell, consecutive 2 weeks rainless (>2.5 mm) period)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moisture conservation measures**
At vegetative stage	Black soils	Cotton	Maintain weed free conditions	• When the crop is 2 weeks old, take up intercultivation to conserve moisture • Spray 2 % urea solution or 1 % water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 to supplement nutrition
		Greengram	As above	Periodical intercultivation at 7-10 days interval to conserve soil moisture
		Redgram		
	Red soils	Cotton	As above	• When the crop is 2 weeks old, take up intercultivation to conserve moisture • Spray 2 % urea solution or 1 % water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 to supplement nutrition
		Redgram (sole crop)	As above	• Periodical intercultivation at 7-10 days interval to conserve soil moisture
		Greengram/ groundnut + redgram (7:1)	Harvest greengram/blackgram as fodder, since there is a chance of much yield reduction	Foliar spray of 2 % urea solution for redgram

Condition	Suggested Contingency measures			
Mid season drought (long dry spell, consecutive 2 weeks rainless (>2.5 mm) period)	**Major Farming situation**	**Normal Crop/cropping system**	**Crop management**	**Soil nutrient & moisture conservation measures**
At reproductive stage	Black soils	Cotton	• Spray 2 % urea solution or 1% KNO3 % or 1% water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 • Life saving irrigation	Intercultivation to create soil mulch to conserve moisture
		Greengram		
		Redgram		
	Red soils	Cotton	As above	As above
		Redgram (sole crop)		
		Greengram/ Groundnut + Redgram (7:1)	Harvest intercrops as fodders as chances of grain yield are poor	• Intercultivation to create soil mulch to conserve moisture • Give supplemental irrigation

Condition	Major Farming situation	Normal Crop/cropping system	Suggested Contingency measures	
			Crop management	**Rabi Crop planning**
Terminal drought (Early withdrawal of monsoon)	Black soils	Cotton	• Spray urea - 2 % or KNO_3 1% or other water soluble fertilizers 1 % to supplement nutrition • Topping to prevent formation of new vegetative and reproductive flush • Supplemental irrigation if available	Plan for groundnut / blackgram / maize Seed multiplication of required pulse varieties can be planned and produced during early rabi season in Uplands Linkage with NFSM
		Redgram	Supplemental irrigation	
	Red soils	Cotton	As above	
		Redgram (sole / base crop)	As above	

2.1.2 Irrigated situation

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/ cropping system	Agronomic measures	Remarks on Implementation
Delayed release of water in canals due to low rainfall	Black soils – Canal irrigated (KED)	Green manure crop – Rice – blackgram/ maize	Green manure crop – Rice – greengram / blackgram / maize	• Promotion of direct seeded rice with drum seeder • Prefer medium duration varieties like NLR 34449, MTU 1010, MTU 1001, JGL-384 • Prefer blackgram varieties like LBG 20, LBG 752, LBG 709 which are early maturing and suitable for delayed sowings • Greengram can be grown in rice fallows under late seasonal conditions • Zero till maize in paddy fields with varieties DHM 117, Trishulatha and other popular hybrids	• Crop plans for *rabi* need to be taken up early by considering the available supply of water • Ensure power supply for efficient use of groundwater in delayed situations
		Paddy – sugarcane (planted) and sugarcane (ratoon)	Paddy – sugarcane planted and ratoon – paddy	• Raising of nurseries with single bud ded sets to save the time and water • Use of drip system in sugarcane for enhanced water productivity • Mulching with sugarcane trash between rows and frequent intercultivations to conserve moisture	

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/ cropping system	Agronomic measures	Remarks on Implementation
	Red soils/ black soils – Canal irrigated (NSP left canal command area / tank fed and lift irrigation)	Greengram – rice – greengram / maize / blackgram / fodder	Rice – greengram/ maize/ blackgram/ fodder	• Management of aged seedlings in long duration varieties • Shallow planting of 20-25 days old seedlings • Prophylactic measures through application of 10% Phorate or 3% Carbofuron granules a week before pulling nursery • Direct sowing of short duration varieties • Prefer short duration varieties of greengram, blackgram, maize and groundnut • Avoid growing rice varieties like BPT 5204 as they are highly susceptible to blast disease under delayed sowings • Select varieties like NLR 34449, NLR 3041, NLR 145, JGL 384 etc. which are resistant to blast and suitable for mid-*kharif* season • If BPT 5204 is grown, adopt plant protection measures for blast	Source of breeder seed : ARS, Machilipatnam, ARS, Garikapadu NLR varieties- ARS, Nellore JGL varieties-RARS, Jagityal BPT varieties-ARS, Pampasagar

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
Limited release of water in canals due to low rainfall	Black soils – Canal irrigated (KED)	Green manure crop – rice – blackgram/maize	Green manure crop – rice – black gram/greengram - aerobic rice	• Management of over aged seedlings • Direct seeding of short duration rice varieties (<135 days) • Adopt alternate wetting and drying upto primordial iinitiation stage to save water • Adopt efficient weed management in aerobic rice • If rice crop cannot be taken up within the season, alternate crops such as pulses/maize may be promoted in *kharif* season • Short duration varieties be selected for delayed sowing: blackgram varieties LBG 20, LBG 752, LBG 709 and maize varieties DHM 117, Trishulatha • Water saving micro irrigation systems like sprinkler irrigation for greengram and blackgram can be followed • If there is a possibility to release water for entire canal command area, nurseries at village level be promoted • Continuous electricity supply be ensured to make use of available groundwater through borewells/wells	• Capacity building of rice farmers for aerobic rice management • Supply of weeders for aerobic rice • Linkage with seed corporations for supply of good quality seed of short duration pulses and maize • Ensure sufficient diesel supply at mandal level for use through motors • Linkage with on-going schemes for supply of good quality PVC pipes for reducing conveyance losses during irrigation

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
		Paddy – sugarcane (planted) and ratoon – paddy	No change	• Raising of nurseries with single budded sets in sugarcane to save time and water • Adopt water conservation practices like intercultivation, earthing-up, alternate row irrigation	Linkage with ARS, Vuyyur
	Black soils/Red soils – Left Canal irrigated (NSP left canal Command area/ Tank fed and lift irrigation)	Greengram – rice – blackgram / greengram/ maize/ fodder	• Green manure – rice – greengram/ blackgram/jowar / fodder • Redgram + greengram /jowar • Cotton (wherever drainage facilities are available)	For rice and crops in rice fallow the agronomic measures as suggested for the above farming situation shall be followed Proper drainage facilities should be created to take up cropping systems as suggested such as redgram and cotton	Linkage with MGNREGS for creation of drainage channels

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
Non release of water in canals under delayed onset of monsoon in catchment	Black soils – Canal irrigated (KED)	Green manure crop – rice – blackgram/maize	Green manure crop – blackgram – maize/ blackgram/ groundnut Plan for early *rabi* cropping	• Green manure crops should be incorporated • Promotion of short duration pulse crops and oilseeds under rainfed condition. Supplemental irrigation to be provided with groundwater wherever feasible	• Linkages with seed supply agencies for large scale supply of seeds of alternate crops as suggested • Ensure continuous power supply for pumping groundwater

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
				• Sowing of ID crops (blackgram and groundnut) can be taken from September second fortnight onwards • Maize / blackgram can be grown from December to February with two to three irrigations after the harvest of early *kharif* crops	• Care to be taken up not to apply poor quality groundwater for crops • Need to mobilize diesel supply for use in diesel engines • Crop holiday may be declared in rabi season
	Black soils/Red soils – Canal irrigation (NSP left canal Command area/ Tank fed and lift irrigation)	• Greengram / green manure crop – rice – blackgram/greengram/ jowar/fodder	• Greengram / green manure crop – blackgram / maize/fodder • Greenmanure crop / greengram – cotton • Greenmanure crop / greengram – redgram	• Green manure crops should be incorporated • Prefer ID crops from September second fortnight onwards • Prefer maize and blackgram from December to February/March raised with two to three irrigations after the harvest of early rabi crops	• Crops like horsegram, blackgram, greengram, be promoted in rabi season for improving soil quality • If there is a possibility to release water during *rabi* season, suitable crops such as short duration maize, chillies etc., be grown utilizing available water

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system	Change in crop/cropping system	Agronomic measures	Remarks on Implementation
Lack of inflows into tanks due to insufficient /delayed onset of monsoon	Black soils - Tankfed situation	Greengram / fodder crops and green manure crops – medium to short duration paddy	Green manure crops - ID crops like maize, jowar, redgram, groundnut	• Green manure crops should be incorporated at the right stage and allowed to decompose with receipt of rain • Sowing of ID crops can be taken from September second fortnight onwards • Maize, blackgram can be taken up from December to February with two to three irrigations after the harvest of early *rabi* crops	Linkage with NFSM & ISOPOM for seed supply and other inputs

Condition	Suggested Contingency measures				
	Major Farming situation	Normal Crop/cropping system[g]	Change in crop/cropping system[h]	Agronomic measures	Remarks on Implementation
Insufficient groundwater recharge due to low rainfall Any other condition (specify)	Not applicable				

2.2 Unusual rains (untimely, unseasonal etc) (for both rainfed and irrigated situations)

Condition - Continuous high rainfall in a short span leading to water logging and Heavy rainfall with high speed winds in a short span				
Crop	Suggested contingency measure			
	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Rice	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water • Take up gap filling either with available nursery or by splitting the tillers from the surviving hills	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water	• Drain excess water as early as possible	• Drain out water and spread sheaves loosely in field or field bunds where there is no water stagnation • Spray common salt 5% on panicles to prevent germination and spoilage of straw from moulds • Thresh after drying the sheaves properly • Ensure proper drying of grains before storing
Cotton	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water • Take up inter cultivation at optimum soil moisture condition to loosen and aerate the soil and to control weeds	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water • Spray 2% urea or DAP or water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 at 1% to support nutrition	• Drain excess water as early as possible • Dry the wet cotton and market immediately • Spray 1% KNO3	Proper storage of picked cotton to avoid wetting and maintaining quality of lint

Condition - Continuous high rainfall in a short span leading to water logging and Heavy rainfall with high speed winds in a short span				
Crop	Suggested contingency measure			
	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Redgram	• Drain excess water as early as possible • Take up intercultivation at optimum soil moisture condition to loosen and aerate the soil and to control weeds	• Drain excess water as early as possible • Take up intercultivation at optimum soil moisture condition to loosen and aerate the soil and to control weeds	• Drain excess water as early as possible • Allow the crop to dry completely before harvesting	• Spread the bundles drenched in rain on field bunds or drying floors to quicken drying • Thresh the bundles after they are dried properly • Dry the grain to proper moisture content before bagging and storing to prevent deterioration in quality during storage
Blackgram	• Drain excess water as early as possible • Take up inter cultivation at optimum soil moisture condition to loosen and aerate the soil and to control weeds	• Drain excess water as early as possible • Take up inter cultivation at optimum soil moisture condition to loosen and aerate the soil and to control weeds	• Drain excess water as early as possible • Allow the crop to dry completely before harvesting	Quick drying followed by threshing
Maize	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water • Intercultivation • Earthing-up the crop for anchorage	• Drain excess water as early as possible • Apply 20 kg N + 10 kg K /ha after draining excess water • Intercultivation • Earthing-up the crop for anchorage	• Drain excess water as early as possible • Harvest green cobs from dislodged plants for immediate marketing	Harvest the cobs after the they are dried up properly. Dry the grain to optimum moisture content before storing

Condition - Continuous high rainfall in a short span leading to water logging and Heavy rainfall with high speed winds in a short span				
Crop	Suggested contingency measure			
	Vegetative stage	Flowering stage	Crop maturity stage	Post harvest
Horticulture crops – Fruits				
Mango	Drain excess water as soon as possible	Drain excess water as soon as possible	• Harvest mature produce on a clear sunny day • Fallen fruit may be collected, graded and marketed, if feasible	• Store fruits in well ventilated temporary structures before marketing • Market the fruits as soon as possible
Banana	• Drain excess water as soon as possible • Inter-cultivation • Topdressing of booster dose of 80 g MOP + 100 g Urea per plant two to three times intervals • Gap filling may be taken up if the plants are two weeks old and sowing window is still available for the crop • If the age of the plant is less than three months and submergence up to three feet, better to replant the garden	• Drain excess water as soon as possible • Topdressing of booster dose of 80 g MOP + 100 g Urea per plant two to three times • If the age of the plant is more than three months and less than seven months allow one sword sucker for ratoon and take up fertilization at monthly intervals for four months • Bamboo staking to prevent uprooting	• Drain excess water as soon as possible • Harvest the marketable bunches on a clear sunny day • Staking with bamboos to prevent further lodging	• Use ripening chambers for quick ripening • Market produce as soon as possbile
Horticultural crops - Vegetables				
Chillies	• Drain excess water as soon as possible • Spray Urea 2% solution 2-3 times	• Drain the excess water as soon as possible • Spray Urea 2% solution 2-3 times.	• Drain excess water as soon as possible • Harvest the matured fruits on a clear sunny day	• Dry the pods on concrete floor immediately after the appearance of sunlight

	• Topdressing of booster dose of 15 kg MOP + 30 kg urea per acre as soon as possible • Gap filling may be taken up if the plants are two weeks old and sowing window is still available for the crop • In case of severe damage (considered as complete economical loss) and the contingency period is between June to August, sowing of best alternative crop must be taken up	• Topdressing of booster dose of 15 kg MOP + 30 kg Urea per acre as soon as possible.		• Use poly house solar driers for quick drying • Grade the pods and market as soon as possible • Do not store such produce for long periods
Plantation crops				
Oil palm	• Planting should be done on ridges • Relief drains [shallow] channels are opened at places where water accumulates and connected with main drain to remove water from the surface	• Drain the excess water as soon as possible • Apply booster dose of NPK fertilizers	• Drain the excess water as soon as possible • Apply booster dose of NPK fertilizers • Harvest the mature nuts as soon as possible	• Store the produce in well ventilated place temporarily before marketing at the earliest opportunity • Immediate processing
Spices and Condiments				
Turmeric	• Drain excess water as soon as possible • Spray 2% urea 2% followed by Ferrous Sulphate 0.5% + Citric Acid 0.1 % solution 2-3 times	• Drain excess water as soon as possible	• Harvest the rhizomes as early as possible	• Dry the rhizomes on concrete floor or use boilers (if available) for processing immediately • Grade and separate the rotten and mould affected rhizomes. • Pack the dried material in disinfected gunny bags • Store in a well ventilated rooms

Condition - Outbreak of pests and diseases due to unseasonal rains				
Rice	• Protection against stem rot and Sheath blight • Foliar application of hexaconazole or propiconazole 1 ml / litre to control blast	• Foliar application of buprofezin @0.6-2ml/l of water Chlorantraniliprole@ 0.1 to 0.3 ml/l of water to control BPH • Hexaconzole or propiconazole @1ml/l to control Blast and Sheath blight	Protect against false smut and grain discolouration	Quick drying to prevent grain discolouration
Cotton	Protect against incidence of wilt and root rot. Apply copper oxy chloride 0.3% or carbendazim 0.1% drench at plant base	• When marginal yellowing of leaves due to jassid injury is seen, apply neem oil 0.3% with sticker or imidacloprid @ 0.6ml/l or acetamiprid @ 0.1-0.2g/l spray • Protect against bacterial leaf blight with foliar application of streptocycline sulphate 6 g + 30 g copper oxy chloride(COC) for every 10 litres of water	To control grey mildew and boll rot, apply carbendazim 1g/l or mancozeb 3 g/l	Proper storage of seed cotton to prevent wetting and incidence of molds
Redgram	Soil application of *Trichoderma harzianum* along with FYM as side dressing to prevent *Fusarium* wilt	• Drenching of carbendazim 0.1% at plant base for controlling wilt • Foliar application of acephate 1.5 g/ l or miticide to prevent sterility mosaic virus	Drench with carbendazim 0.1% at plant base to control wilt	Quick drying to prevent molds
Blackgram	-	Foliar application of carbendazim @ 0.5g/l or sulphur @ 3g/litre against leaf spots, and powdery mildew respectively	-	-

Condition - Outbreak of pests and diseases due to unseasonal rains				
Maize	Spray imidacloprid 0.3 ml/l or dimethoate 1 ml/ litre to control leaf hopper	Foliar application of mancozeb @ 0.25 to 0.4% at 8-10 days interval to control *Turcicum* leaf blight spray	Trichoderma mixed with FYM @10g/kg at 10 days prior to its use in the field can be applied to control Stalk rot incidence which is likely during post flowering	-
Mango	• Spray imidacloprid 0.3 ml or dimethoate 1 ml/litre to control leaf hopper • Drench the seedlings with COC 0.3% against root rot	Spray imidacloprid 0.3 ml/l or dimethoate 1 ml/ litre to control leaf hopper	• Spray dithane M-45 3g/l or carbendazim 1g/ l against anthracnose • Spray sulphur 0.5% to control powdery mildew	Maintain aeration in storage to prevent fungal infection and blackening of fruits
Chillies	Drench nursery beds with COC 3 g/l to prevent damping off	Spray COC 30 g + 1g streptocycline in 10 litres of water, 2-3 times against the bacterial leaf spot and blight	Spray carbendazim 0.1% to control fruit rot	Quick drying of produce to prevent fruit rot
Banana	Soil drenching with COC @ 3 g/litre to avoid rhizome rotting	Spray Dithane M-45 0.3% or propiconazole 1ml /litre, 2-3 times against *Cercospora* leaf spots.	Soil drenching with COC @ 3 g/litre to avoid rhizome rotting	-
Turmeric	Spray Propiconazole 1 ml/ l, 2-3 times against leaf spot/ blotch	Spray propiconazole 1 ml/l, 2-3 times against leaf spot/blotch	Soil drenching with COC 3g or ridomil 2g/l to prevent rhizome rot	Quick drying of rhizomes to prevent rot

2.3 Floods

Condition	Transient water logging/ partial inundation and Continuous submergence for more than 2 days			
	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Rice	• Drain out the excess water at the earlies • Apply booster dose of 50 kg/ha • Micronutrient deficiency corrections for Zn and Fe foliar application of 0.2 % $ZnSO_4$, $FeSO_4$ two to three times at 4 -5 days interval • Maintain weed free conditions	• Drain out the excess water at the earliest • Take up gap filling either with available nursery or by splitting the tillers from the surviving hills • Apply booster dose of 50 N kg/ha • Spray $ZnSO_4$ 0.2 % if it is less than 45 days after transplanting	• Drain out the excess water at the earliest • Take up need based plant protection measures	• Drain out water. Spread sheaves loosely in field or field bunds where there is no water stagnation • Spray common salt at 5% on panicles to prevent germination and spoilage of straw from moulds • Thresh after drying the sheaves properly • Ensure proper grain moisture before storing
Cotton	• Drain out the excess water at the earliest and apply 20 kg N + 10 kg K /ha after draining excess water. • Takeup the gap filling • Intercultivation at optimum field moisture condition. • Spray water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 at 1% to support nutrition. • In case of severe damage, prefer resowing with short duration hybrids and adopt closer spacing of 90X45 or 90X30 cm.	• Drain out the excess water at the earliest and apply 20 kg N + 10 kg K /ha after draining excess water • Takeup the gap filling • Intercultivation at optimum field moisture condition. • Spray water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 at 1% to support nutrition. • Need based correction of the micronutrients like Zn by spraying $ZnSO_4$ two times at 7-10 days interval.	• Drain out the excess water at the earliest • Spray water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 at 1% to support nutrition.	• Proper storage of picked cotton to avoid wetting and maintaining quality of lint

Condition	Transient water logging/ partial inundation and Continuous submergence for more than 2 days			
	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Sugarcane	Drain out the excess water at the earliest	• Drain out the excess water at the earliest • Apply 50 kg N /ha and 25 K / ha	• Drain out the excess water at the earliest • Apply 50 kg N /ha and 25 K / ha	Drain out the excess water at the earliest Harvest as early as possible
Horticulture				
Mango	• Drain the excess water in the orchad • Spray urea 2% solution for 2-3 times at 7-10 days interval	• Drain the excess water in the orchad • Spray urea 2% solution for 2-3 times at 7-10 days interval	• Drain the excess water in the orchad • Spray urea 2% solution for 2-3 times at 7-10 days interval	• Drain the excess water as soon as possible. • Harvest the mature fruits as soon as possible. • Store the fruits in well ventilated place temporarily before it can be marketed. • Market the fruits as soon as possible. • Spray Dithane M-45 3.0% or bavistin 1.0% against Anthracnose

Condition	Transient water logging/ partial inundation and Continuous submergence for more than 2 days			
	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Banana	• Open deep trenches between the plant rows for improving the drainage condition of the field • Give foliar spray of 2% potassium sulphate. After 15 days, give another foliar spray of 1% potassium nitrate. • There is possibility of outbreak of Sigatoka leaf spot . As a prophylactic measure, give 0.1 per cent Propiconozole, 0.2 per cent Carbendazim and 0.25 per cent Mancozeb foliar sprays with 15 day intervals between each spray.	• Open deep trenches between the plant rows for improving the drainage condition of the land. • Give foliar spray of 2% potassium sulphate. After 15 days, give another foliar spray of 1% potassium nitrate. • At the bunch development stage, give 2% potassium sulphate spray on bunch, two times with 15 day interval. • There is possibility of outbreak of Sigatoka leaf spot. As a prophylactic measure, give 0.1 per cent Propiconozole, 0.2 per cent Carbendazim and 0.25 per cent Mancozeb foliar sprays with 15 day intervals between each spray.	• Open deep trenches between the plant rows for improving the drainage condition of the land. • At the bunch development stage, give 2% potassium sulphate spray on bunch, two times with 15 day interval • There is possibility of outbreak of Sigatoka leaf spot. As a prophylactic measure, give 0.1 per cent Propiconozole, 0.2 per cent Carbendazim and 0.25 per cent Mancozeb foliar sprays with 15 day intervals between each spray	• Grade the good quality fruit bunches for ripening and marketing • Use ripening chambers for quick ripening • Market produce as soon as possbile

2.4 Extreme events: Heat wave / Cold wave / Frost / Hailstorm / Cyclone

Extreme event type	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Heat Wave	Not applicable in the district			
Cold wave	Not applicable			
Frost				
Hailstorm				
Cyclone				
Rice	• Drain out the excess water at the earliest Apply booster dose of 0.2 kg N/40 sq. m • Spray micronutrients like Zn, Fe 2-3 times at 4 -5 days interval • Take up proper weed control measures	• Drain out the excess water at the earliest • Apply booster dose of 20 kg N/acre • Spray $ZnSO_4$ 0.2 % if it is less than 45 days after transplanting • Takeup need based plant protection measures	• Drain out the excess water at the earliest • Takeup need based plant protection measures • Lodged plants to be lifted and tied together to make them stand erect	• Drain out water spread sheaves loosely in field or field bunds where there is no water stagnation • Spray common salt at 2% to prevent germination of seed and spoilage of straw from moulds • Thresh after drying the sheaves properly • Ensure proper grain moisture before storing
Cotton	• Drain out the excess water at the earliest • Intercultivation at optimum field moisture condition • Apply 20 kg N + 10 kg K /ha after draining excess water	• Drain out the excess water at the earliest • Intercultivation at optimum field moisture condition • Earthing up to be done to provide anchorage to plants • Apply 20 kg N + 10 kg K /ha after draining excess water	• Drain out the excess water at the earliest • Spray KNO_3 @1 % or water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 @ 1% to support nutrition • Earthing up to be done to provide anchorage to plants	• Kapas picking should be done carefully to prevent admixtures with extraneous material

Extreme event type	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
		• To spray KNO_3@1 % or water soluble fertilizers like 19-19-19, 20-20-20, 21-21-21 @ 1% to support nutrition • Spray of micronutrients two times at 7-10 days interval • Take up plant protection measures against possible pests and disease incidence	• Spray of micronutrients two times at 7-10 days interval • Take up plant protection measures against possible pests and disease incidence	
Maize	• Drain out the excess water at the earliest and apply 20 kg N + 10 kg K /ha • Intercultivation and earthing up to be done • In case of severe damage, plan for resowing	• Drain out the excess water • Maintain weed free condition • Take up intercultivation followed by earthing up • In case of severe damage, take up resowing or plan for sowing with alternate crop	• In case of severe damage, plan for alternate crop • If possible, harvest good quality plants for fodder • Pick god quality green cobs	• Drain out excess water • Take up picking of completely dried cobs
Horticulture crops – Fruits				
Mango	If the damage is severe, go for replanting	• Propping and earthing up of the damaged and fallen trees • Manuring in the basins • Broken and damaged branches may be pruned and applied with 1 % Bordeaux paste	• Broken and damaged branches may be pruned and applied with 1 % Bordeaux paste • Manuring in the basins	• Drain the excess water as soon as possible. • Harvest the mature fruits • Collect the fallen fruits , grade for good quality and marketing or go for preparation of processed products • For storage, keep the fruits in well-ventilated place temporarily before marketing • Broken and damaged branches may be pruned and applied with Bordeaux paste

Extreme event type	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Banana	• Gap filling may be taken up if the plants are two weeks old and sowing window is still available for the crop.	• Banana crop can not tolerate the wind speed of even 30 km/hr. Being a shallow rooted crop requires proper propping with bamboo or Casuarina poles to avoid lodging . The supporting poles should be tied against the peduncle of developing bunches, so that it protects the plant from lodging. • Wind damaged plants should be pruned using disinfected secateurs and cut ends must be smeared with 1 % Bordeaux paste • Drain the excess water as soon as possible • The fallen plants may be cut leaving two suckers • Topdressing of booster dose of 80 g MOP + 100 g Urea per plant at two to three times intervals. • There is possibility of outbreak of Sigatoka leaf spot . As a prophylactic measure, give 0.1 per cent Propiconozole, 0.2 per cent Carbendazim and 0.25 per cent Mancozeb foliar sprays with 15 day intervals between each spray. • Soil drenching with COC @ 3 g/litre to avoid rhizome rotting. • If the age of the plant is less than three months and submergence is upto three feet better to replant the garden.	• Wind damaged plants should be pruned using disinfected secateurs and cut ends must be smeared with 1 % Bordeaux paste • Drain the excess water as soon as possible • The fallen plants may be cut leaving two suckers • Topdress with booster dose of 80 g MOP + 100 g Urea per plant at two to three times intervals • Covered the matured bunches completely with the leaves and harveste with in 15-20days • There is possibility of outbreak of Sigatoka leaf spot. As a prophylactic measure, give 0.1 per cent Propiconozole, 0.2 per cent Carbendazim and 0.25 per cent Mancozeb foliar sprays with 15 day intervals between	• Drain the excess water as soon as possible. • Harvest the mature bunches as soon as possible. use ripening chambers for quick and uniform ripening • Store the harvested bunches in well ventilated place temporarily before it can be marketed. • Market the produce as soon as possible. • 3-4 foliar application of KNO_3 on immature/developing bunches and leaves at weekly intervals. • Staking with bamboo for support

Extreme event type	Suggested contingency measure			
	Seedling / nursery stage	Vegetative stage	Reproductive stage	At harvest
Horticulture crops vegetables				
Chillies	Grow nursery on raised beds.	• Drain the excess water at the earliest • Take up gap filling • In case of severe damage, take up replanting • Application of urea 2% solution at 5 days interval 2 to 3 times • Topdressing of booster dose of 15 kg MOP + 30 kg Urea per acre as soon as possible. • In case of severe damage, adopt clean cultivation and plan for alternate crop	• Drain the excess water at the earliest • Application of urea 2% solution at 5 days interval for 2 to 3 times • Topdressing of booster dose of 15 kg MOP + 30 kg Urea per acre as soon as possible. • In case of severe damage, adopt clean cultivation and plan for alternate crop	• Drain the excess water as soon as possible. • Dry the pods on concrete floor/ tarpaulins immediately • Use poly house solar driers for quick drying • Grade the good quality pods for marketing .
Spices and Plantation crops				
Oil Palm	• Drain out excess water particularly from the root zone • Relief drains [shallow] channels are opened at places where water accumulates and connected with main drain to remove water from the surface	• Drain the excess water • Cut and remove damaged leaves • Palms fallen with root system but still having contact with soil need to be brought to position and provided with soil mound and support	• Drain the excess water • Hanging bunches may be provided with supports wherever possible . • Palms fallen with root system but still having contact with soil need to be brought to position and provided with soil mound and support	• Cut and remove damaged leaves • Hanging bunches may be provided with supports wherever possible • Harvest the mature nuts • Market the produce as soon as possible.

2.5 Contingent strategies for Livestock, Poultry & Fisheries

2.5.1 Livestock

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Drought			
Feed and Fodder availability	• Available paddy straw and sugarcane tops should be stored properly • Preserving the green maize fodder and sugar cane tops as silage • Encourage progressive farmers to grow perennial fodders like APBN-2, APBN-3, Co-2, CO-3 etc., as small blocks in cultivated lands • Development of silvopastoral models with *Leucaena* as fodder tree and *Cenchrus* and *Stylo* as under storey grass in CPRs • Encourage green fodder production with multi-cut sorghum/ bajra/ maize (UP chari, MP chari, HC-136, HD-2, GAINT BAJRA, L-74, K-677, Ananad/African Tall, Kisan composite, Moti, Manjari, B1-7 etc.) • Promotion of cultivation of short-duration fodder crops like sunhemp, cow pea, horse gram during Rabi • Promote Azola cultivation in a backyard • Chopping of fodder should be made as mandatory in every village through supply and establishment of good quality chaff cutters.	• Harvest and use biomass of dried up crops (rice, maize, greengram, blackgram etc.) and wetted grain as feed for livestock • Use *azolla* and silage as supplement for milch animals during drought • Offer green fodder to pregnant and high productive milch animals • Harvest all the top fodder (neem, subabul, *Acacia*, Pipal, *Leucaena* etc.,) availablein and around the village and feed the small ruminants during drought • Available feed and fodder should be cut from CPRs and stall fed in order to reduce the energy requirements of the animals • Encourage mixing available kitchen vegetable waste with dry fodder while feeding to the milch animals • All the hay should be enriched with 2% Urea molasses solution or 1% common salt solution and fed to livestock • Continuous supplementation of mineral mixture to prevent infertility	• Supply of quality seeds of multi cut sorghum, bajra, maize and slips of napier, guinea grass, CO-3, APBN-2 etc., well before monsoon • Short duration fodder crops like sunhemp, cow pea, horse gram should be sown in unsown and crop failed areas where no further routine crop sowing is not possible

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Floods/ cyclones	• As the district is chronically prone for cyclone, store minimum required quantity of hay and concentrates at level for feeding the livestock a week's time • In case of early forewarning (EFW), harvest all the crops (Rice/maize/backgram/green gram) that can be useful as fodder in future (store properly) • Protect the stored paddy straw from inundation of flood water • All the large ruminants are immunized for the endemic diseases like HS and BQ during the month of May and FMD in July • Procure and stock emergency medicines and vaccines for important contagious diseases. • Surveillance and disease monitoring network to be established at Joint Director (Animal Husbandry) office in the district • Arrangement for transportation of animals from low lying area to safer places and also for rescue animal health workers to get involve in rescue operations	• Transportation of animals to elevated areas • Proper hygiene and sanitation of the animal shed • In severe storms, un-tether or let loose the animals • Use stored grain/bran along with crop residues for feeding of livestock. • Avoid soaked and mould infected feed / fodders to livestock • Emergency outlet establishment for required medicines or feed in each village • Spraying of fly repellants in animal sheds and relief camps • Carryout deworming to all animals entering into relief camps • Identification and quarantine of sick animals • Perform ring vaccination (8 km radius) in case of any disease outbreak • Restrict movement of livestock in case of any epidemic • Sprinkle lime in animal sheds and relief camps	• Repair of animal shed • Bring back the animals to the shed • Cleaning and disinfection of the shed • Bleach (0.1%) drinking water / water sources • Deworm the animals through mass camps • Encouraging farmers to cultivate short-term fodder crops like cow pea, horse gram, sunhemp etc. • Proper disposable of the dead animals / carcasses by burning / deep burying (4-8 feet) with lime powder (1kg for small ruminants and 5kg for large ruminants) in pit • Drying the harvested crop and fodder material and proper storage • Vaccinate the small ruminants against blue tongue disease and large ruminants against HS, BQ and FMD • Collect drowned crop and fodder material, dry it and store properly

	Suggested contingency measures		
	Before the event	**During the event**	**After the event**
Drinking water	• Adopt water conservation methods at village level to improve the ground water level for adequate water supply. • Identification of water resources • Desilting of ponds • Rainwater harvesting in farm ponds create water • Construction of drinking water tanks in herding places/village junctions/relief camp locations • Community drinking water trough can be arranged in shandies /community grazing areas	• Provide wholesome clean drinking water to all the livestock during the day time • Restrict wallowing of animals in water bodies/resources • Add alum in stagnant water bodies	• Watershed management practices shall be promoted to conserve the rainwater. • Bleach (0.1%) drinking water / water sources • Make available wholesome clean drinking water throughout the year
Health and Disease management	• Store vaccines for the endemic diseases like HS, BQ, FMD (large ruminants) and ET, BT, PPR (small ruminants) • All the stock must be immunized for endemic diseases like Hemorrhagic septicemia (HS), Black quarter (BQ) (large ruminants) and Enterotoxaemia (ET) (small ruminants) etc. before the onset of monsoon • Procure and stock multivitamins & area specific mineral mixture • Surveillance and disease monitoring network to be established at Joint Director (Animal Husbandry) office in the district	• Carryout deworming to all animals entering into relief camps • Identification and quarantine of sick animals • Performing ring vaccination (8 km radius) in case of any disease outbreak • Restrict movement of livestock in case of any epidemic • Tick control measures be undertaken to prevent tick borne diseases in animals • Organize with community, daily lifting of dung from relief camps	• Keep close surveillance on disease outbreak. • Undertake the vaccination depending on need • Keep the animal houses clean and spray disinfectants
Insurance	• Encouraging insurance of livestock	• Listing out the details of the dead animals	• Submission for insurance claim and availing insurance benefit

2.5.2 Poultry

	Suggested contingency measures		
	Before the event [a]	**During the event**	**After the event**
Drought			
Shortage of feed ingredients	• Storing of household grain like maize, broken rice etc, in to use as feed in case of severe drought	• Supplementation only for productive birds with household grain • Supplementation of shell grit (calcium) for laying birds • Culling of weak birds	• Supplementation to all surviving birds
Drinking water	• Rain water harvesting	• Use water sanitizers or offer cool hygienic drinking water	• Make available adequate hygienic and wholesome water during the day time
Health and disease management	• Culling of sick birds. • Deworming and vaccination against Ranikhet Disease (RD) and Infectious Bursal Disease (IBD)	• Mixing of Vit . A,D,E, K and B -complex including vit C in drinking water (5ml in one litre water)	• Hygiene and sanitation of poultry house • Disposal of dead birds by burning / burying with lime powder in pit
Floods/cyclones			
Shortage of feed ingredients	• In case of early forewarning, shift the birds to safer place • Storing of house hold grain like maize, broken rice, bajra etc,	• Use stored feed as supplement • Don't allow for scavenging • Culling of weak birds	• Routine practices are followed • Deworming and vaccination against RD
Drinking water	• Store sufficient drinking water	• Use water sanitizers or offer cool hygienic drinking water	• Make available adequate hygienic and wholesome water during the day time
Health and disease management	• In case of EFW, add antibiotic powder (Terramycin/Ampicilline/ Ampiclox etc., 10g in one litre) in drinking water to prevent any disease outbreak	• Prevent water logging surrounding the sheds through proper drainage facility • Assure supply of electricity by generator or solar energy or biogas • Sprinkle lime powder to prevent ammonia accumulation due to dampness	• Sanitation of poultry house • Treatment of affected birds Disposal of dead birds by burning / burying with line powder in pit • Disposal of poultry manure to prevent protozoal problem • Supplementation of coccidiostats in feed • Vaccination against RD
Heat and Cold wave	**Not applicable in the district**		

[a] based on forewarning wherever available

2.5.3. Fisheries

1) Drought	Suggested Contingency Measures		
	Before the event	During the event	After the event
A. Capture			
Marine	No intervention	No intervention	No intervention
Inland			
(i) Shallow water depth due to insufficient rains / inflow	Stocking of advanced fingerlings in half or even less than the normal stocking density or stocking of common carp seed	Immediate harvesting or decreasing the density commensurate with the water quantity.	De weeding and deepening of tank to ensure retention of water for a longer period and provision of employment under MGNREGP
(ii) Changes in water quality	Regular monitoring of water quality parameters and application of geolites, soil probiotics, etc to maintain water quality	Immediate harvesting or changing the water quality by application of sanitizers.	Removal of top layer, deep ploughing of tank and application of lime
B. Aquaculture			
(i) Shallow water in ponds due to insufficient rains / inflow	Crop holiday or going for stocking of yearlings by reducing the density according to availability of water	Harvesting of fish and leaving the pond fallow till next season	Removal of top layer, deep ploughing of tank and application of lime
(ii) Impact of salt load build up in ponds / changes water quality	Stocking of salinity tolerant fish / shrimp, application of geolites and other buffers	Frequent change of water with fresh water	Frequent draining of the pond with fresh water, removal of top layers
2) Floods			
A. Capture			
Marine	No intervention	No intervention	No intervention

	Suggested Contingency Measures		
	Before the event	During the event	After the event
Inland			
(iv) Loss of stock	Avoidance of surface species like catla, silver carp since they are vulnerable in tanks prone to floods, erection of nets across the spill way or just beyond it	Erection of nets at spill ways	Taking up compensatory stocking
(v) Changes in water quality	Not applicable	When dissolved oxygen levels go down, aerators, recirculation of water, etc are to be attempted to maintain dissolved oxygen (DO) levels, going for partial harvest, etc	No intervention
(vi) Health and disease	Sometimes there may be heavy accumulation of nutrients and organic matter.	There may be break out of diseases. Addition of antibiotics like Chloro Tetra Cycline or Oxy Tetra Cycline to the feed to constrol the disease	Removal of weeds, top layer of soil, deep ploughing of tank and application of lime, exposing to sun light
B. Aquaculture			
(i) Inundation of ponds with flood water	Raising and riveting the bunds, construction of spill way to release excess water, erection of nets to avoid escape of fish	Continuous pumping of excess water, erection of nets low lying areas	Strengthening of bunds, excavating channels along the sides of the ponds for free escape of water
(ii) Water contamination and changes in water quality		When dissolved oxygen levels go down, aerators, recirculation of water, etc are to be attempted to maintain DO levels, going for partial harvest, etc	
(iii) Health and disease	Sometimes there may be heavy accumulation of nutrients and organic matter.	There may be break out of Heamorrhagic septicimea. Addition of antibiotics like Chloro Tetra Cycline or Oxy Tetra Cycline to the feed to control the disease	Removal of weeds, top layer of soil, deep ploughing of tank and application of lime, exposing to sun light
(iv) Loss of stock and inputs (feed, chemicals, etc)	Advance erection of nets, strengthening of bunds where they are prone to breaches, harvesting or reducing the density	Suspension of feeding, application of organic manures	Compensatory stocking, assessment of values and payment of subsidy on inputs

	Suggested Contingency Measures		
	Before the event	During the event	After the event
(v) Infrastructure damage (pumps, aerators, huts, etc.)	• Insuring pond, accessories, etc., • Shifting of aerators, pumps soon after warnigs are issued	Relocating pumps, aerators to elevated places	Assessment of damages and providing on subsidy
3) Cyclone / Tsunami			
A. Capture			
Marine			
Damage to boats, nets etc.	Avoid of fishing when warnings are issued, shifting of boats and nets to safer places	• Shifting and relocating boats and nets to safer places • Shifting of fishermen by relief boats to the relief camps	Assessment of damages to boats, nets etc. and provide loans/subsidy to procure boats, nets etc. for restoration of livelihoods
Inland			
Loss of fish and infrastructure	• Erection of protective nets across the surplus weir to prevent fish loss due to overflows • Shifting of nets, aeration pumps etc. to safer place	Continuous monitoring to prevent or minimize escape of fish along with surplus water	Compensatory stocking of seed Rearranging the shifted infrastructure
B. Aquaculture			
(i) Overflow / flooding in ponds	The design of the pond must a manner as to bail out surplus water and to prevent loss of standing crop	Continuous monitoring to prevent or minimize escape of fish along with surplus water	Compensatory stocking of seed
(ii) Changes in water quality (fresh water / brackish water ratio)	Recirculating water to replenish and ensure sufficient dissolved oxygen levels in the pond. Maintenance of salinity levels by pumping in water from creeks.	Continuation of the same process.	Restoration of physical and chemical parameters
(iii) Health and disease	Removal of stress causing factors to maintain the health of the animal	Removal of stress causing factors to maintain the health of the animal	Restoration of physical and chemical parameters
(iv) Loss of stock and inputs (feed, chemicals, etc)	Preventive nets must be erected to minimize loss of stock	Continuation of the same process.	Compensatory stocking of seed
(v) Infrastructure damage (pumps, aerators, huts, etc.)	Pumps, aerators, etc must be protected by moving them to safe locations	Avoid use of aerators, pumps and other appliances	Overhauling of the equipment to prevent from being damaged
4) Heat and Cold wave conditions	Not applicable in the district		

Chapter V

Impact of Extreme Events on Key Crop Commodities

Though extreme events (heat wave, cold wave, storms, unseasonal rains and flash floods) cause extensive damage to crops and livestock, the impact on some key commodities which are important for food security or exports is severe on the economy and public life. In this chapter the available information to deal with the above extreme events on wheat, onion, mango and apple are described stage wise.

1. Wheat

India is the major wheat growing country with a production of about 70 million tonns per annum accounting about 12 per cent of the world production. The major wheat growing states Uttar Pradesh, Punjab, Haryana, Madhya Pradesh, Rajasthan and Bihar which account for 93% of the total production in the country. The normal sowing window for wheat crop is during November - January and crop is harvested during March -April.

Wheat requires a cool and wet climate during the growing season and a dry warm climate at the time of grain development stage with an annual average rainfall of 500 to 750 mm. Wheat prefers diurnal temperature range of 15-25^{0}C with longer cool growing season. The optimum temperature range during vegetative phase and reproductive phases are 15-20^{0}C and 20-25^{0}C, respectively, although the base temperature for growth is much lower. High temperature at the time of sowing hampers tillering of the early sown crop and sudden rise in temperature at the time of ripening shrivels the grain especially when sown late. The farmers of Amritsar incurred losses of over 20 percent of their wheat harvest due to intense heat wave that swept across Punjab in April, 2010.

Studies indicated that increase in maximum temperature during the mid-anthesis period (76-79 days) and particularly 1 day after 50% anthesis (78th day) led to a decline in number of grains per earhead at maturity (up to 40% reduction), grain yield (reduction of 350 g per m^2) and harvest index (Ferris *et al*., 1998). Similarly, simulated grain yield of wheat estimated through CERES-Wheat model under incremental units of maximum temperature (1 to 3^{0}C) showed a gradual decrease in yield ranging from 3546 to 2646 kg/ha (8 to 31%) (Vyas Pandey and Patel, 2008). Also wheat yield showed a negative relationship with minimum temperature at Varanasi (Vijayakumar and Rao, 2010). At the initiation of stem elongation, cold weather decreases photosynthesis, damages growing tips and spikes and injures to culms (Acevedo *et al*., 2006). Wheat yields in India were lower by 10-40% due to cold wave in 2002-03 as compared to the normal year of 2001-02. Due to prolonged low temperature, foggy and cloudy weather that prevailed in Central Uttar Pradesh and nearby places during the winter of 2002-03 resulted in a reduction in wheat yields up to 40% (Samra *et al.*, 2003). Wheat crop in over 8,000 acres was damaged due to rain and hailstorm yield losses up to 35% in Moga and Kapurthala districts of Punjab.

Impact of extreme events on wheat and suggested measures

Extreme event	Effects	Suggested measures
Heat Stress Shriveled grains due to heat stress	• Emerging seedlings rapidly get desiccated if soil temperatures reaches 40°C or greater. • Reduction in number and formation of spikelets and florets and also poor grain filling • Late-boot and seed-set stages are mostly vulnerable to heat stress • High temperature can shorten the grain filling period, hasten maturity, and reduce grain yield	• Protect seedlings with mulching which lowers the soil temperature during the day by insulating it from incoming solar radiation and conserving water. Further, mulch also reduces soil cooling at night. • Sprinkler irrigation at seedling stage during evening lowers the soil temperature
Cold wave Yellowing of leaves and tip burning due to cold wave	• Leaf chlorosis, burning of leaf tips and silage odour. Leaves turn dark green/ blue and look water soaked • Stems may split or develop lesions and form additional tillers (low yielders)	• To late sown wheat, apply second dose of N with first irrigation • Apply light and frequent irrigation to the standing crop and arrange for smoking around the field to prevent cold / frost injury
Hail storm Crop damage due to hailstorm	• Pronounced physical damage resulting in broken stems or damaged heads. • Hail damage at the boot stage may cause the head to be entrapped in the boot, not allowing it to emerge normally, which results in deformation of the head • Damage to the whole head or only slight damage to areas struck by hail stones • Damage to spikelets, knocking of grain out of florets, breaking of stems over and usually results in shriveled kernels in these heads	Apply light and frequent irrigation
Frost White ring in peduncle due to frost	• Frost affected plants remain high in sugars and nutrients because they have no grain to fill thus encourage microbial infection which turn the crop black or brown. • Most characteristic symptoms of frost injury is the 'White ring' on the peduncle. • Frost damage during grain filling can cause grain shrivelling.	Apply light and frequent irrigation to the standing crop and arrange for smoking around the field to prevent from frost injury.

2. Rice

India is the world's second largest producer of rice with a production of 99.2 m tonnes from an area of 45.5 m ha, accounting for 20% of the world's rice production. It is the main staple food for the people in southern and eastern India. Irrigated rice which is cultivated in about 22 million ha and contributes 61% of the total rice production, is extensively grown in the states of Punjab, Haryana, Uttar Pradesh, West Bengal, Andhra Pradesh, Tamil Nadu and Karnataka. Rainfed low land rice is cultivated in about 14.4 million ha in the states of eastern Orissa, West Bengal, UP, Bihar, Jharkhand, Chattisgarh and Assam contributes about 30% of the rice production. About 7% of the rainfed rice area is also flood prone and contributes only 4% of the total production.

Rice cultivation at nursery, vegetative, reproductive and maturity stages is prone to extreme events such as flash floods due to excess and intense rains in a short span of time and cyclones. In coastal areas, cyclonic storms during October to December are responsible for submergence of the crop at ripening stage (Fig. 22) or just after harvested leading to enormous loss in production and hardship to farmers. As an illustration, contingency measures for flood and cyclone are described for Andhra Pradesh, Tamil Nadu and Orissa.

Fig. 22. Rice crop heavily damaged due to floods

Extreme Event	State (affected districts)	Suggested measures
Floods	**Andhra Pradesh (Coastal districts)**	**In case of partial damage at nursery stage:** • Take up gap filling. If the seedlings are not available for gap filling, the survived hills are to be split into i ndividual tillers and used for gap filling. Swarna and Chaitanya varieties perform well with split tillers and can withstand submergence for about one week. • A booster dose of 25 kg of urea and 15 kg of MOP / acre are to be applied to hasten establishment a nd promotion of tillering. Later, nitrogen may be applied at panicle initiation stage as per normal recommendations **In case of complete damage:** • Take up replanting before 31st August after raising nurseries with short duration varieties like NLR-34449, NLR-145, Swarna, MTU-1010, IR-64, IET-1444 and JGL-1798. • Adopt spacing of 15x15cm (44 hills/m^2) • N, P, K @ 180:40:30 kg/ha is recommended. 50% of the recommended N and entire P & K should be applied as basal dose before planting and 25% of the N should be appl ied at 20 days after planting . If needed, spray 0.2% Zinc sulphate solution 2-3 times at 4 -5 days interval. • A final dose of 25% recommended N should be applied at panicle initiation stage **In standing crop (vegetative and reproductive stages):** • Drain out excess water • Apply booster dose of 50 kg N/ha • Spray Zinc sulphate 0.2% up to 45 days after transplanting
Cyclones	**-Do-**	**At crop maturity and harvest:** • Drain out excess water • Spray common salt 5% solution to prevent seed germination (and grain discolouration) and spoilage of straw from moulds • If the crop is caught in cyclone or tidal waves at maturity stage, the fields need to be drained and paddy sheaves threshed immediately • For raising the second crop of paddy after sea water inundates the field (Tsunami) , salination can be reclaimed by flooding with good quality irrigation water, leaching the salts and draining the fields followed by puddling • Application of superphosphate (100 kg/ha) in the field will hasten up the decomposition of stubbles for preparing the field for second crop • Nurseries need to be raised in non-saline areas • In nurseries which come up after a week's submergence, apply about 50 - 100 kg superphosphate mixed and pre-incubated with 200 kg soil/ha • If the case of complete damage , fodder shortage would be severe. Fodder crops like cowpea, sunhemp may be grown in marginal lands unfit for rice cultivation • Catch crops like oilseeds and pulses can be raised wherever rice is not possible
Floods	**Orissa (eg. Puri)**	• Adoption of varieties tolerant to submergence such as Swarna sub-1 and IR.64 sub-1 • Drain out excess water • Avoid beushening in partially damaged fields • Apply a fertilizer dose of 40:20:20 NPK / ha • Gap filling to be done by redistributing the tillers. • If damage is more than 50%, re-transplant with medium duration varieties. Raise nursery by Dapog method for transplanting, wherever possible. • Maintain a buffer nursery in the backyard/highland area to ensure adequate plant population in the field after flood damage • Plan for *rabi* crops such as blackgram / greengram or groundnut

3. Onion

Onions is cultivated in three distinct seasons across the country namely *kharif* (June to October), late *kharif* (September to February) and *rabi* / summer (December to April). Early *kharif* is also practiced in some parts of the country. The major onion growing states in the country Maharashtra, Orissa, Bihar, Andhra Pradesh, Karnataka and Tamilnadu. The share of Maharashtra in the total production is around 30%. (Samra *et al.,* 2006). The state produces bulbs around 1.2 million metric tons annually. Within Maharashtra, Nashik is the leading producer followed by Pune, Ahmednagar and Satara. Excess and unusual rains cause profound influence on the performance of the crop leading to considerable losses to farmers, resulting in exorbitant prices for consumers due to short supply immediately following the post-rainy season. Excess rainfall events during September and October adversely affected (Fig. 23 and 24) the *kharif* crop in Nashik district during 1998, 2005 and 2010 (Fig. 25)

Fig. 23: Purple blotch

Fig. 24: Basal rot

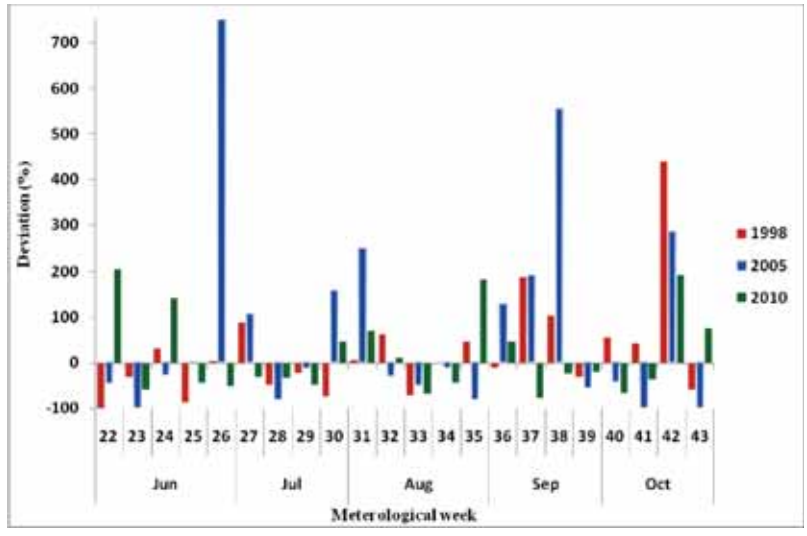


Fig. 25 Rainfall deviations from normal during 1998, 2005 and 2010 at Nashik

Impact of excess rainfall on onion and suggested measures

Condtion	Effects	Suggested measures
Unusual excess rains during September - October	• When excess rainfall coincides with bulb initiation and bulb development phases causes major damage to *kharif* and late *kharif crops* as well as nursery stages of *rabi* crops • 2-3 days continuous wet spell damages onion bulbs and triggers diseases like purple blotch, basal rot, twister, *Colletotrichum* blight, and brown rot	• Staggered sowing of Nursery facilitates gap filling / re-sowing in case of crop failure due to excess rains • Provide proper drainage to remove excess water and avoid surface ponding • Practice broad bed furrow planting • Apply foliar spray of mancozeb @ 0.25% or carbendazim @ 0.1% to protect against purple blotch, twister, blight and rot diseases • Rotate onion with aster or other crops in field patches to reduce disease inoculum load • Setting up of quick hot drying facilities for pumping dry hot air in storage structures

4. Mango

Mango occupies 22% of the total area under fruits comprising of 1.2 million hectares, with a total production of 11 million tonnes and accounts for 40% of total fruit exports from India. Uttar Pradesh and Andhra Pradesh have the largest area under mango covering 25% of the total area followed by Bihar, Karnataka, Kerala and Tamil Nadu. Andhra Pradesh ranks first in mango production with a share of 24% of total production. Mango is grown extensively in Krishna, Vijayanagaram, Visakhapatnam, West and East Godavari, Kadapa, Warangal, Nalgonda, Adilabad, Medak and Rangareddy districts of Andhra Pradesh.

Mango production and productivity in Andhra Pradesh have undergone serious fluctuations due to weather aberrations such as unseasonal/ untimely rains and heat wave conditions. The untimely rains in 2010 during November and December and in 2011 during February have hit mango production badly. The second spell of rain during February led to partial flower drop and further dip in production. Untimely rains were responsible for outbreak of pests and diseases such as hopper, powdery mildew and *Anthracnose* (Fig. 26 and 27). Similarly, excessive heat and rain during the month of November, 2009 destroyed flowering during the months of November and December in A.P. Also the second round of flowering yielded only 10% of the crop in the year 2010.

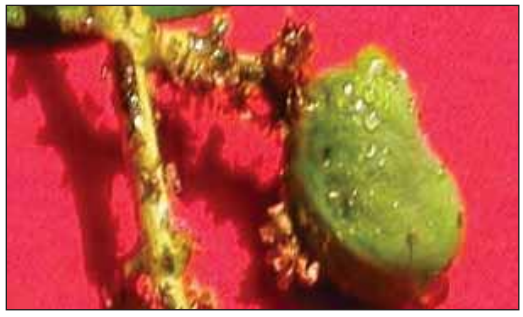


Fig. 26: Fruit drop due to hoppers

Fig. 27: Mangoes badly damaged due to *Anthracnose*

Extreme event	Effects	Suggested measures
Untimely rains	**Vegetative Phase (young plantation)** • High water table and soil moisture affects root growth and functioning of active roots • Lodging of young seedlings may occur • Increased susceptibility to thrips and powdery mildew • Broken and damaged branches	• Provide proper drainage to remove the excess water from the active root zone • Staking of young seedlings to prevent lodging • Adopt plant protection measures • Pruning of damaged branches and application of Bordeaux paste (0.1%)
	Flowering & Fruiting Phase : • Mango needs a dry period of 3 to 5 months to induce maturity of vegetative parts and flowering. Rains during this period hinder flowering and lead to more vegetative growth. Night temperatures around 15°C during December and January, are congenial for flowering, this conducive temperature is interrupted due to untimely rains • Rain water during flower bloom washes off pollen affecting pollination • Normal flower and fruit drop may be accentuated under excess rainfall • Creates favourable conditions for the spread of hoppers and diseases like powdery mildew and *Anthracnose*	• Provide proper drainage to avoid water logging • Protect from strong winds especially during flowering and fruiting stage by growing windbreaker trees around the orchard • 2% potassium nitrate spray at full bloom improves fruit set and fruit retention **Control measures for hopper** • Use light traps during early stages of flower development to attract and kill adult hoppers. One light trap is required per hectare of mango plantation. • Prune the crowded branches to allow good light penetration and make the habitat unfavorable for hopper development • Apply three sprays of 0.15% carbaryl or 0.04 % monocrotophos. First spray should be given at the early stage of panicle formation. Apply the second spray at full length stage of panicles but before full bloom and the third spray after fruit set and fruits have attained pea stage **Control measure for powdery mildew** • Apply three sprays of fungicides: wettable sulphur 2g / l or tridemorph 1ml/l or dinocap 1 ml/ l at 15 days interval for effective control of the disease **Control measures for *Anthracnose*** • Trees may be sprayed twice with carbendazim (1g/l) at 15 days interval during flowering to control blossom infection. Application of copper fungicides (0.3%) controls foliar infection • Postharvest dip treatment of fruits in hot water at 52±1°C for 15 minutes
	Harvesting phase: • Drop of fully matured fruits • Incidence of powdery mildew and *Anthracnose*	Collect fallen fruits and use in the preparation of processed products

Extreme event	Effects	Suggested measures
Heat wave	• For pollination and fruit maturation, 30-36ºC temperature is ideal. 41ºC temperature burns the flowers and reduces yield • Fruits, leaves and trunk when exposed to direct sun and excessive heat show signs of scorching (sunscald) (Fig.28 & 29).	• Light and frequent irrigation during flowering • Light summer pruning after fruit set ensures good vegetative growth which can protect fruits from sunscald • Trunk may be whitewashed with lime to protect from sunscald • Mulching • Post harvest exposure to low temperature of 10-15ºC for 10-18 h

Fig. 28: Dried up panicle due to heat wave

Fig. 29: Sunscald on fruits

5. Apple

India is the seventh largest producer of Apples. Himachal Pradesh has an area of 1 lakh ha under apple production and is the second largest producer after Jammu and Kashmir. Shimla district ranks as the largest producer of apples, followed by Kullu, Kinnaur, Mandi, Chamba, and Solan districts. Apple is being grown at altitudes of 1500 to 2700 m in the Himalayan range which experiences 1000-1500 hours of chilling (the number of hours during which temperature remains at or below 7°C during the winter season). The temperature during the growing season is around 21-24°C. For optimum growth and fruiting, apple trees need about 100-125 cm of annual rainfall evenly distributed during the growing season. The optimum temperature for pollen germination and fruit is 21.1-26.7°C. The apple crop, in its tender stage of flowering, requires warm and sunny days with a temperature above 18°C.

Weather aberrations such as hailstorms, unseasonal rains and thunderstorms are being experienced in the apple growing states. In recent years the number of days with hail is about 6-7 per year over Himachal Pradesh. Hailstorms resulted in a loss of Rs 300 crores during 2010 in the state out of which loss in Shimla district alone was Rs 248 crores. Also, prevalence of warmer temperatures during winter season adversely affects the chilling requirements of the crop in some blocks viz., Theog, Jubbal, Narkanda, Rohru and Chirgaon. Temperature fluctuations such as acute cold wave and frost, deficit or excess rainfall are some of the other weather related contingencies the crop faces in Shimla district. Hail and frost cause extensive physical damage (Fig. 30 & 31).

Fig. 30: Physical damage to fruits due to hail

Fig. 31: Leaf damage due to frost

Extreme event	Effects	Suggested measures
Hail storm	• Damages apple bloom and causes bruises on fruits	• Use hail nets to prevent physical damage • Use of anti-hail gun technology which is currently under pilot testing • Damaged fruits to be marketed for preparing processed products • After the event, foliar spray of urea (1%), carbendazim (0.1%) and micronutrients to reinvigorate the trees
Cold wave and frost	• Low temperatures below 15°C during flowering restricts bee activity while complete inhibiton occurs at temperatures below 4.4°C, affecting fruit set • Cold injury to fully opened blossoms at temperatures below zero	• Smoking of trees by burning organic wastes in the orchards to raise temperature • If the frost is not severe, the tree can be wetted with water spray before the morning sun hits the flowers
Unseasonal Rains	**Vegetative stage:** • Excessive soil moisture slows root growth, leaches nitrogen, sulfur and boron out of the root zone leading to nutrient deficiencies. • Excessive soil moisture also provides ideal environment for crown and collar rots and scab on leaves. Also induces excessive vegetative vigor **Flowering and fruiting**: • Continuous rains washes away pollen and also hinders bee activity resulting in reduced fruit set	• Excessive vegetative growth can be checked by foliar application of 500 - 1000 ppm Ppaclobutrazol • Apply mancozeb at 600g/200 litres of water to protect from scab
Rise in winter temperatures	• Warmer winters lead to reduction of chilling hours. A winter chilling of less than 800 h below 7°C is insufficient for breaking dormancy of delicious varieties	• Choose varieties with lower chilling requirements while going for new plantations

Chapter VI

Operationalization of Contingency Plans

To implement the contingency plans described in Chapter 4 requires extensive planning both at the district and state level which need to be coordinated and facilitated by Government of India. Currently, at the Government of India level, the crop-weather watch group under the Ministry of Agriculture monitors the weather situation and helps the ministry to coordinate the preparedness for droughts and other contingencies. At the state level, the Commissioner, Department of Agriculture monitors the weather situation at block level through the State planning department cells located at the district level along with progress of sowings, storage in different reservoirs and prepares a weekly report. In most states, the Relief Commissioners coordinate the overall implementation of the contingency plans during droughts, floods and other natural hazards after notification by the Government. However, the responsibility of planning agriculture contingencies including organizing seed and other inputs wrests with the Agriculture Production Commissioner. At the district level, the Collector or District Magistrate chairs such contingency planning meetings involving all senior officers from agriculture, irrigation, animal husbandry and fishery departments.

The current efforts are basically to save the crop season and minimize the losses with some broad interventions at the agro-climatic zone or district level. Though the Government may wait for certain time before declaration of drought, the district team needs to gear up as the information on rainfall and progress of sowings are collected continuously during the season. Similarly, in case of irrigation commands, delays in release can be assessed based on inflows into reservoirs. In case of flood, the district team needs to coordinate with irrigation authorities to avoid excess release from reservoirs leading to submergence of crop lands. In case of cyclones the available time for response would be less and precautions need to be taken up to avoid flooding etc. The DLCPs presented in this report are quite exhaustive giving details of the alternate crops and varieties in case of delay in monsoon for each major farming situation within the district. With advance planning and organizing inputs in time, these plans can help the farmers and reduce the losses in the affected areas.

Role of Weather Forecasting

Currently, IMD gives an advance forecast of monsoon in the month of April for the country as a whole. In this forecast, the overall quantum of rainfall whether surplus or deficit is announced at the national level. The likely date of onset in Kerala is announced in the middle of May. The forecast is revised in the month of June where quantitative forecast is given for four broad geographical regions of India (North-West India, North-East India, Central India and Southern Peninsular) and also the monthly

rainfall for July and August. At the State level, short and medium range forecast is given by the regional meteorological centres for the next 2 to 5 days period during the monsoon season. This forecast information can be used as a base to plan the contingencies at different levels as the season progresses.

Identifying Vulnerable Districts

Based on long term historical data, the districts can be categorized as drought prone, flood prone, cyclone prone and vulnerable to heat wave or cold wave. This information is also given in part I of the DLCPs. Such districts should be put on alert every year irrespective of the pattern of monsoon so that in case of any rainfall aberration, action can be immediately initiated and the plans put into action.

Special Pre-*kharif* Meetings

Currently, the Government of India and the states hold *kharif* and *rabi* planning meetings to draw out the overall strategy and production targets for the year. However, it is suggested that immediately after this, a separate meeting may be held at the State level exclusively focused on coping with weather aberrations and how to tackle the contingencies, if arisen including the preparedness of the departments with inputs and the strategies to dovetail funds under different schemes. Representatives from IMD, State Agricultural Universities and local ICAR institutes may be invited for this meet. Such a meeting is also to be held at the district level, where officers from agriculture, animal husbandry, fisheries, power distribution and irrigation may attend. The copies of the DLCPs are to be distributed to the participants and preparedness discussed in detail. Joint Director, Agriculture may act as Convener of this group. It should meet as frequently as possible and draw up the action plans as the season progresses and the weather situation unfolds.

In season Review and Implementation

Once a given district or agro-climatic zone is identified either for drought or floods, the State and district administration may constantly review the situation during the season and take appropriate steps for effective implementation of the DLCPs. In case of droughts, the progressive rainfall deficits and the information on crop situation coming from the blocks/districts will be used to execute the plan at appropriate locations. Mobilizing inputs particularly seeds of alternative crops and varieties as listed in the DLCPs is critical. For this, a district-wise seed production and storage plan need to be put in place as a medium term strategy. The details are discussed later in the chapter. In case of cyclone warning, suitable agro-advisories are to be given by the agricultural universities and the department of Agriculture through the available dissemination channels so that farmers can take appropriate and timely decisions regarding either harvesting a crop or postponing, giving or not giving irrigation etc., so that the loss is minimized. Ensuring power or fuel supplies and need based release of irrigation water are part of this strategy.

Seed Production and Storage

While the above steps guide the implementation of DLCPs in the current season, some medium term strategies are required for enabling this implementation. The most important input is the seeds of the crops and varieties that can be sown in case of delayed monsoon. Though, the agricultural universities and ICAR institutes have recommended several such crops and varieties for delayed sowing, the seeds of these crops and varieties are not readily available as there is no regular demand for them and also they are not in the normal seed chain. In many cases, the breeder seed is not available easily. In the DLCPs, efforts are made to provide the source of breeder seed wherever alternative crops/varieties are suggested.

To address this problem, the seeds of crops and varieties to be used in case of delayed monsoon have to be produced every year whether they are eventually used or not. In case the monsoon is normal and the seeds are not used, they may be disposed off as grain. Appropriate seed storage mechanism including godowns and fumigation systems have to be evolved along with the locations where seeds are to be stored depending on the cropping pattern in a region and the weather vulnerability. These seed storage hubs are key to the successful implementation of the DLCPs. The financial modalities of the cost of production of such seeds, transport, storage and the loss incurred when disposed as grain need to be worked out and budgeted.

Other Associated Measures

There are a number of medium and long term strategies which are critical for sustained implementation of DLCPs as and when monsoon aberrations occur. These include restoration and renovation of drains in flood prone, delta areas, flat lands and cyclone affected areas to minimize the damage of floods and inundation of either the standing or harvested crop, and use of MGNREGA and IWMP programs for undertaking drought proofing works in a planned manner. Others include, rainwater harvesting for supplemental irrigation and soil organic carbon build up through biomass recycling so that the water holding capacity of the soils increases and the crops can cope with the dry spells. River bank protection is another important step to avoid flooding and sand casting of the crop fields due to heavy unseasonal rains. In chronically drought prone districts, agroforestry helps in building resilience to drought both by improving soil fertility and providing fodder for the livestock. Lastly, promoting weather insurance is another step which will minimize the risk to farmers due to monsoon aberrations.

References:

Acevedo, E., Silva, P. and Silva, H. 2006. Wheat growth and physiology, FAO Corporate Document Repository.

Chattopadhyay, N., and Lal, B. 2007. Agrometeorological Risks and Coping strategies-Perspective from Indian Subcontinent. *In*: Mannava V.K.Sivakumar and Raymond P. Motha (editors), Managing Weather and Climate Risks in Agriculture. Springer Publications, New York. Pp: 83-98.

De, U.S., Dude, R.K. and Prakasa Rao, G.S. 2005. Extreme Weather Events over India in the last 100 years, Indian Geophysical Union 9: 173-187.

Ferris, R., Ellis, R. H., Wheeler T. R. and Hadley. P. 1998. Effect of High Temperature Stress at Anthesis on Grain Yield and Biomass of Field grown Crops of Wheat. Annals of Botany. 82: 631-639.

Ghosh, S.P.1991. Agro Climatic Zone specific Research-Indian Perspective under NARP. Publication and information Division, ICAR, New Delhi. p.4.

Guha, G.S. and Basu, D.N.1996. Agro-Climatic Regional Planning in India. Vol. 1. Concepts and Applications. p. 38.

Livestock Census. 1989. Govt of Rajasthan.

Orissa Disaster Mitigation Mission (ODMM) 2000. Super Cyclone and Rehabilitation. Occasional Paper. Orissa Rehabilitation, Social Development Consultants (SDC), New Delhi. p 412.

Prakash Rao and Vivekanandan, E. 2008. Impact of Climate Change on Indian Marine Fisheries. Bay of Bengal News, March – June, Pp: 32-37.

Rathore, L.S. and Stigter, C.J. 2007. Challenges to coping strategies with agrometeorological risks and uncertainties in asian regions. *In*: Mannava V. K.Sivakumar and Raymond P. Motha (editors), Managing Weather and Climate Risks in Agriculture. Pp: 53-66.

Samra, J.S., Gurubachan Singh and Ramakrishna, Y.S., 2003. Cold Wave of 2002-03 - Impact on Agriculture. Natural Resource Management Division, ICAR, New Delhi. p. 16.

Samra, J.S., Ramakrishna, Y.S., Desai, S., Subba Rao, A.V.M., Rama Rao, C.A., Reddy, Y.V.R., Rao, G.G.S.N., Victor, U.S., Vijaya Kumar, P., Lawande, K.E., Srivastava, K. and Krishna Prasad, V.S.R. 2006. Impact of excess rains on yield, market availability and prices of onion. Information Bulletin - Central Research Institute for Dryland Agriculture (ICAR), Hyderabad. p. 52.

Sanjoy Hazarika. 2006. Living intelligently with floods. Background Paper No. 5. Centre for North East Studies, India. p. 2.

Sehgal, J., Mandal, D.K. and Mandal, C.1999. Agro-ecological subregion map of India. NBBSS & LUP, Nagpur.

Vijayakumar, P. and Rao, G.G.S.N. 2010. Climate change and its impact on wheat in two extreme geographical locations of Indo Gangetic plains. In Extended Summaries (Vol.I). National Symposium on Climate Change and Rainfed Agriculture, Indian Society of Dryland Agriculture,18-20 February, 2010. CRIDA, Hyderabad. p. 448.

Vyas Pandey and Patel, H. R. 2008. Impact of climate change on agriculture over Gujarat. *In:* G.S.L.H.V. Prasada Rao, G.G.S.N. Rao, V.U.M. Rao and Y.S. Ramakrishna (editors), Climate Change in Agriculture over India. Pp: 163-181.

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Acronyms

AAU	Anand Agricultural University
ACR	Agro climatic Region
ACWC	Area Cyclone Warning Center
ACZ	Agro Climatic Zone
AESR	Agro Ecological Sub Region
AICRP	All India Coordinated Research Project
ANGRAU	Acharya NG Ranga Agricultural University
APHU	Andhra Pradesh Horticultural University
APSSDC	Andhra Pradesh State Seeds Development Corporation
ARS	Agricultural Research Station
BAU	Birsa Agricultural University
BCKVV	Bidhan Chandra Krishi Viswavidyalaya
BPH	Brown Plant Hopper
BQ	Black Quarter
BSKKV	Dr Balasaheb Sawant Konkan Krishi Vidyapeeth
BT	Blue Tongue
CAU	Central Agricultural University
CAZRI	Central Arid Zone Research Institute
CCSHAU	Chaudhary Charan Singh Haryana Agricultural University
CGWB	Central Ground Water Board
CLDP	Community Land Development Programme
CRIDA	Central Research Institute for Dryland Agriculture
CRRI	Central Rice Research Institute
CSAUA&T	Chandra Shekar Azad University of Agriculture & Technology
CSKHPKVV	CSK Himachal Pradesh Krishi Vishvavidyalaya
CSWCRTI	Central Soil And Water Conservation Research and Training Institute
CWC	Central Water Commission
CWC	Cyclone Warning Centers
CWDS	Cyclone Warning Dissemination Systems
DAATTC	District Agricultural Advisory and Transfer of Technology Centre
DAC	Department of Agriculture and Cooperation
DACNET	Department of Agriculture and Cooperation Network
DARE	Department of Agricultural Research and Education
DAS	Days After Sowing
DG	Director General
DLCP	District Level Contingency Plan
DM	Disaster management
DoS	Department of Space
DWM	Directorate of Water Management
EFW	Early Fore Warning
ENSO	El-Nino Southern Oscillation
EWS	Early Warning Systems
FMD	Foot and Mouth Disease
GADV&ASU	Guru Angad Dev Veterinary and Animal Sciences University

GBPUA&T	Govind Ballabh Pant University of Agriculture & Technology
GDP	Gross Domestic Product
GIS	Geographical Information System
GoI	Government of India
HS	Haemorrhagic Septicemia
HYV	High Yielding Varieties
IBD	Infectious Bursal Disease
ICAR	Indian Council of Agricultural Research
ICAR RCER	ICAR Research Complex for Eastern Region
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IIT	Indian Institute of Technology
IMD	India Meteorological Department
INCOIS	Indian National Center for Ocean Information Services
IPM	Integrated pest management
IRAD	Institute of Agricultural Research for Development
ISOPOM	Integrated Scheme on Oilseeds, Pulses, Oil palm and Maize
IWMP	Integrated Watershed Management Programme
JAU	Junagadh Agricultural University
JNKVV	Jawaharlal Nehru Krishi Viswavidyalaya
KAU	Kerala Agricultural University
KED	Krishna Eastern Delta
KVAFSU	Karnataka Veterinary, Animal and Fisheries Sciences University
KVK	Krishi Vigyan Kendra
MAFSU	Maharashtra Animal Sciences & Fishery University
MAU	Marathwada Agricultural University
MGNREGS	Mahatma Gandhi National Rural Employment Guarantee Scheme
MoA	Ministry of Agriculture
MPKV	Mahatma Phule Krishi Vidyapeeth
MPUA&T	Maharana Pratap Univ. of Agriculture & Technology
MSL	Mean Sea level
MSSC	Maharashtra State Seed Corporation
NAA	Napthalene Acetic Acid
NAIS	National Agricultural Insurance Scheme
NARP	National Agriculture Research Project
NAU	Navsari Agricultural University
NBSS&LUP	National Bureau of Soil Survey and Lane Use Planning
NDMA	National Disaster Management Authority
NDUA&T	Narendra Dev University of Agriculture & Technology
NE/SW monsoon	North East/South West monsoon
NEH	North Eastern Region
NFSM	National Food Security Mission
NGO	Non Governmental Organisation
NHM	National Horticulture Mission
NLM	National Livelihood Mission

NRC	National Research Centre
NRM	Natural Resource Managemenet
NSC	National Seed Corporation
NSKE	Neem Seed Kernal Extract
NSP	Nagarjuna Sagar Project
ODMM	Orissa Disaster Mitigation Mission
OUA&T	Orissa University of Agriculture & Technology
PAU	Punjab Agricultural University
PDFSR	Project Directorate on Farming Systems Research
PDKV	Dr Panjabrao Deshmukh Krishi Vidyapeeth
PPR	Peste des Petits Ruminants
PSB	Phosphate Solubilizing Bacteria
RADP	Rainfed Area Development Programme
RARS	Regional Agricultural Research Station
RAU	Rajasthan Agricultural University
RD	Ranikhet Disease
RF	Rainfall
RKVY	Rashtriya Krishi Vikas Yojana
RRS	Regional Research Station
RRTTS	Regional Research & Technology Transfer Station
RSC	Residual Sodium Carbonate
RVSKVV	Rajmata Vijayraje Scindia Krishi Vishwa Vidyalaya
SAU	State Agricultural University
SBBPUA&T	Sardar Ballabh Bhai Patel University of Agriculture & Technology
S-E-KUAS&T, J&K	Sher-E-Kashmir University of Agricultural Sciences & Technology
SGWB	State Ground Water Board
SKDAU	Sardarkrushinagar Dantiwada Agricultural University
SRI	System of Rice Intensification
SVVU	Sri Venkateswara Veterinary University
TNAU	Tamilnadu Agricultural University
TNVASU	Tamil nadu Veterinary and Animal Sciences University
UAS	University of Agricultural Sciences
UBKVV	Uttar Banga Krishi Viswavidyalaya
UHS	University of Horticultural Sciences
UP	Uttar Pradesh
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VC	Vice Chancellor
WBCIS	Weather Based Crop Insurance Scheme
WBUA&FS,Kolkata	West Bengal University Of Animal & Fishery Sciences
WIHG	Wadia Institute of Himalayan Geology
WMO	World Meteorological Organisation
YSPUH&F	Y.S.Parmar University of Horticulture & Forestry
ZARS	Zonal Agricultural Research Station
ZPD	Zonal Project Directorate
ZRS	Zonal Research Station